resource_util vs resource_util_max

Side-by-Side Comparison

COMP 597 — Energy Profiling of Whisper Fine-Tuning

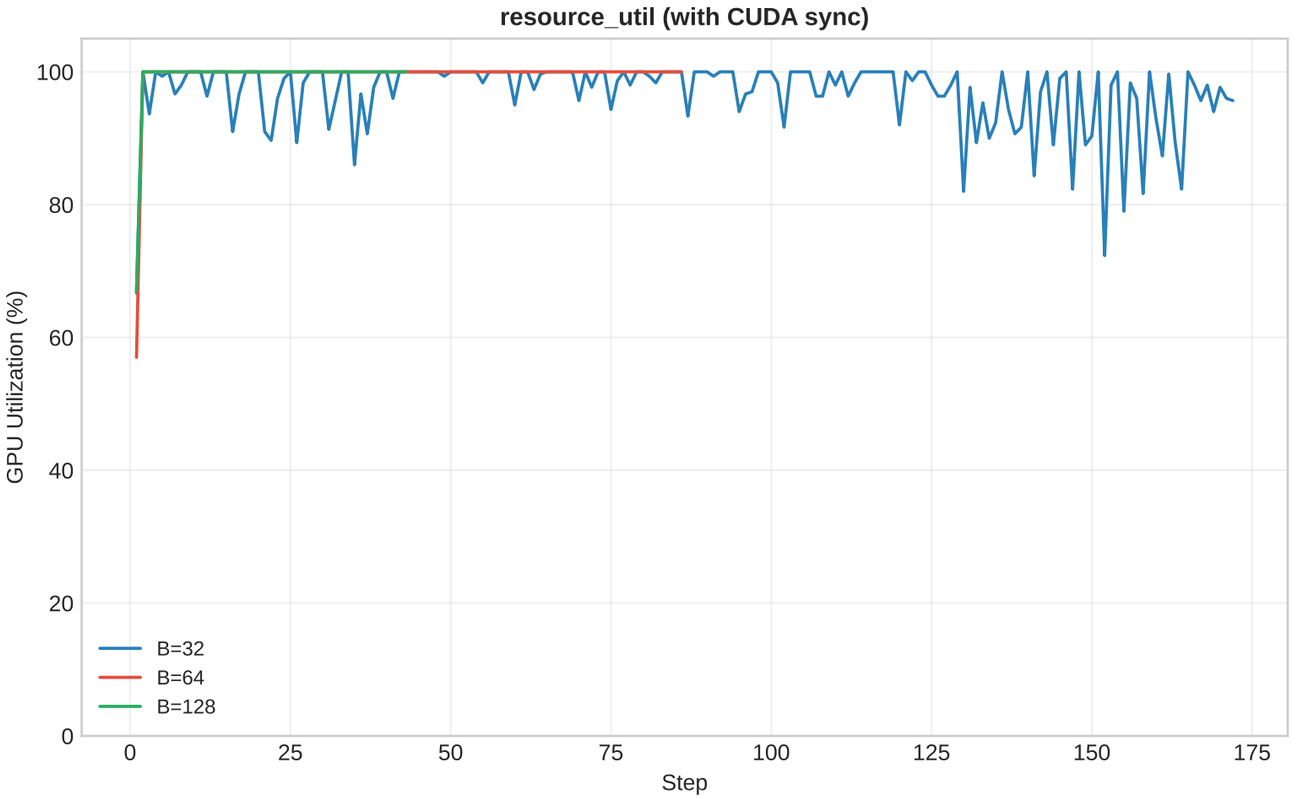
```
resource_util: per-step GPU/CPU metrics with CUDA synchronization
 • GPU Memory = torch.cuda.memory_allocated() (tensor memory only)

resource_util_max: per-step GPU/CPU metrics without CUDA sync
• GPU Memory = pynvml.nvmlDeviceGetMemoryInfo().used (total GPU memory)
```

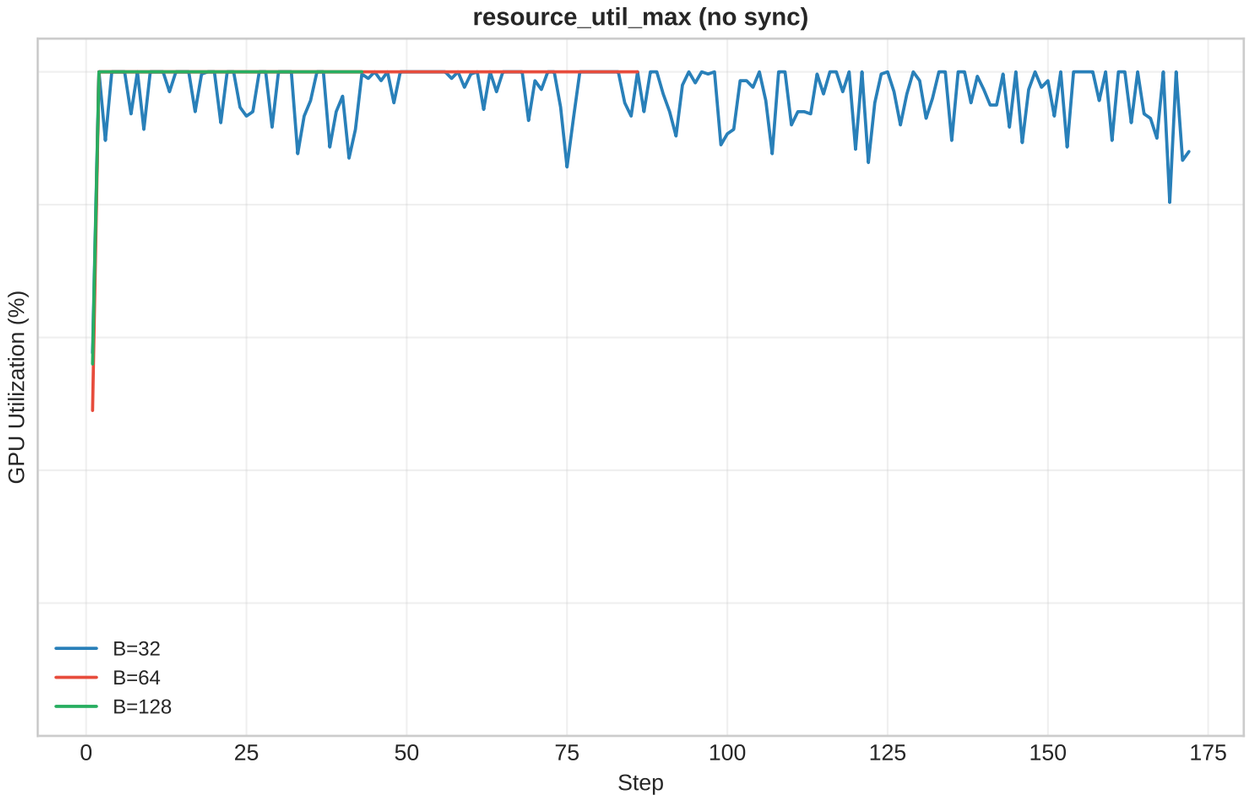
Blue = resource_util | Orange = resource_util_max

Summary: resource_util vs resource_util_max
(GPU Mem columns measure different things — see header labels)

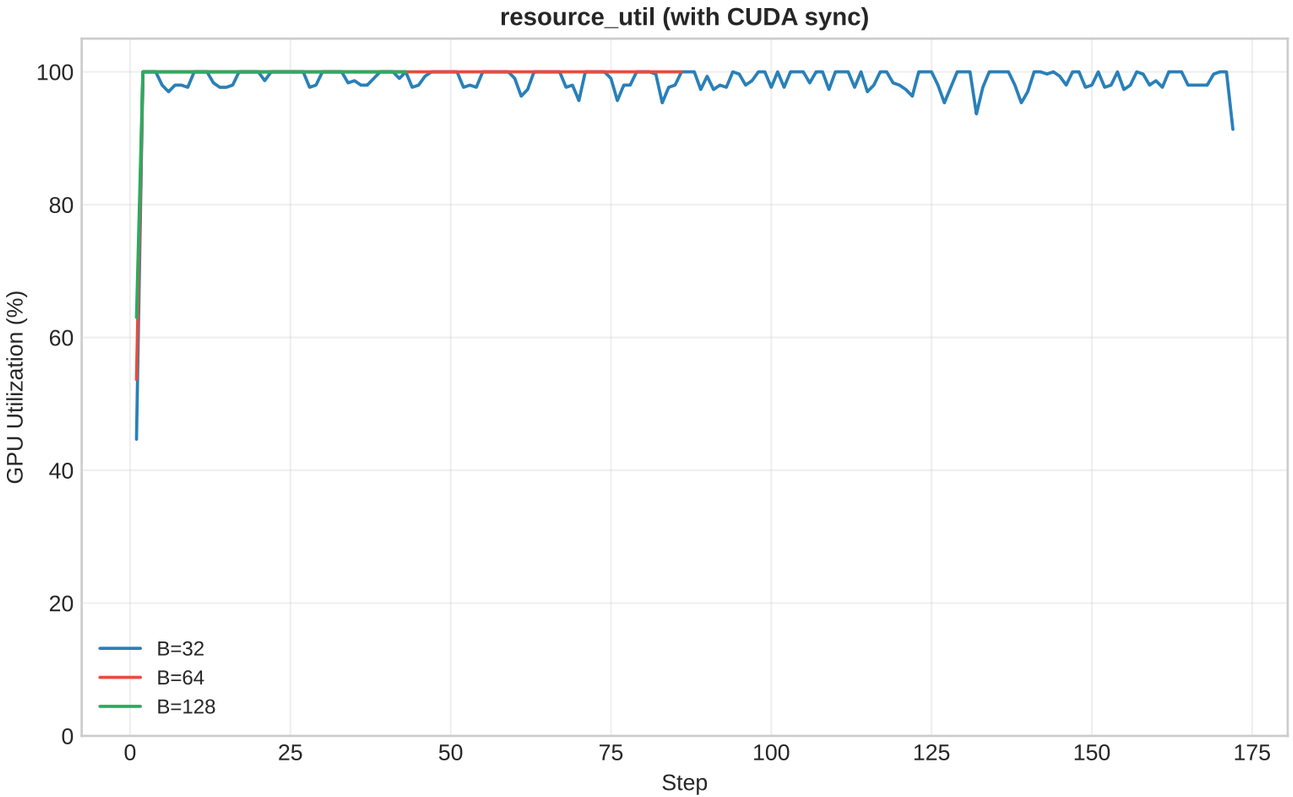
Data	W	B	Trainer	Steps	GPU Util Mean	GPU Util Std	CPU Util Mean	GPU Mem (torch alloc)	GPU Mem (NVML used)
Disk	0	32	res_util	172	97.1%	4.8	132.1%	0.14 GB	nan
Disk	0	32	res_util_max	172	96.9%	4.1	127.1%	nan	6.81 GB
Disk	0	64	res_util	86	100.0%	0.0	128.6%	0.14 GB	nan
Disk	0	64	res_util_max	86	100.0%	0.0	124.4%	nan	12.49 GB
Disk	0	128	res_util	43	100.0%	0.0	128.2%	0.14 GB	nan
Disk	0	128	res_util_max	43	100.0%	0.0	122.0%	nan	23.94 GB
Disk	4	32	res_util	172	98.9%	1.4	98.8%	0.14 GB	nan
Disk	4	32	res_util_max	172	98.6%	1.7	98.8%	nan	6.81 GB
Disk	4	64	res_util	86	100.0%	0.0	98.6%	0.14 GB	nan
Disk	4	64	res_util_max	86	100.0%	0.0	98.7%	nan	12.49 GB
Disk	4	128	res_util	43	100.0%	0.0	95.3%	0.14 GB	nan
Disk	4	128	res_util_max	43	100.0%	0.0	96.8%	nan	23.94 GB
Milabench	0	32	res_util	500	99.6%	0.6	110.1%	0.14 GB	nan
Milabench	0	32	res_util_max	500	99.6%	0.6	109.9%	nan	6.81 GB
Milabench	0	64	res_util	250	100.0%	0.0	109.7%	0.14 GB	nan
Milabench	0	64	res_util_max	250	100.0%	0.0	110.1%	nan	12.49 GB
Milabench	0	128	res_util	125	100.0%	0.0	107.1%	0.14 GB	nan
Milabench	0	128	res_util_max	125	100.0%	0.0	107.5%	nan	23.94 GB
Milabench	4	32	res_util	500	99.1%	1.3	98.8%	0.14 GB	nan
Milabench	4	32	res_util_max	500	99.1%	1.3	99.9%	nan	6.81 GB
Milabench	4	64	res_util	250	100.0%	0.0	100.0%	0.14 GB	nan
Milabench	4	64	res_util_max	250	100.0%	0.0	100.1%	nan	12.49 GB
Milabench	4	128	res_util	125	100.0%	0.0	100.0%	0.14 GB	nan
Milabench	4	128	res_util_max	125	100.0%	0.0	100.0%	nan	23.94 GB

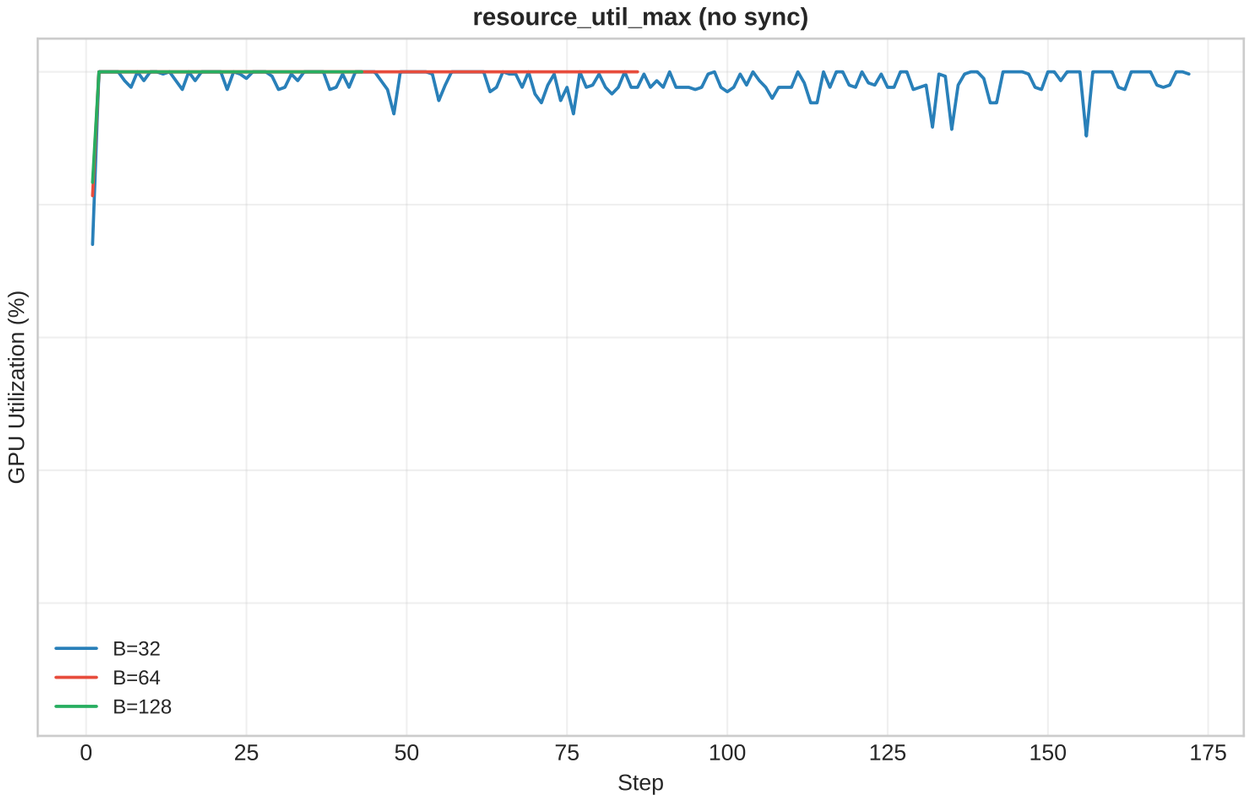
GPU Utilization Across Batch Sizes — Disk, w=0



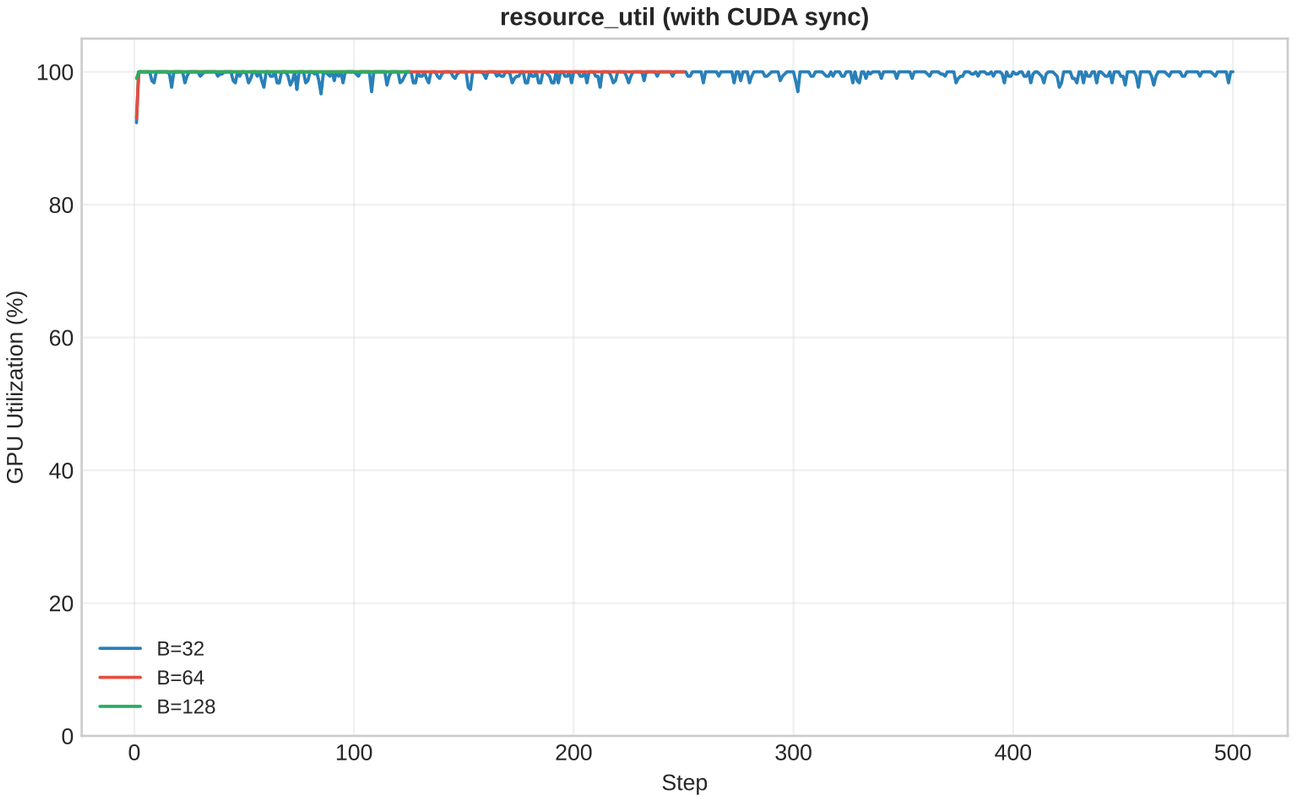


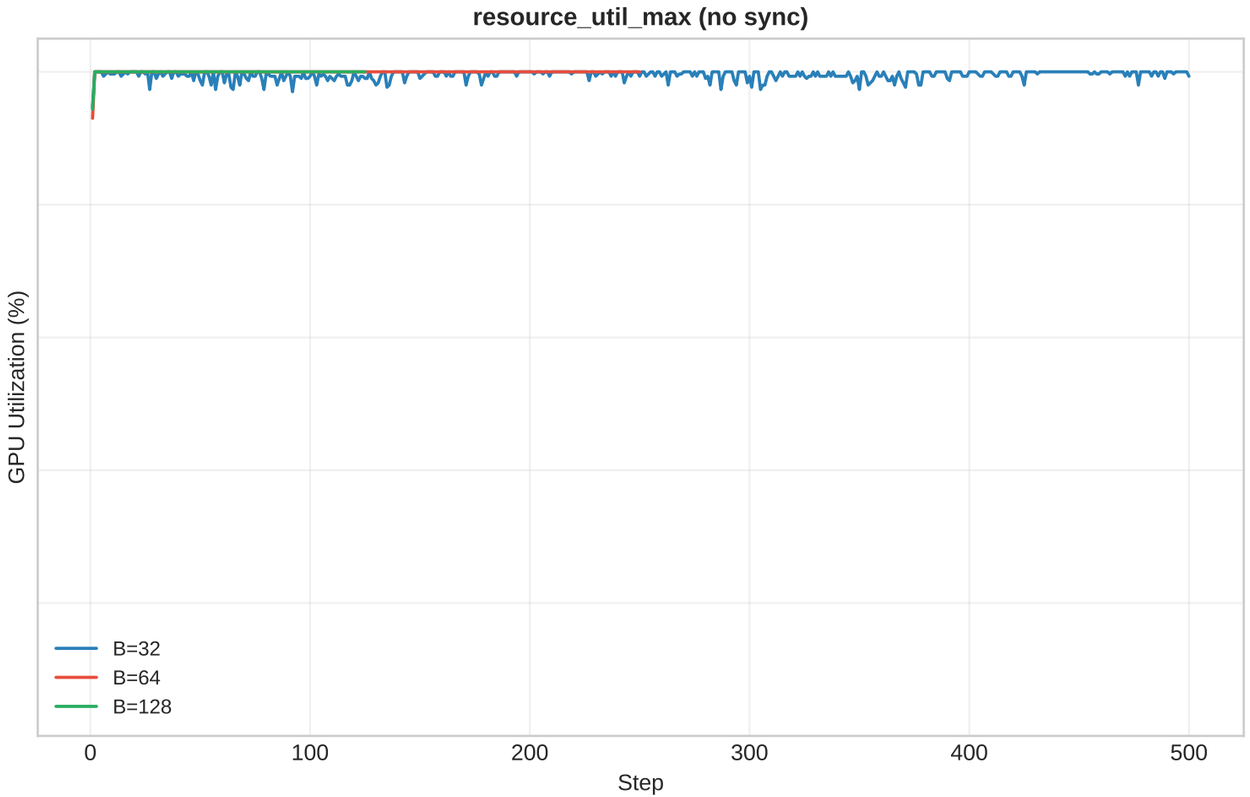
GPU Utilization Across Batch Sizes — Disk, w=4



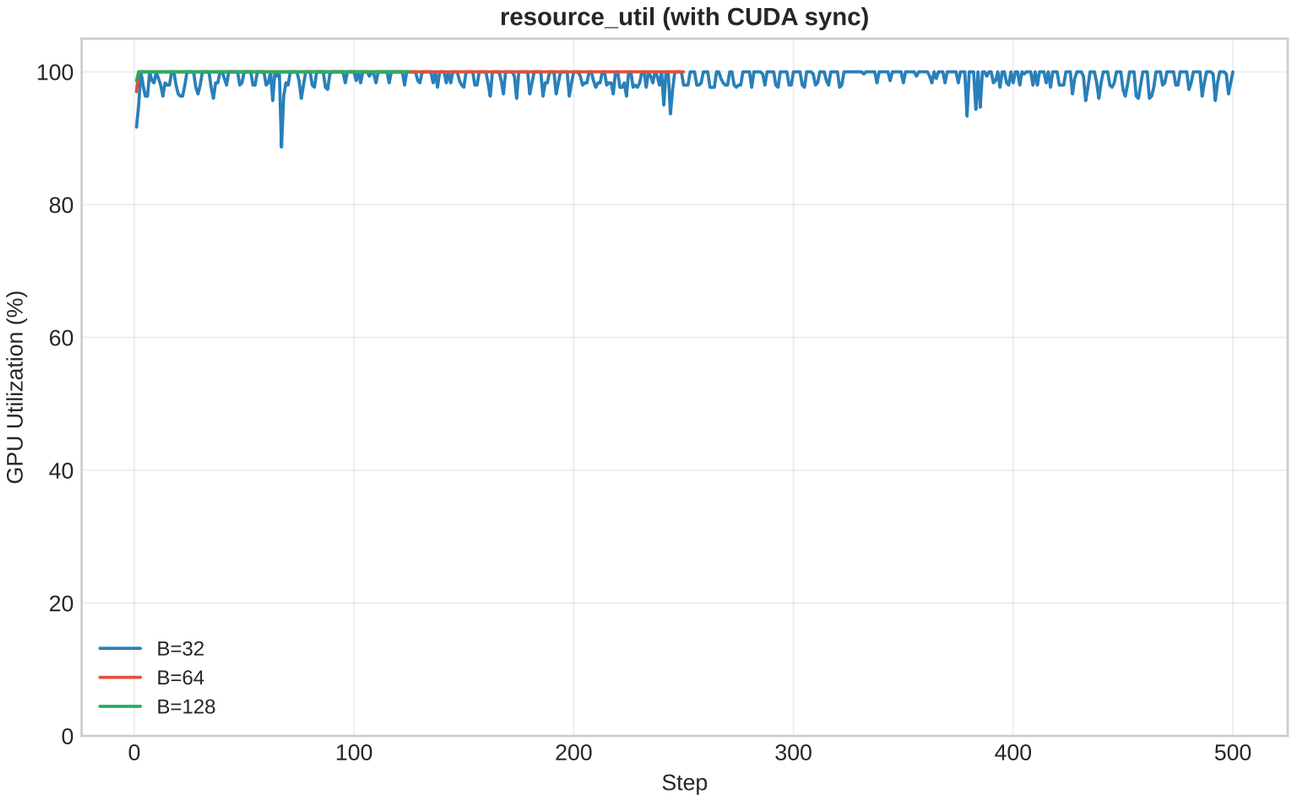


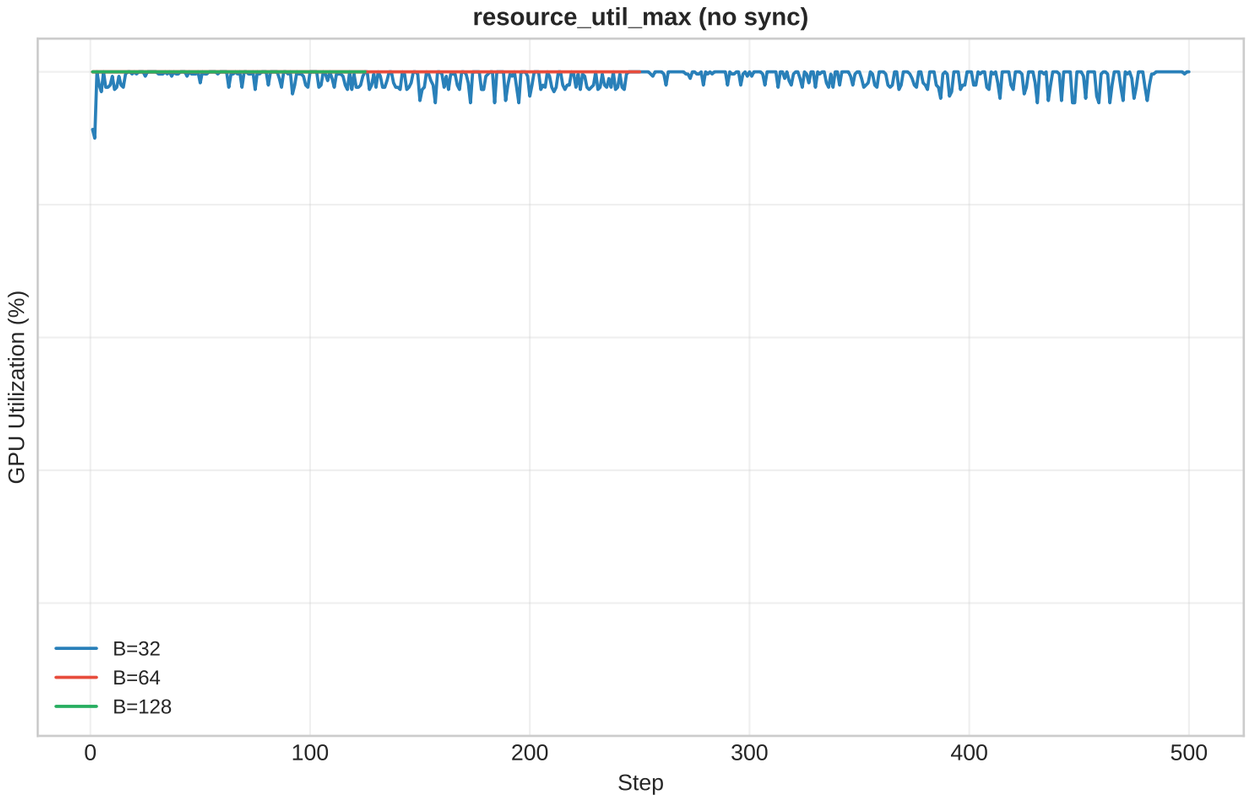
GPU Utilization Across Batch Sizes — Milabench, w=0



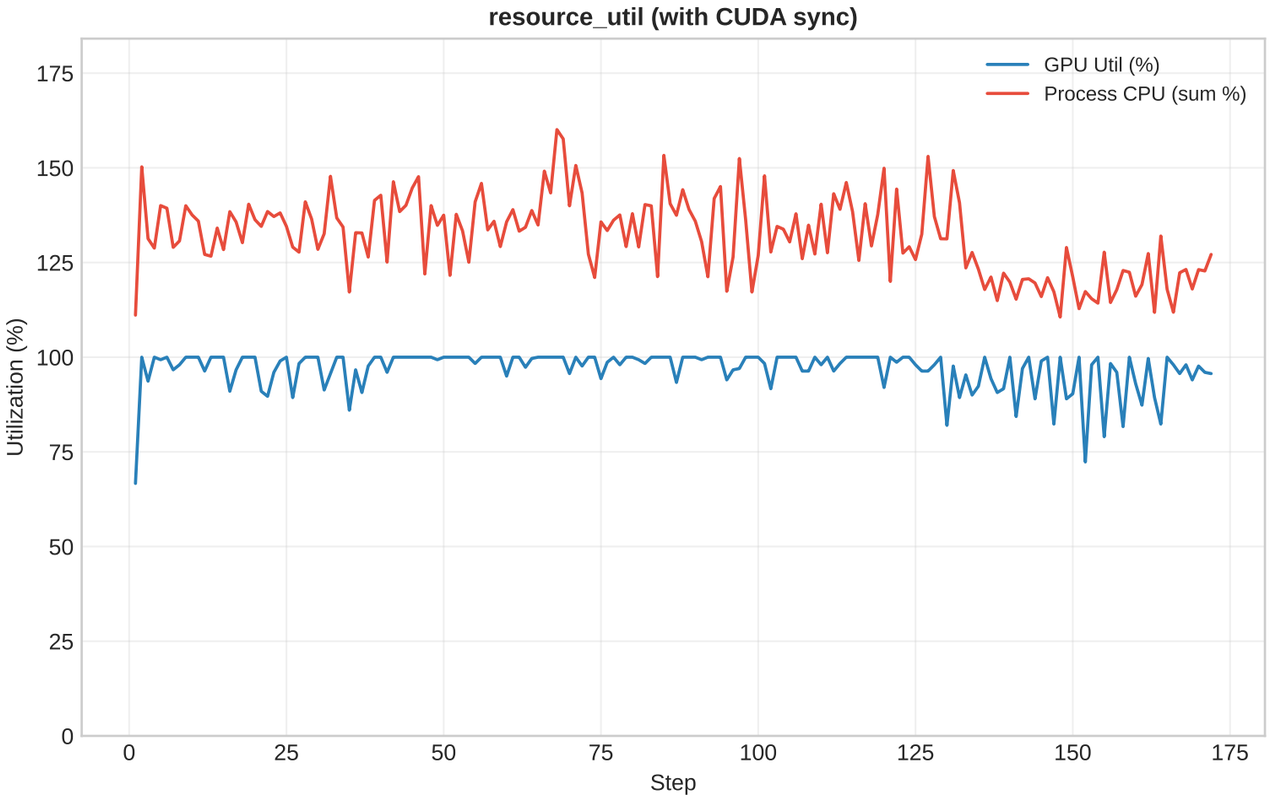


GPU Utilization Across Batch Sizes — Milabench, w=4

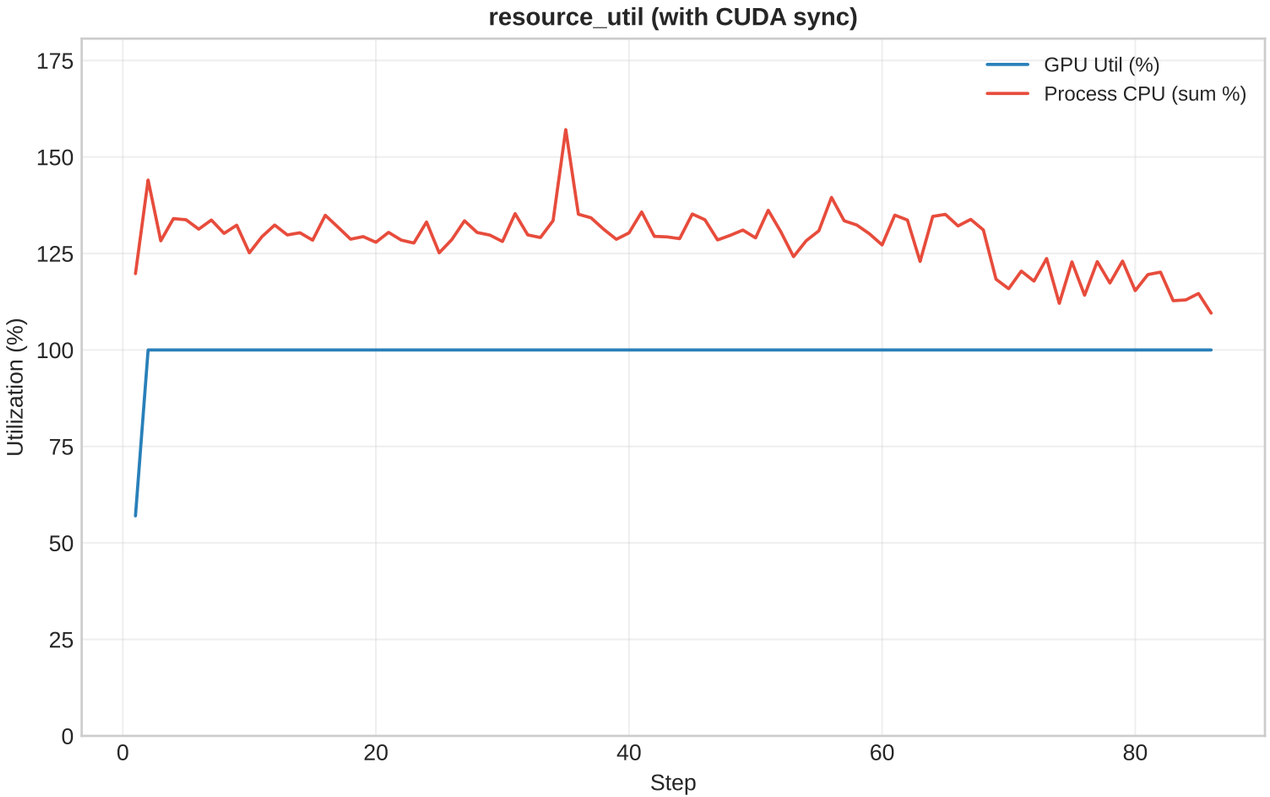


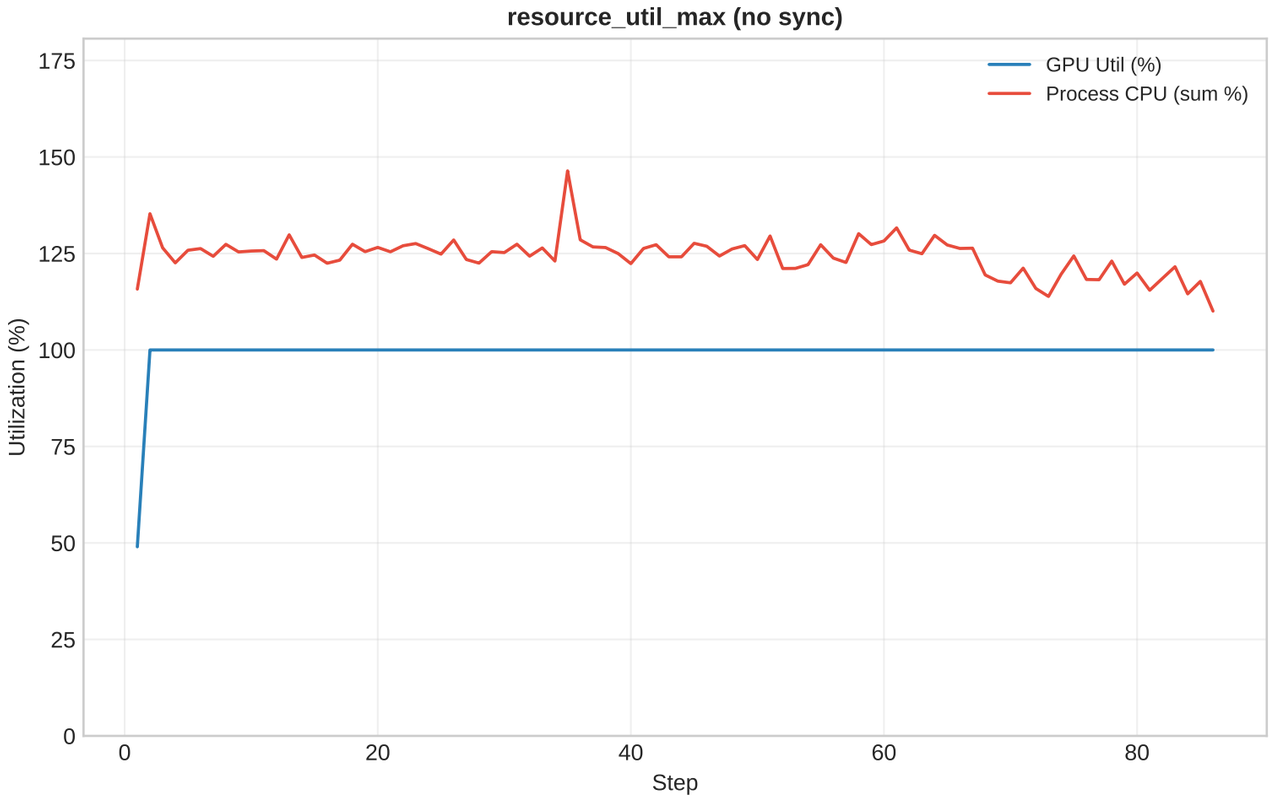


GPU vs CPU Utilization — Disk, w=0, B=32

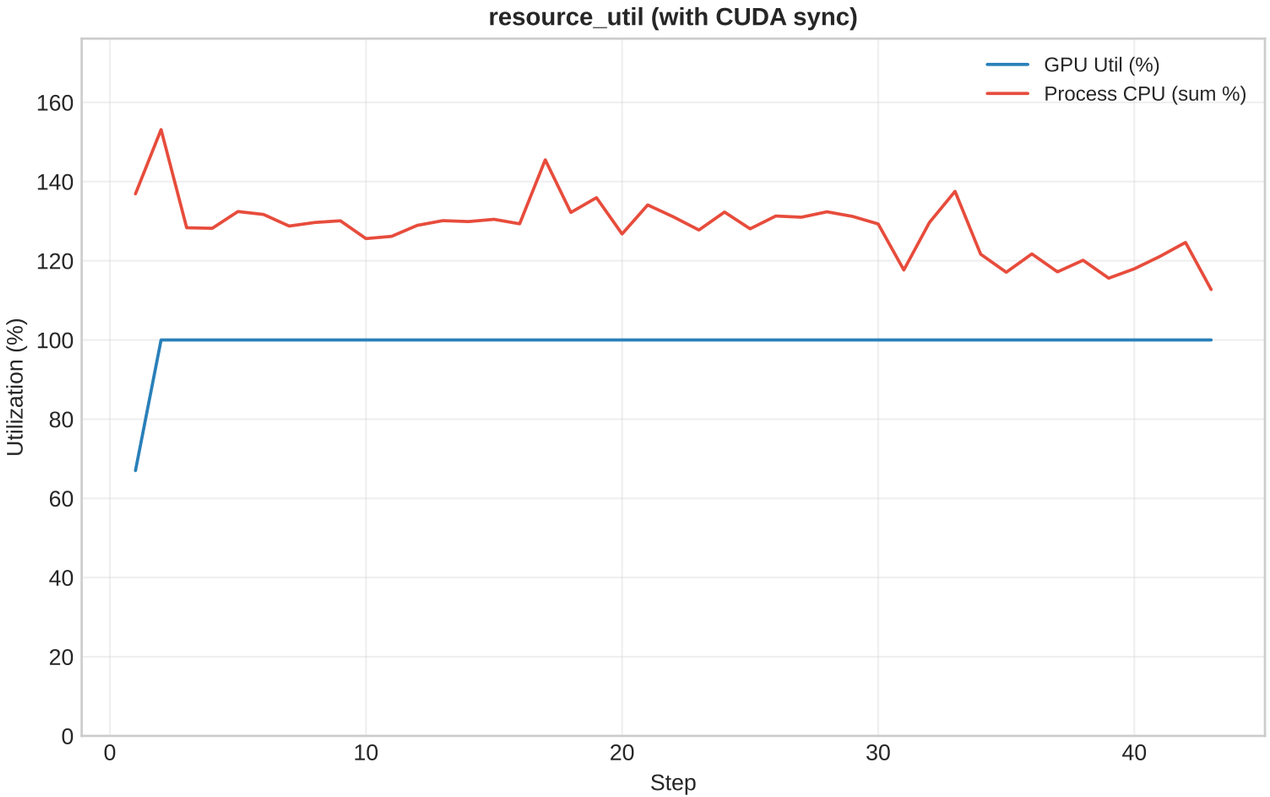


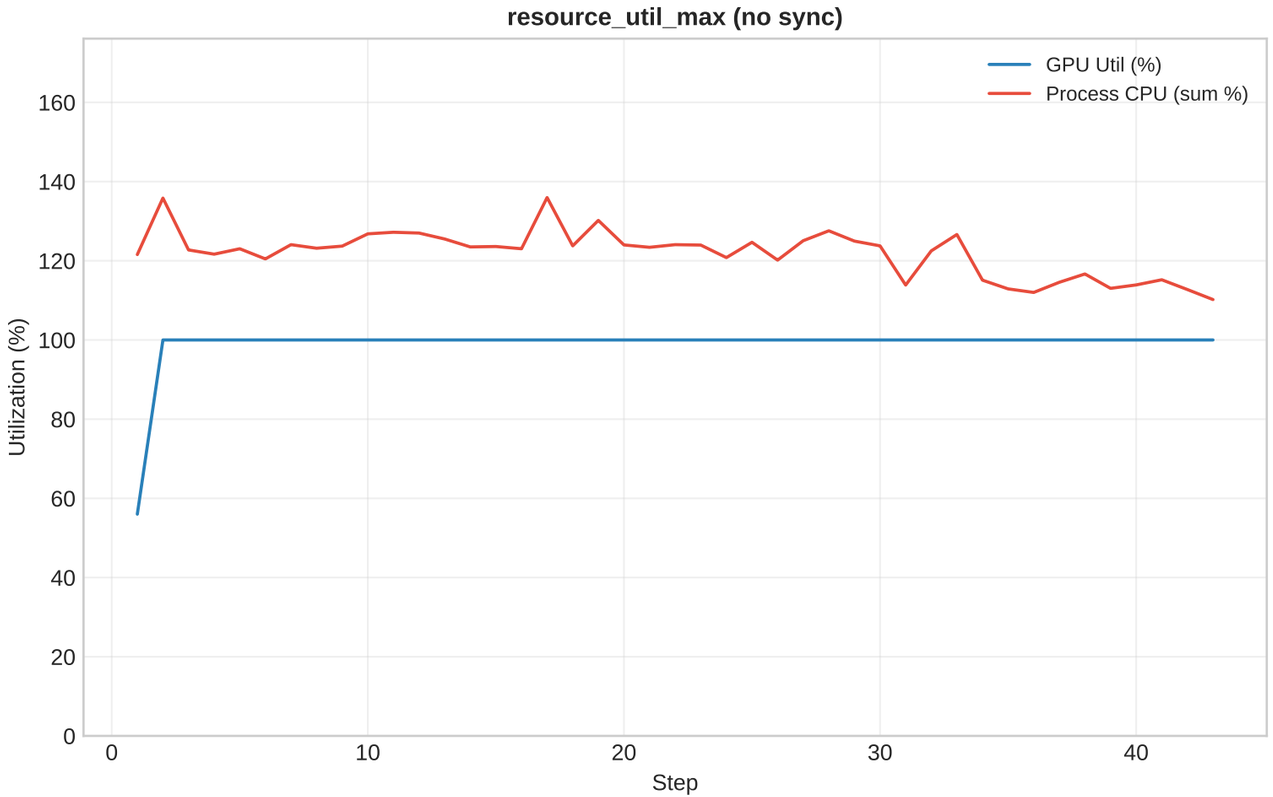
GPU vs CPU Utilization — Disk, w=0, B=64



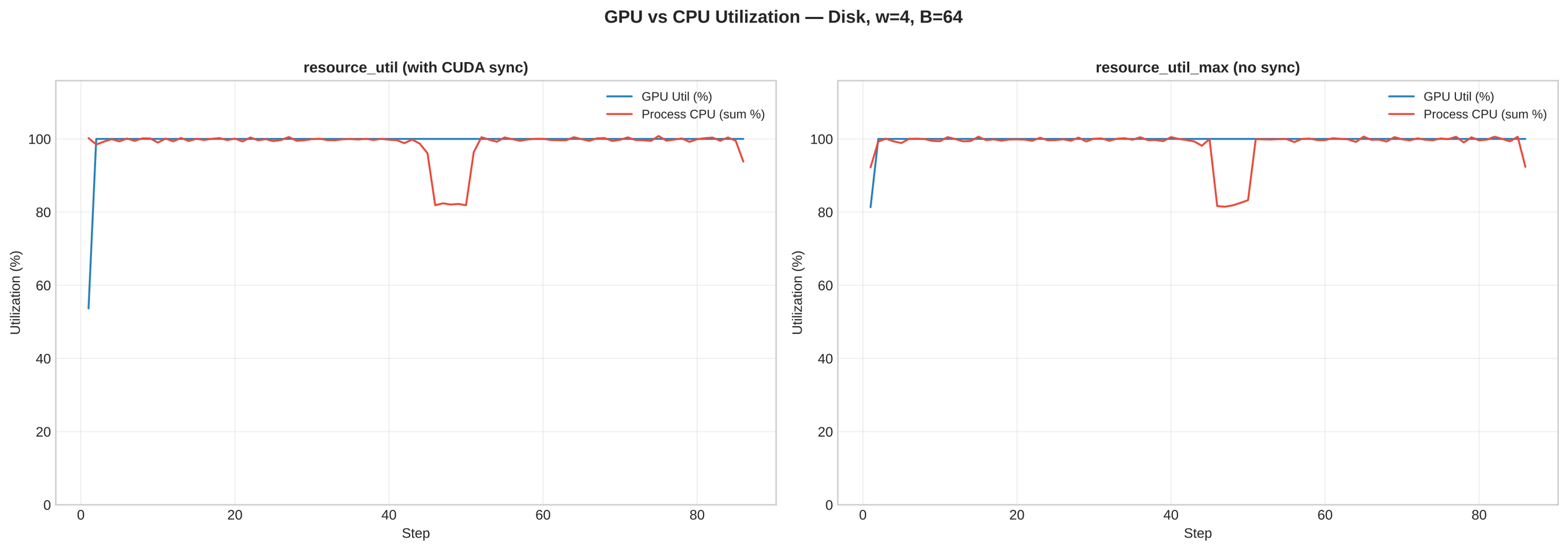


GPU vs CPU Utilization — Disk, w=0, B=128

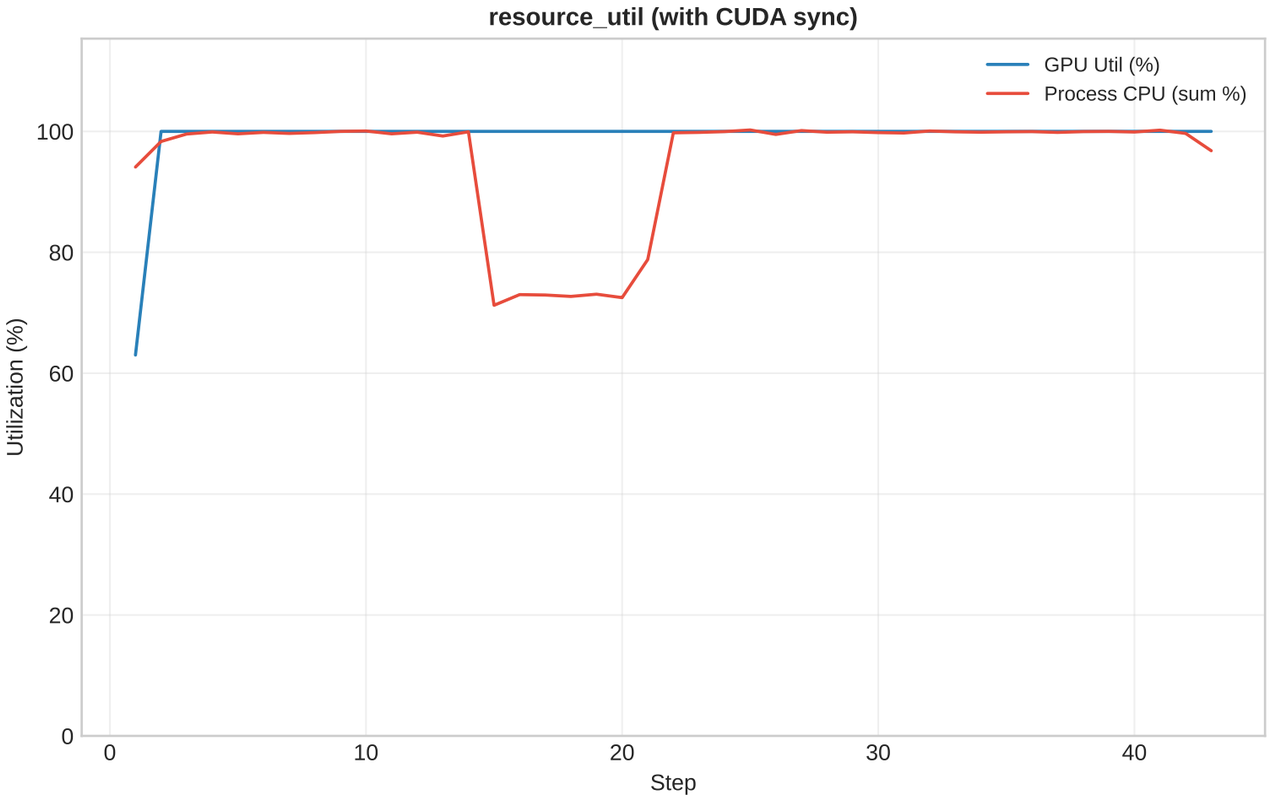


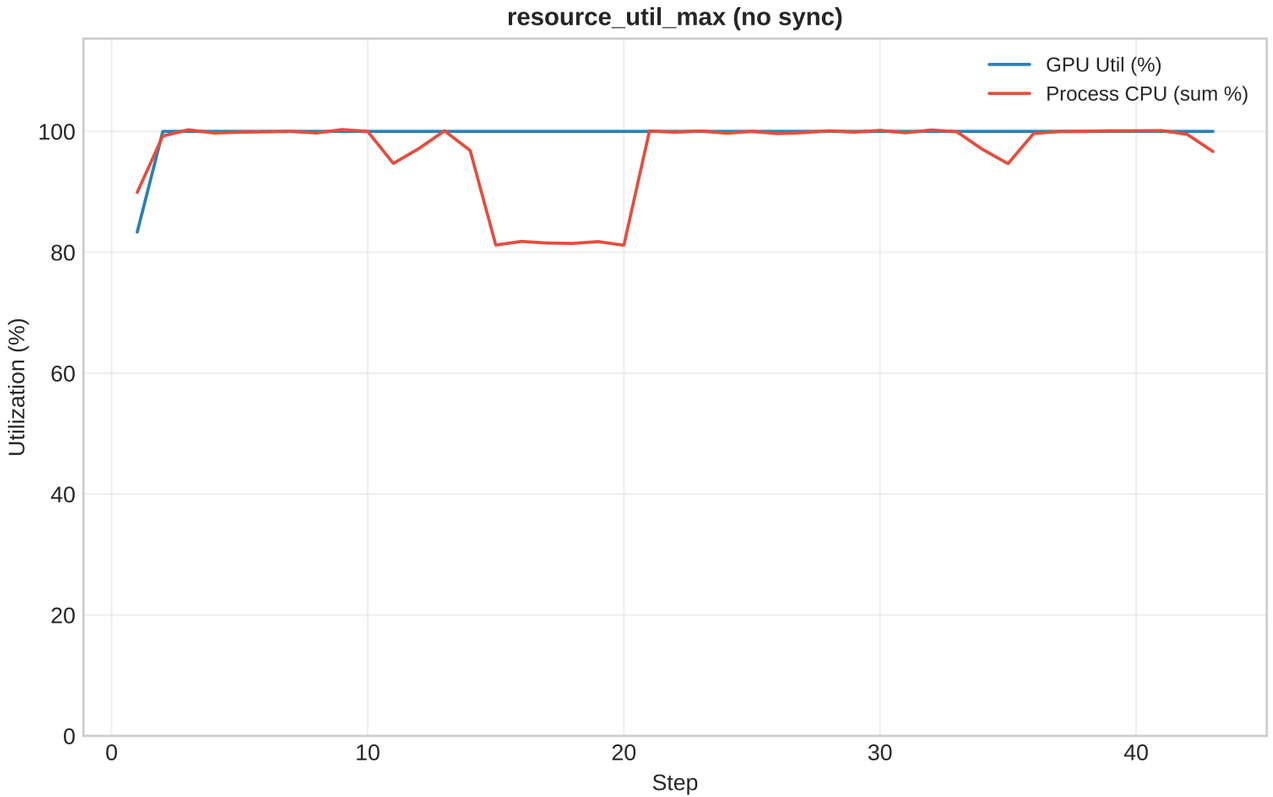


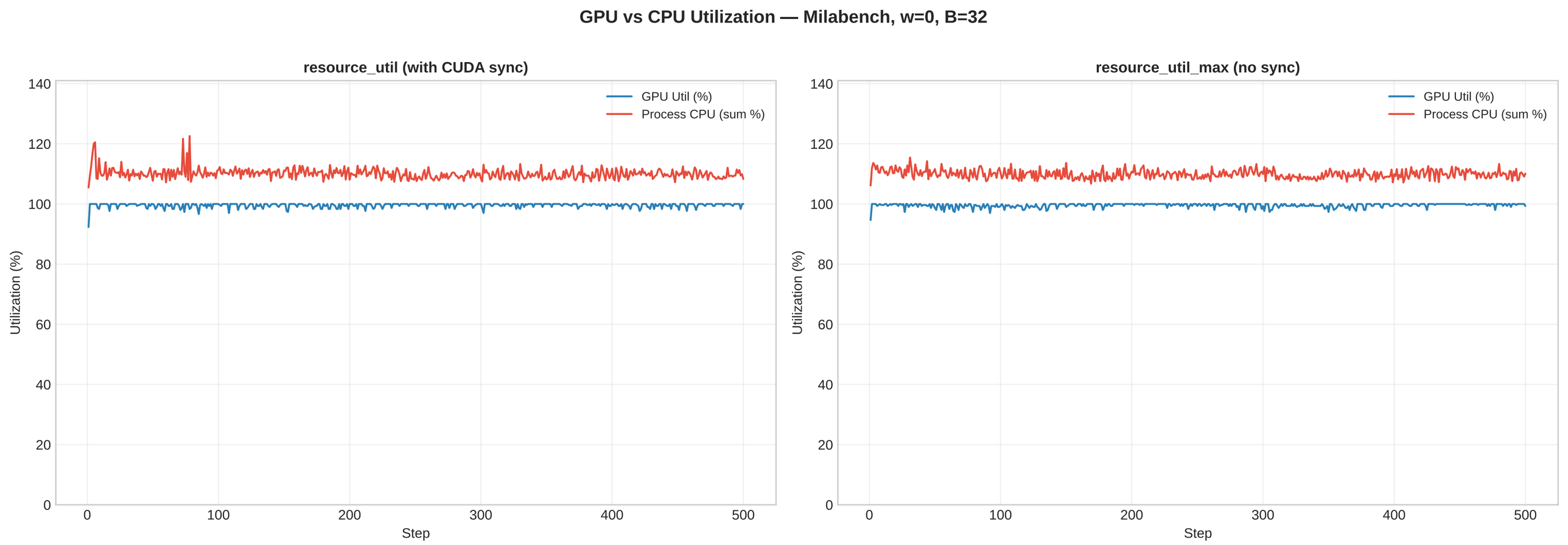
GPU vs CPU Utilization — Disk, w=4, B=32


GPU vs CPU Utilization — Disk, w=4, B=64
resource_util (with CUDA sync)
resource_util_max (no sync)
GPU Util (%)
Process CPU (sum %)
Utilization (%)
Step
0
20
40
60
80
100

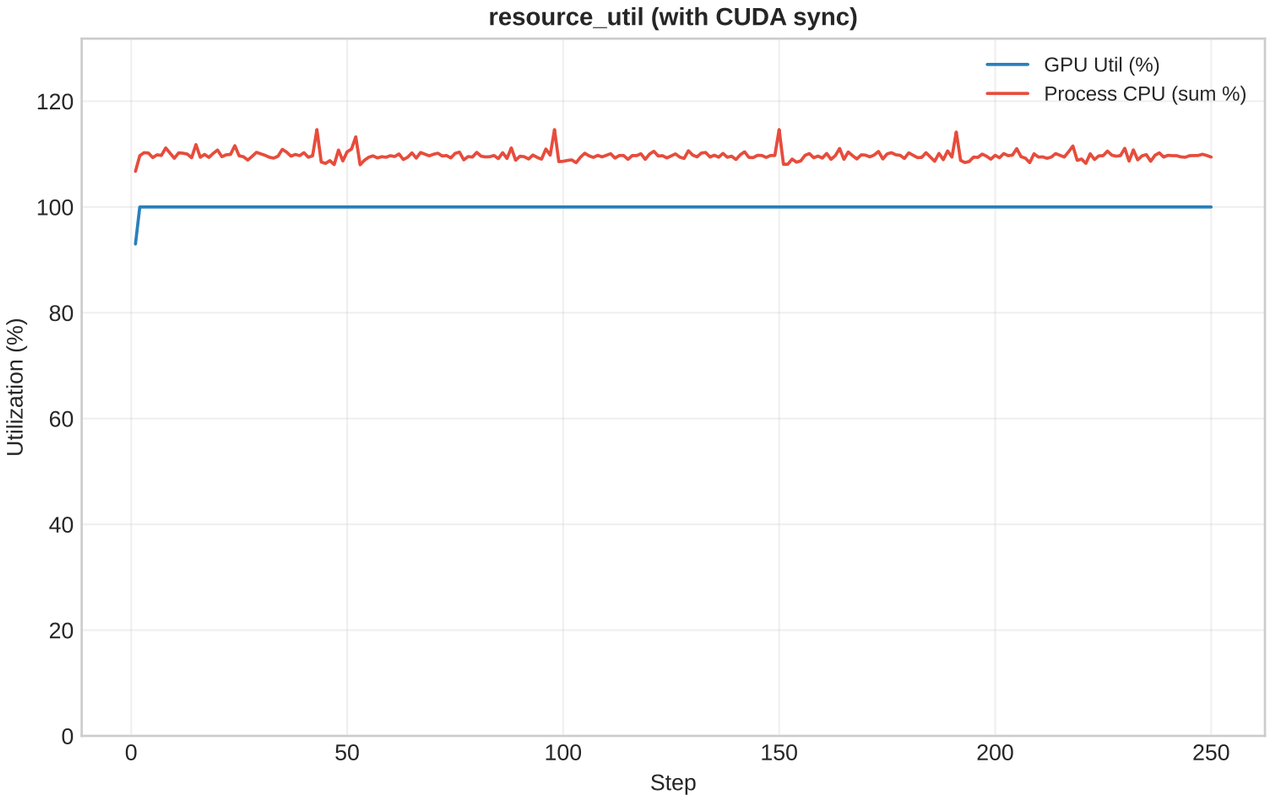
GPU vs CPU Utilization — Disk, w=4, B=128

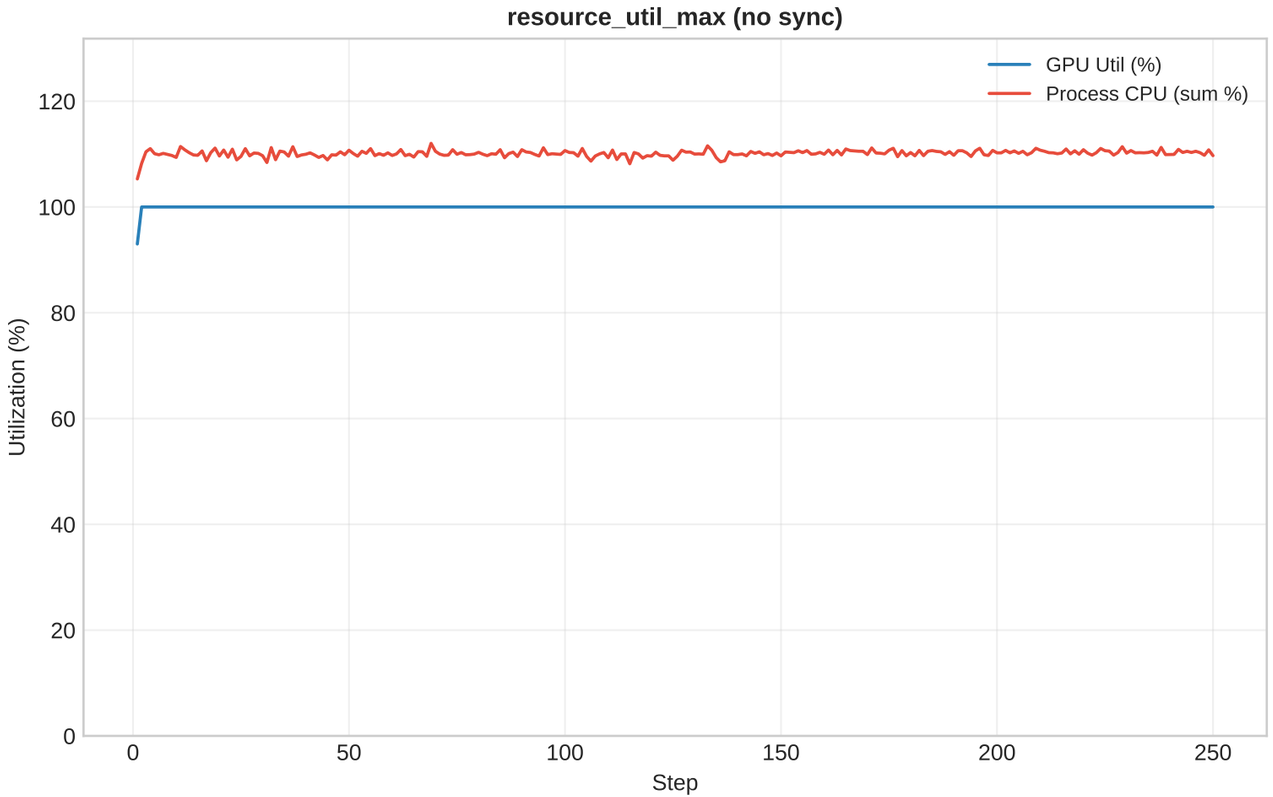


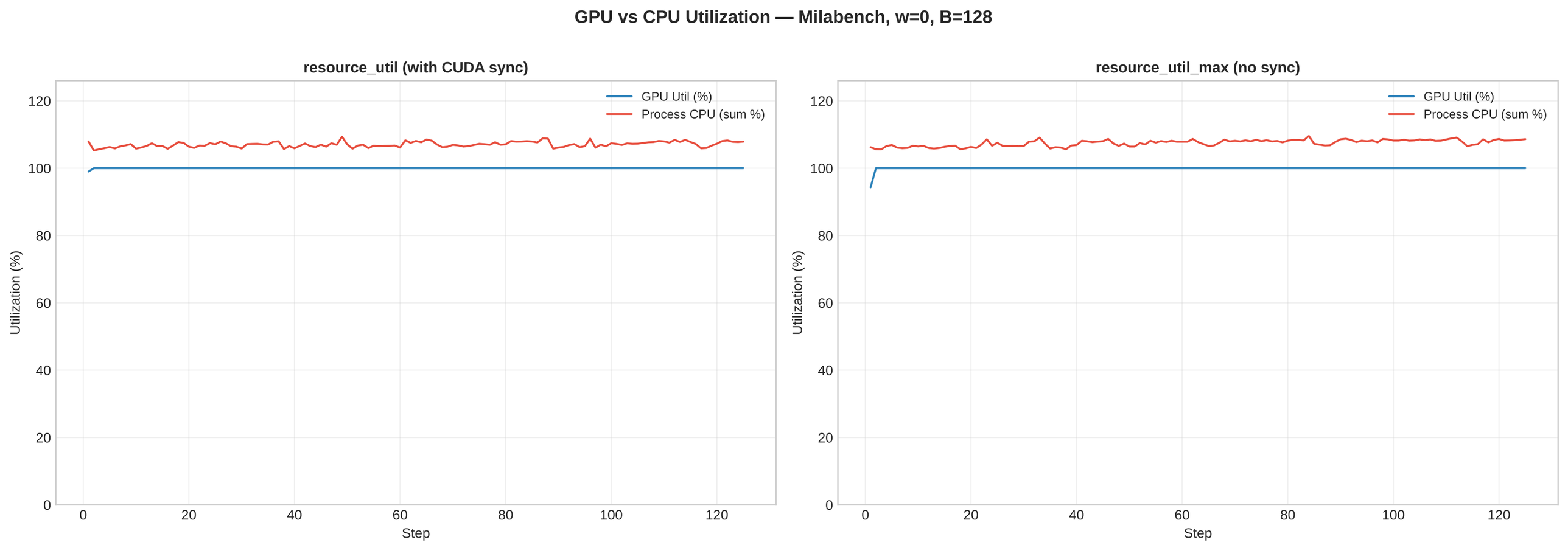



GPU vs CPU Utilization — Milabench, w=0, B=32
resource_util (with CUDA sync)
resource_util_max (no sync)
GPU Util (%)
Process CPU (sum %)
Utilization (%)
Step
0
20
40
60
80
100
120
140
0
100
200
300
400
500

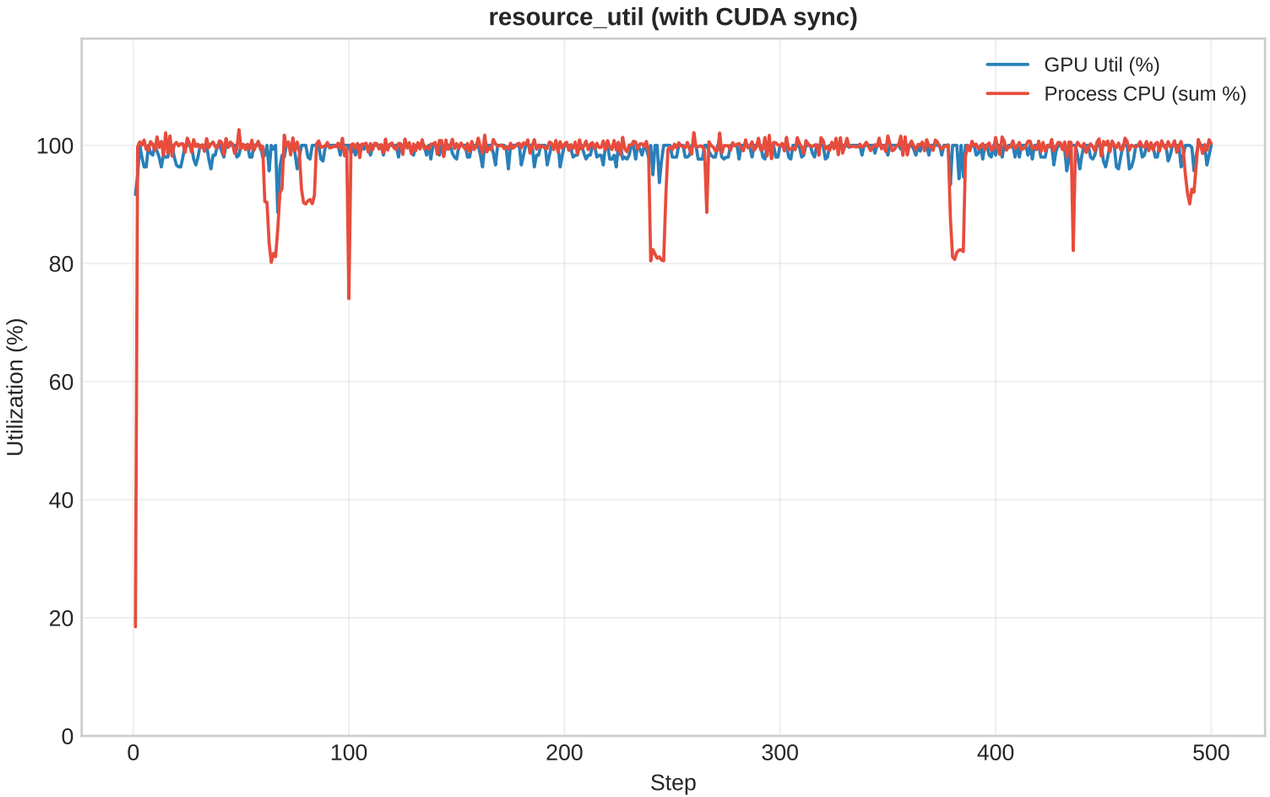
GPU vs CPU Utilization — Milabench, w=0, B=64

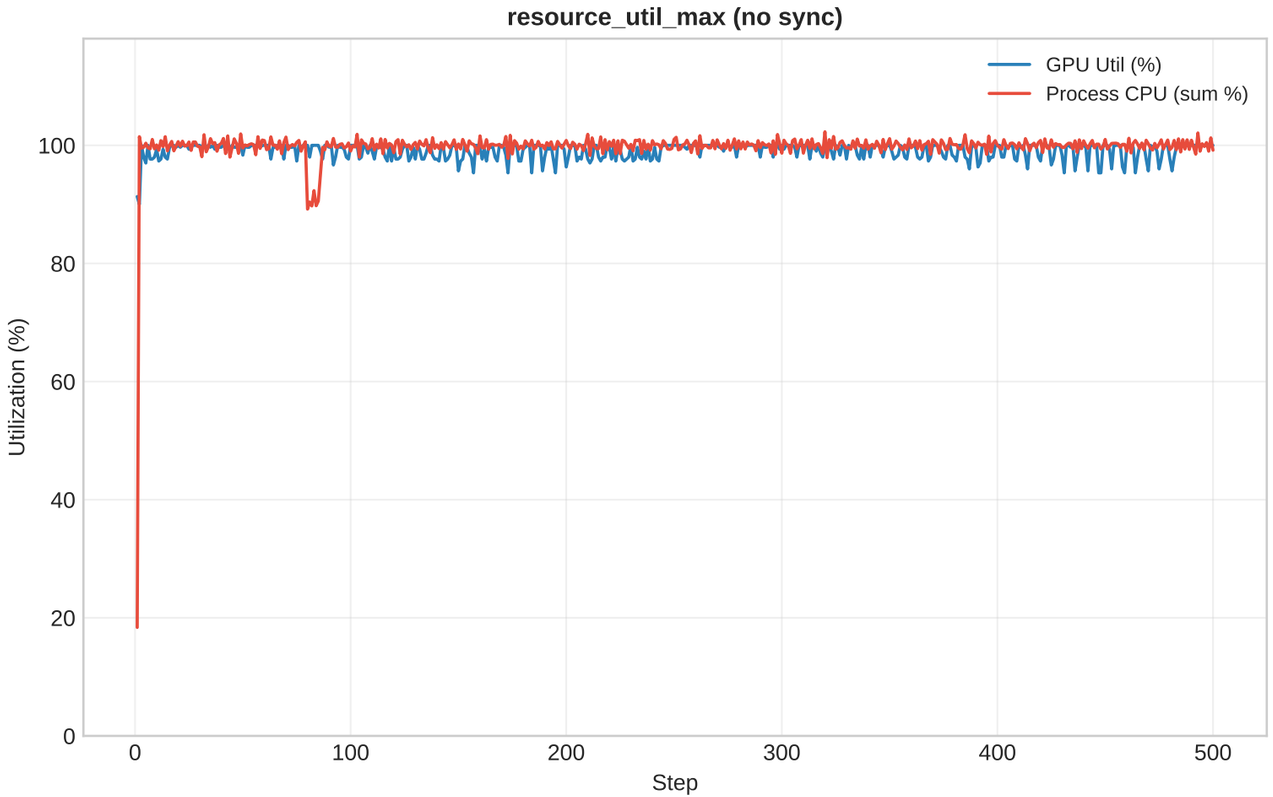




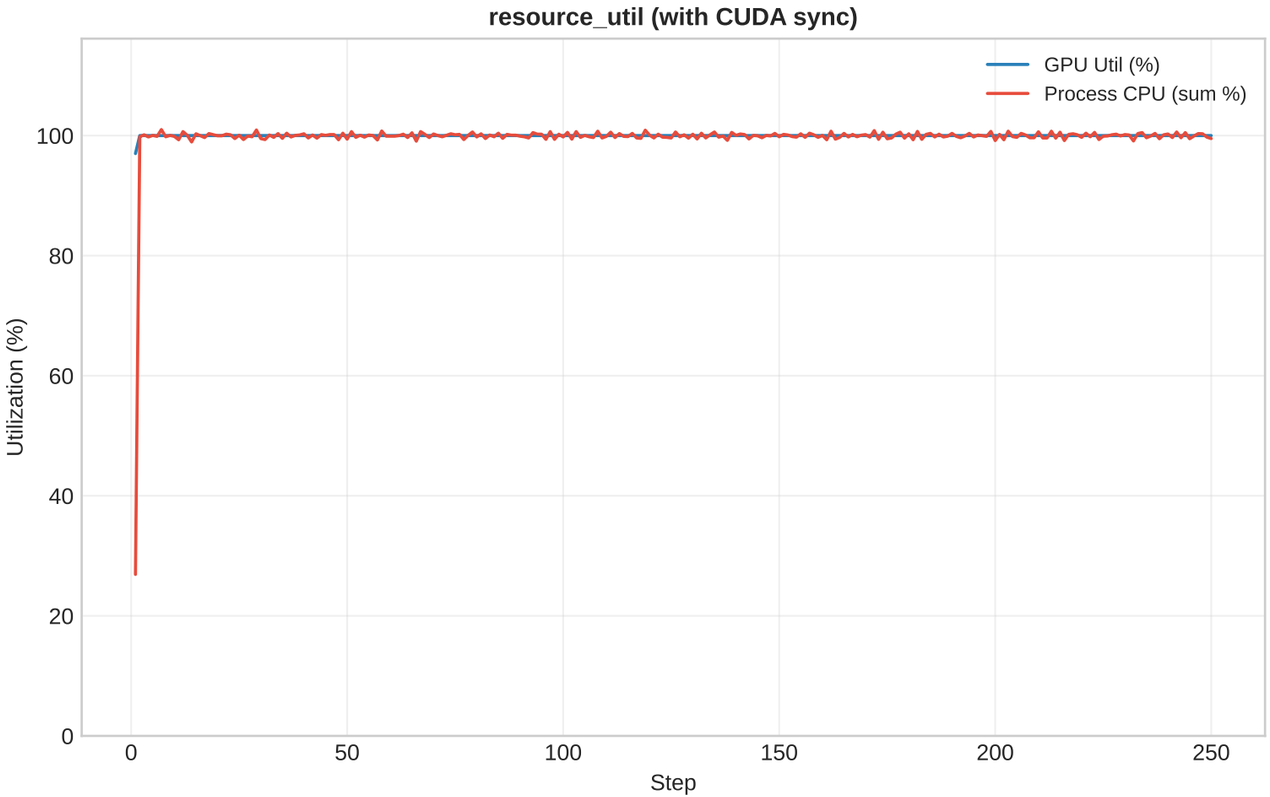

GPU vs CPU Utilization — Milabench, w=0, B=128
resource_util (with CUDA sync)
resource_util_max (no sync)
GPU Util (%)
Process CPU (sum %)
Utilization (%)
Step
0
20
40
60
80
100
120

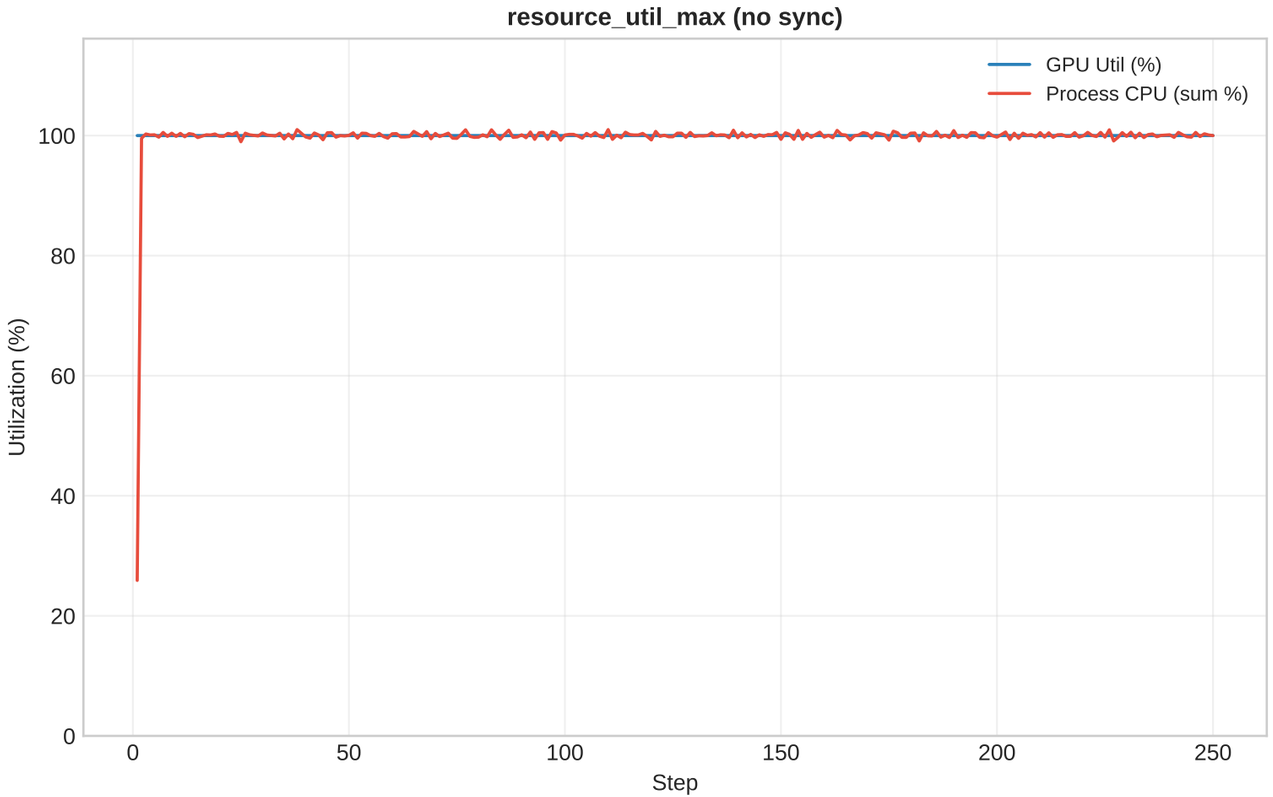
GPU vs CPU Utilization — Milabench, w=4, B=32

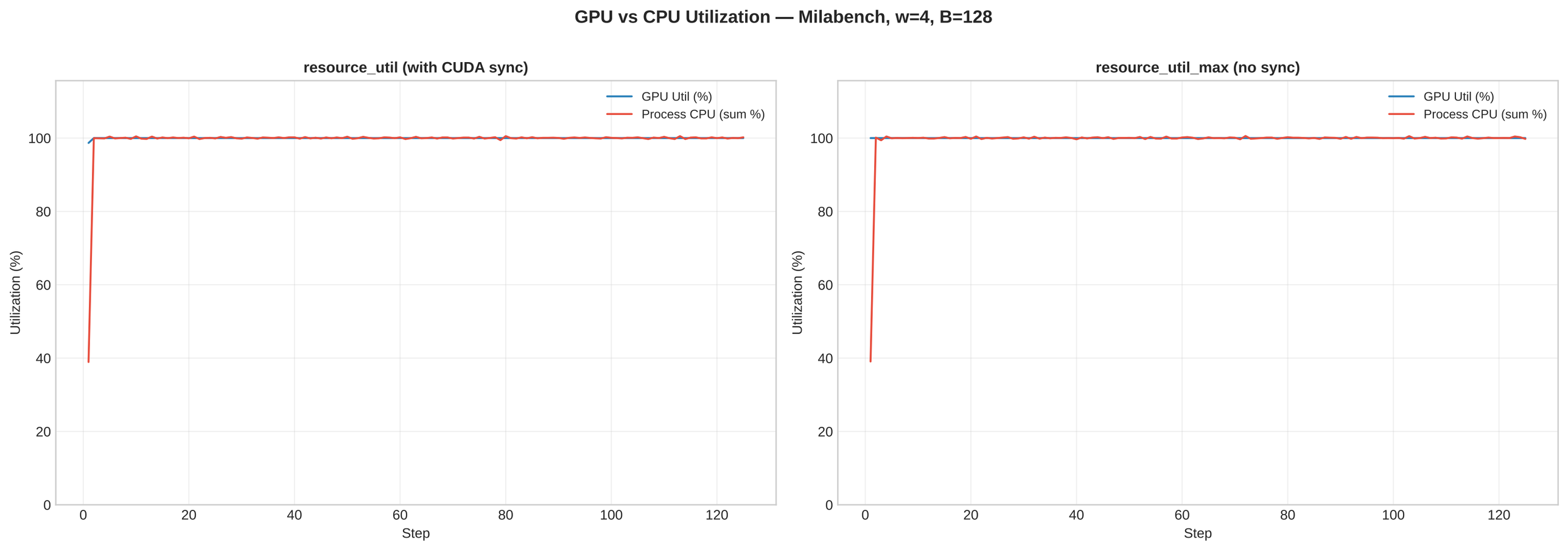




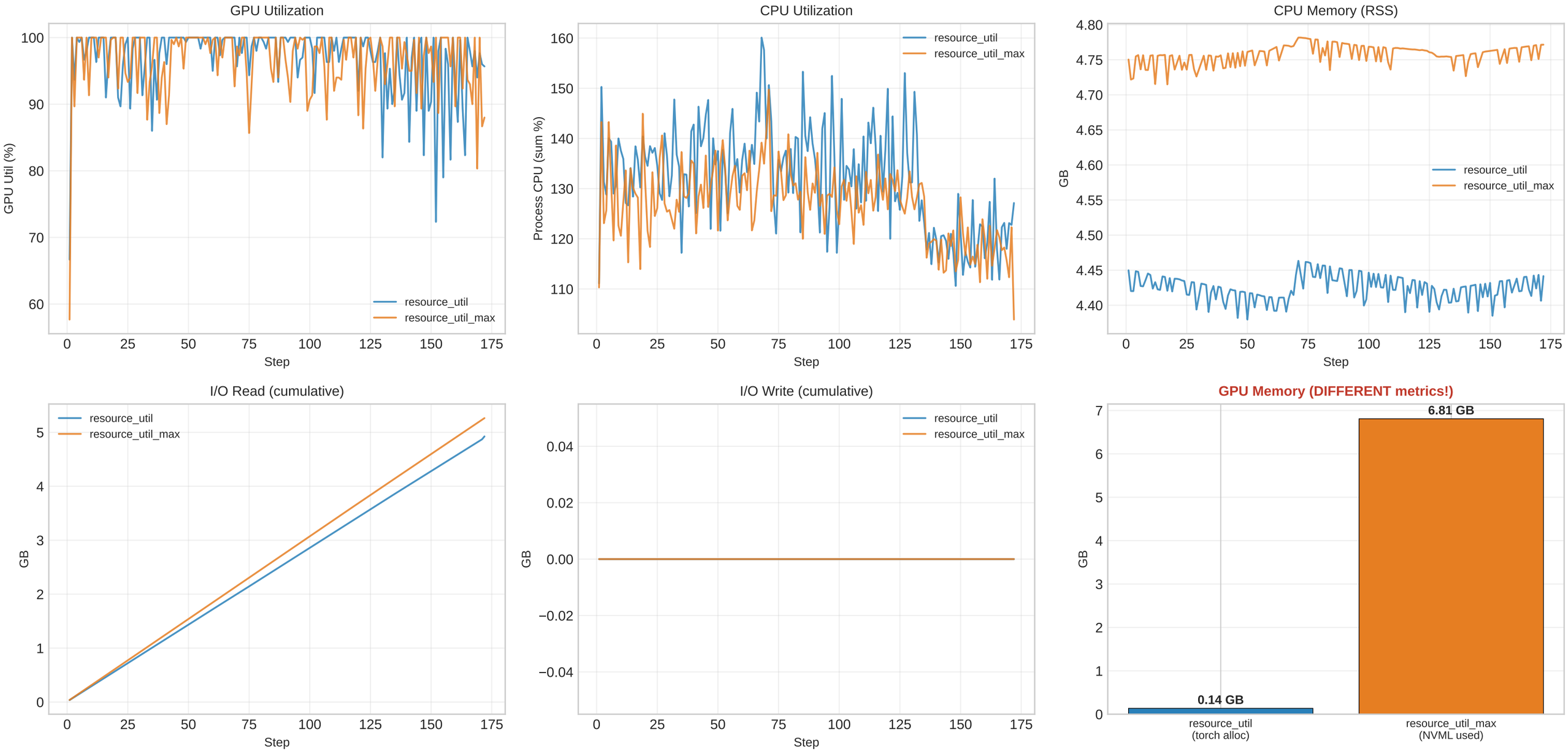
GPU vs CPU Utilization — Milabench, w=4, B=64

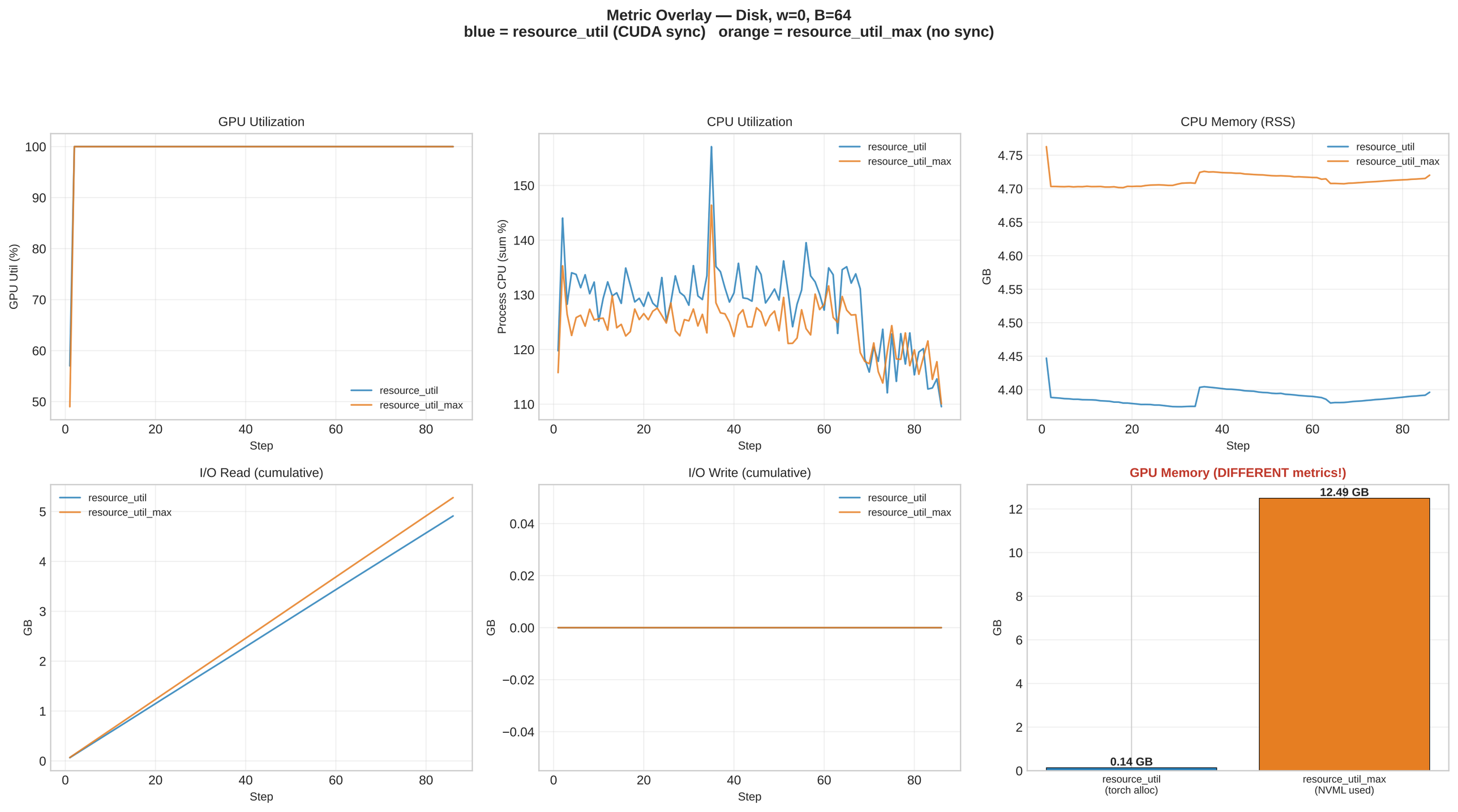


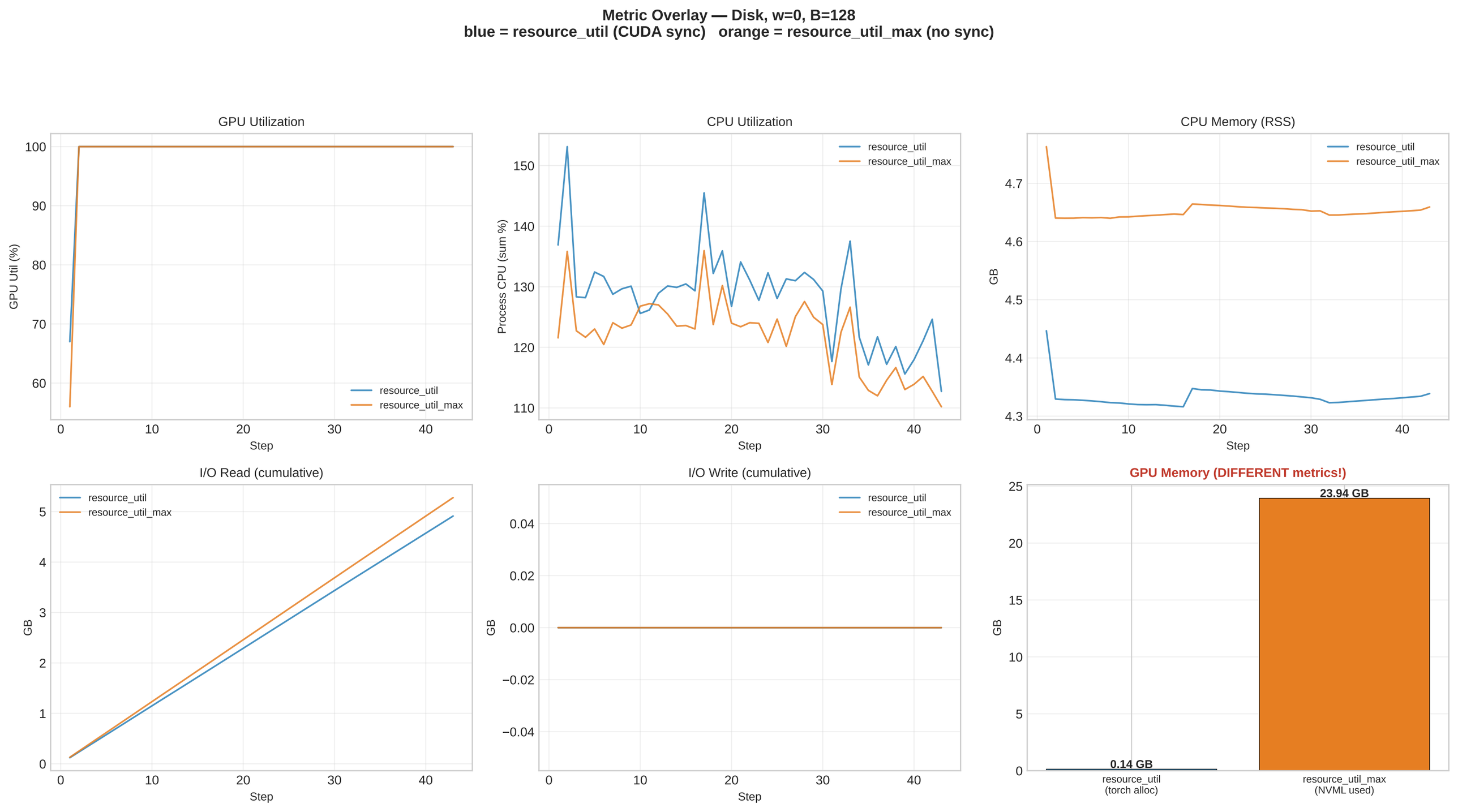


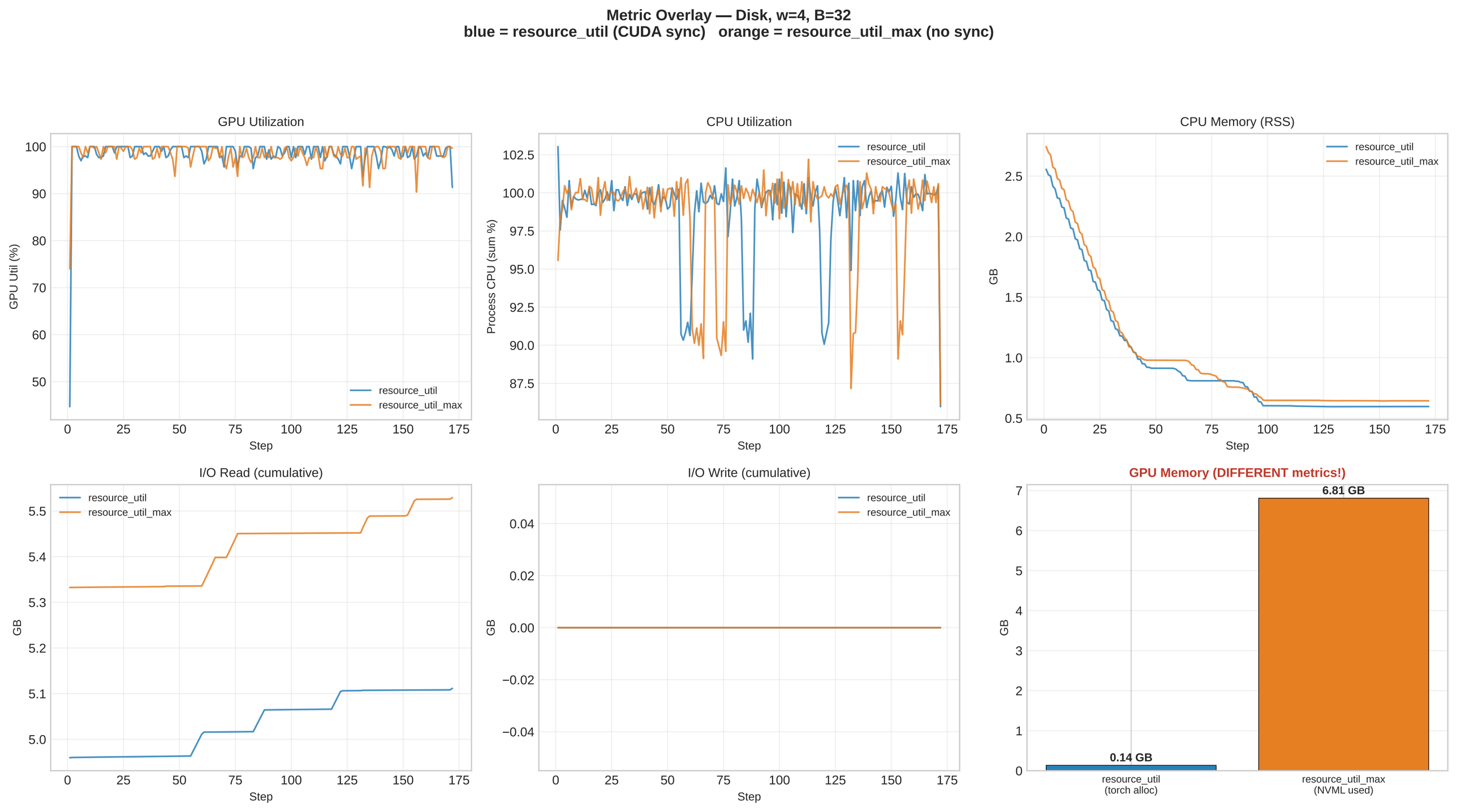

GPU vs CPU Utilization — Milabench, w=4, B=128
resource_util (with CUDA sync)
resource_util_max (no sync)
GPU Util (%)
Process CPU (sum %)
Utilization (%)
Step
0
20
40
60
80
100
120

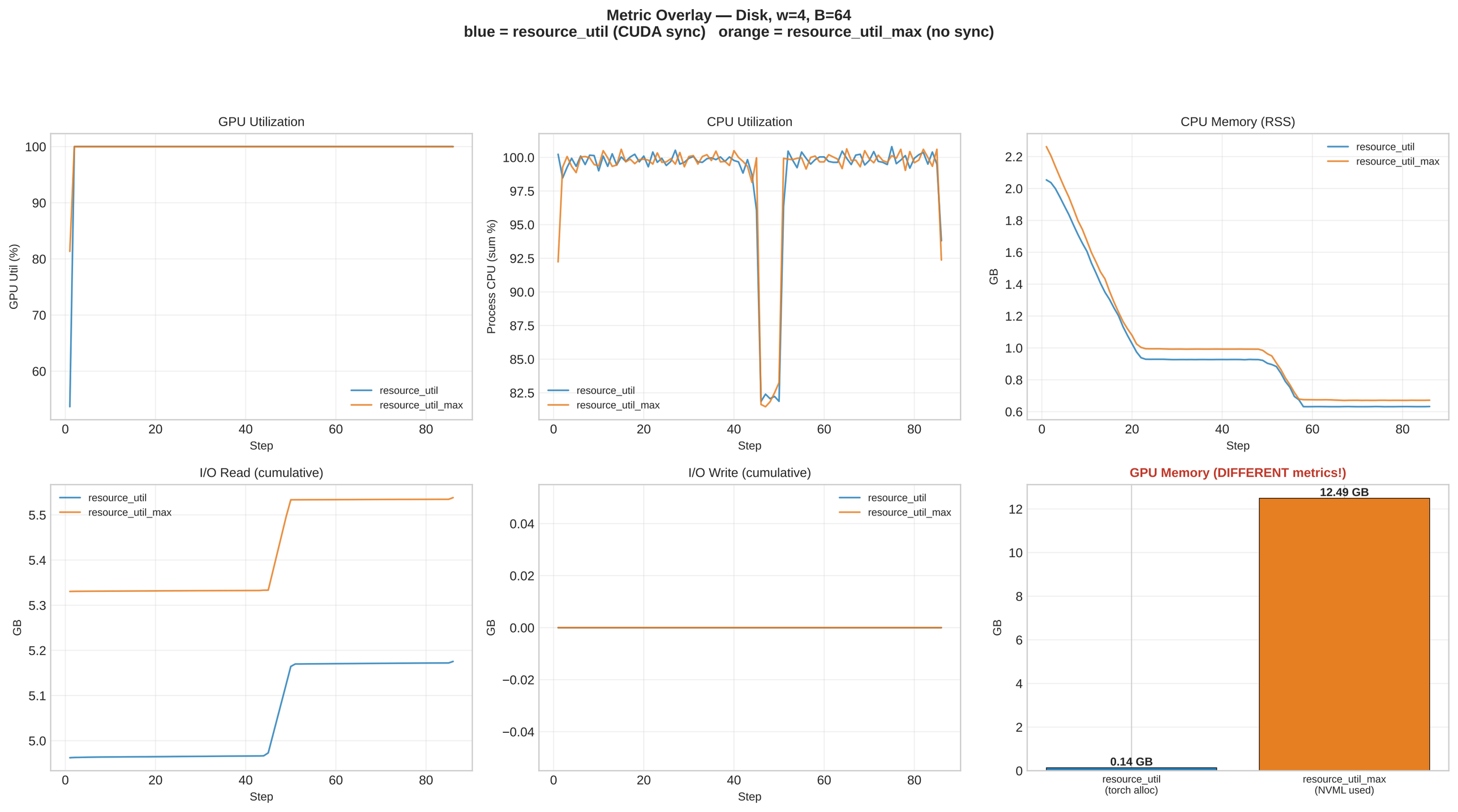
Metric Overlay — Disk, w=0, B=32
blue = resource_util (CUDA sync) orange = resource_util_max (no sync)



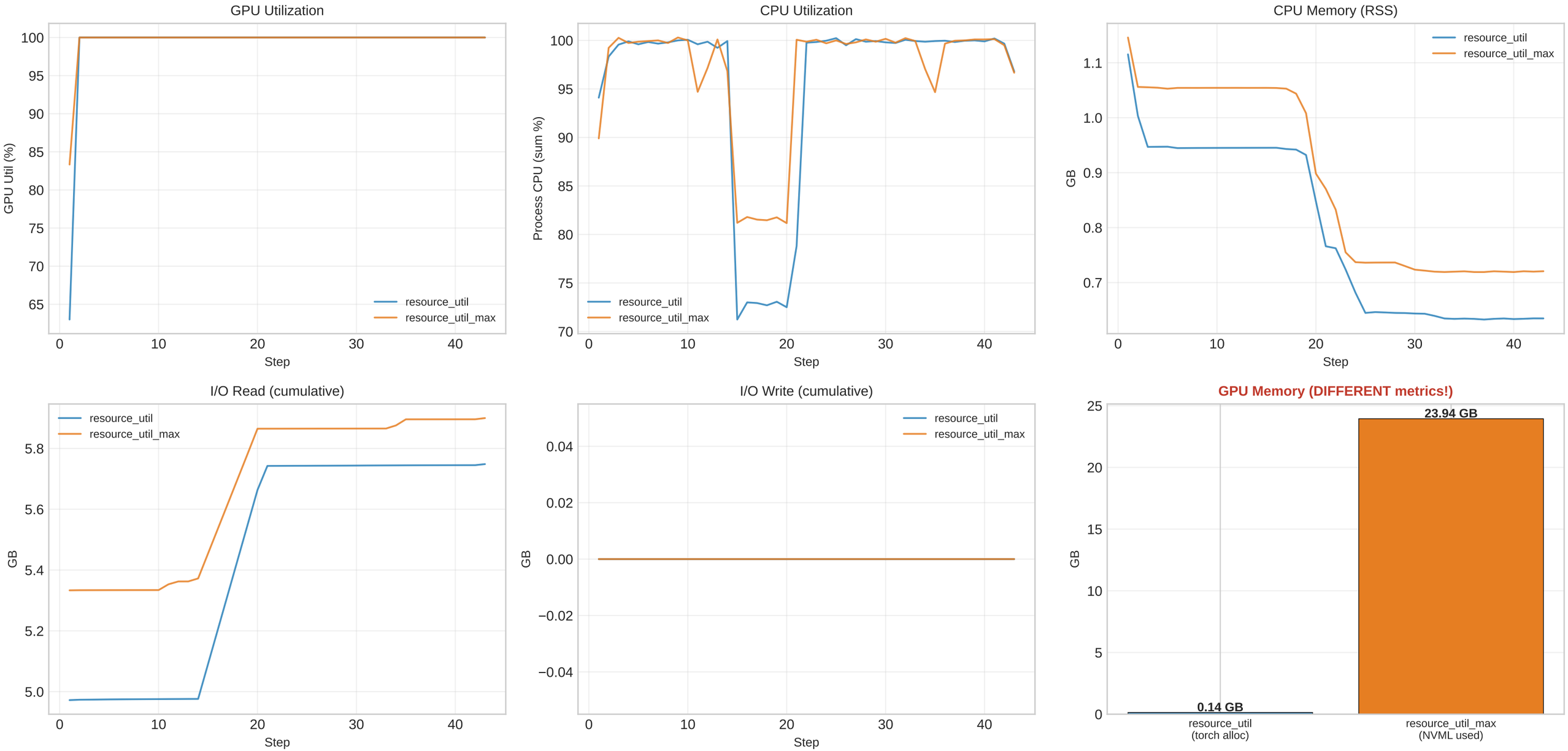

Metric Overlay — Disk, w=0, B=64
blue = resource_util (CUDA sync) orange = resource_util_max (no sync)
GPU Utilization
GPU Util (%)
Step
resource_util
resource_util_max
CPU Utilization
Process CPU (sum %)
Step
resource_util
resource_util_max
CPU Memory (RSS)
GB
Step
resource_util
resource_util_max
I/O Read (cumulative)
GB
Step
resource_util
resource_util_max
I/O Write (cumulative)
GB
Step
resource_util
resource_util_max
GPU Memory (DIFFERENT metrics!)
GB
12.49 GB
0.14 GB
resource_util
(torch alloc)
resource_util_max
(NVML used)

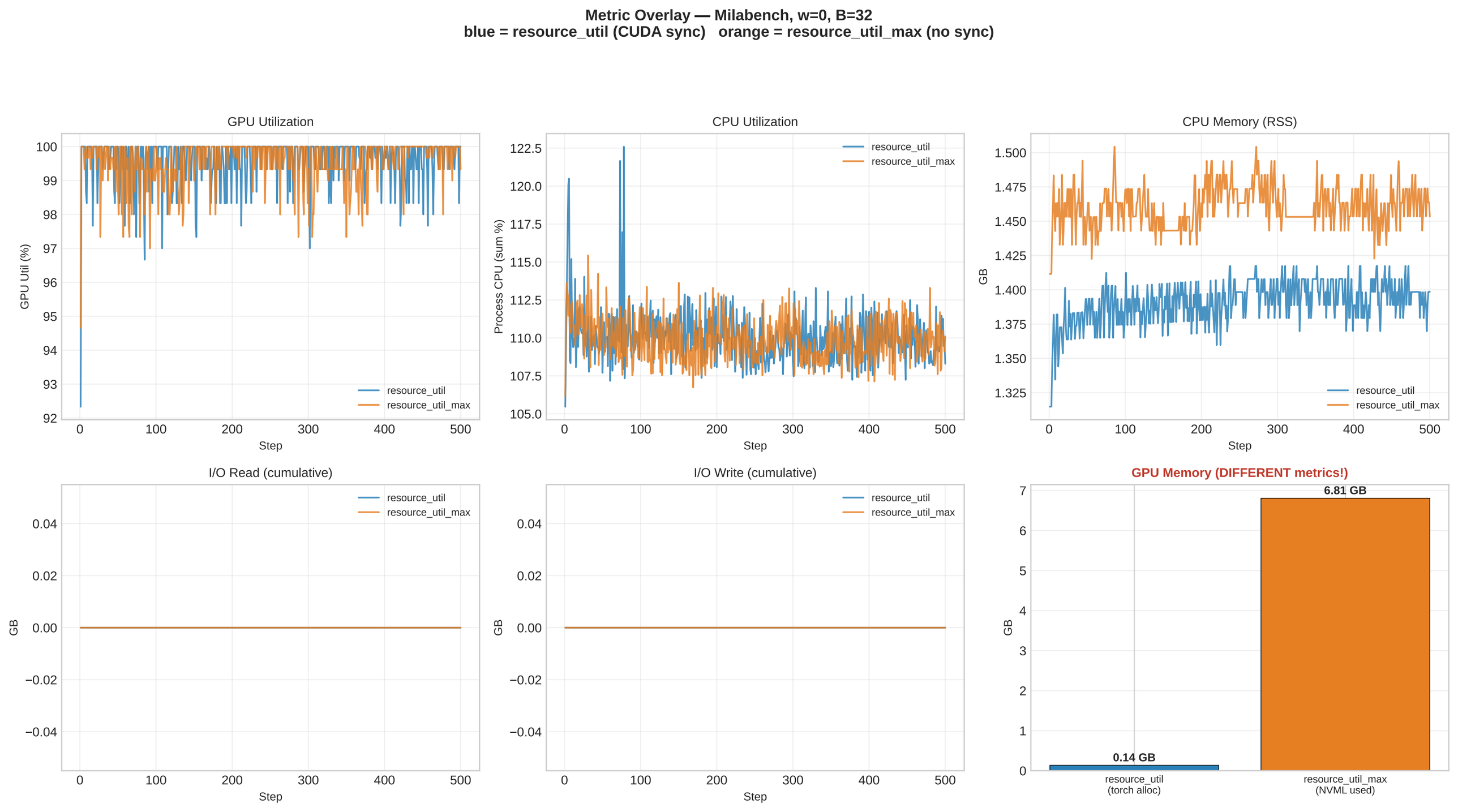

Metric Overlay — Disk, w=0, B=128
blue = resource_util (CUDA sync) orange = resource_util_max (no sync)
GPU Utilization
GPU Util (%)
Step
resource_util
resource_util_max
CPU Utilization
Process CPU (sum %)
Step
resource_util
resource_util_max
CPU Memory (RSS)
GB
Step
resource_util
resource_util_max
I/O Read (cumulative)
GB
Step
resource_util
resource_util_max
I/O Write (cumulative)
GB
Step
resource_util
resource_util_max
GPU Memory (DIFFERENT metrics!)
GB
23.94 GB
0.14 GB
resource_util
(torch alloc)
resource_util_max
(NVML used)


Metric Overlay — Disk, w=4, B=32
blue = resource_util (CUDA sync) orange = resource_util_max (no sync)
GPU Utilization
GPU Util (%)
Step
resource_util
resource_util_max
CPU Utilization
Process CPU (sum %)
Step
resource_util
resource_util_max
CPU Memory (RSS)
GB
Step
resource_util
resource_util_max
I/O Read (cumulative)
GB
Step
resource_util
resource_util_max
I/O Write (cumulative)
GB
Step
resource_util
resource_util_max
GPU Memory (DIFFERENT metrics!)
GB
0.14 GB
6.81 GB
resource_util
(torch alloc)
resource_util_max
(NVML used)

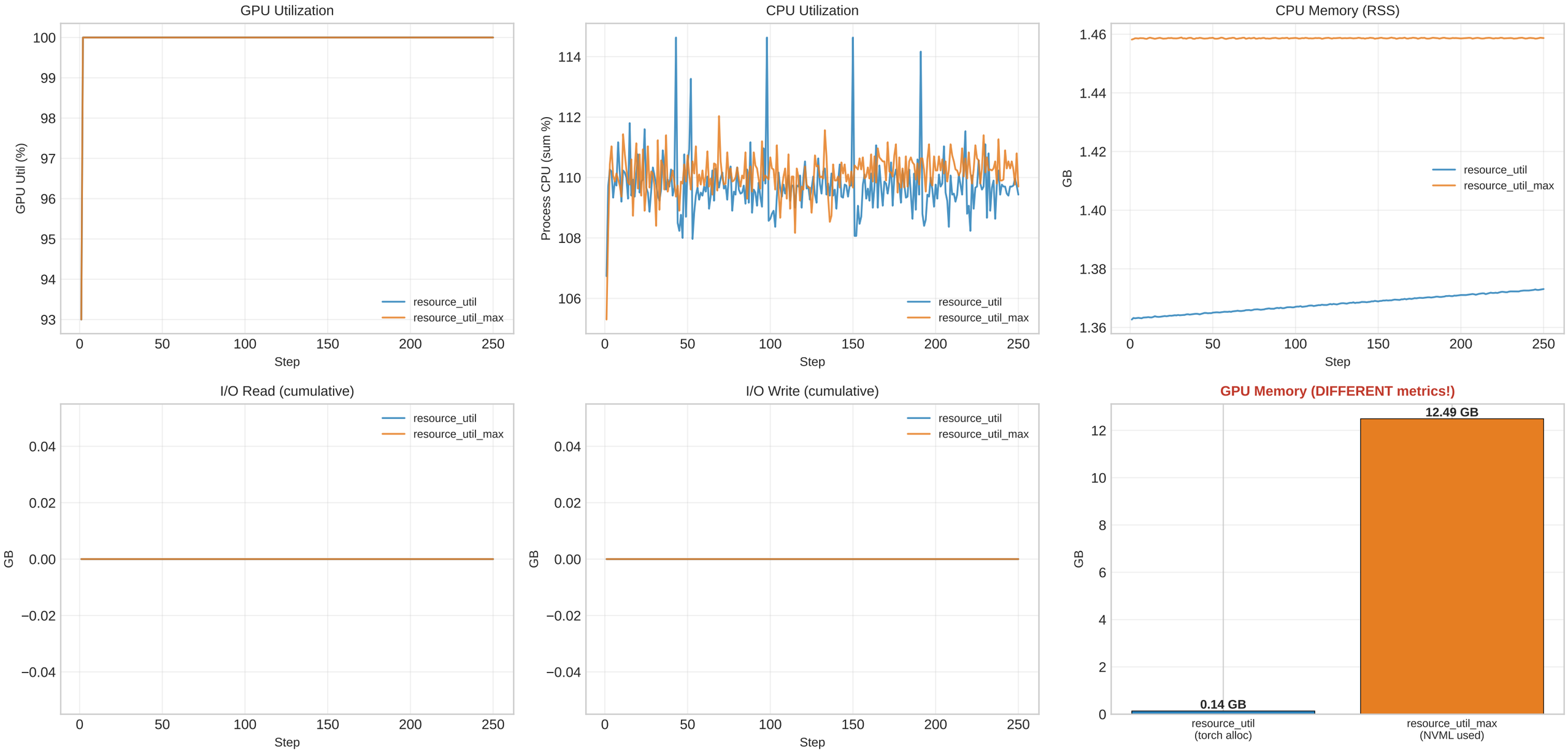

Metric Overlay — Disk, w=4, B=64
blue = resource_util (CUDA sync) orange = resource_util_max (no sync)
GPU Utilization
GPU Util (%)
Step
resource_util
resource_util_max
CPU Utilization
Process CPU (sum %)
Step
resource_util
resource_util_max
CPU Memory (RSS)
GB
Step
resource_util
resource_util_max
I/O Read (cumulative)
GB
Step
resource_util
resource_util_max
I/O Write (cumulative)
GB
Step
resource_util
resource_util_max
GPU Memory (DIFFERENT metrics!)
GB
12.49 GB
0.14 GB
resource_util
(torch alloc)
resource_util_max
(NVML used)

Metric Overlay — Disk, w=4, B=128
blue = resource_util (CUDA sync) orange = resource_util_max (no sync)

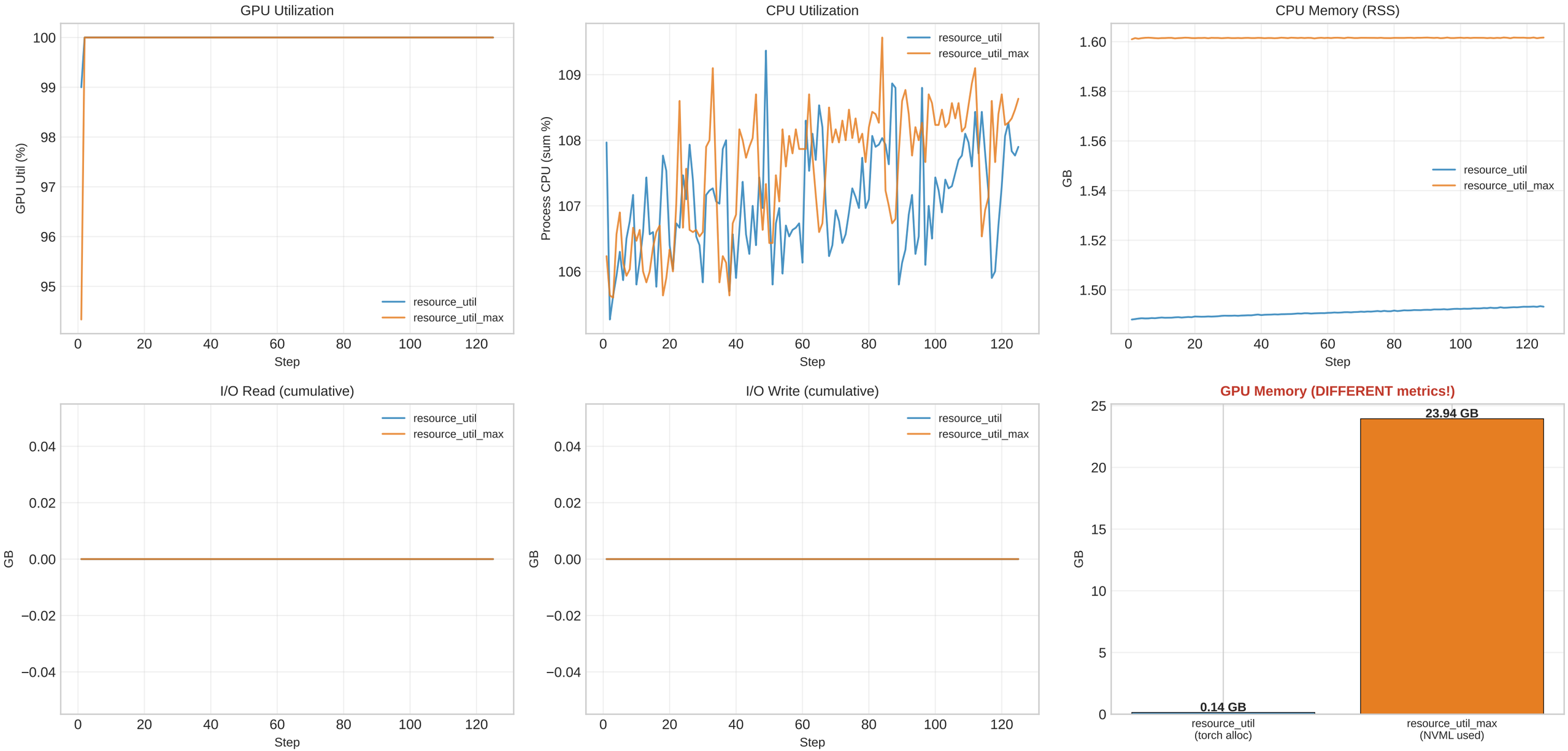



Metric Overlay — Milabench, w=0, B=32
blue = resource_util (CUDA sync) orange = resource_util_max (no sync)
GPU Utilization
GPU Util (%)
Step
resource_util
resource_util_max
CPU Utilization
Process CPU (sum %)
Step
resource_util
resource_util_max
CPU Memory (RSS)
GB
Step
resource_util
resource_util_max
I/O Read (cumulative)
GB
Step
resource_util
resource_util_max
I/O Write (cumulative)
GB
Step
resource_util
resource_util_max
GPU Memory (DIFFERENT metrics!)
GB
6.81 GB
0.14 GB
resource_util
(torch alloc)
resource_util_max
(NVML used)

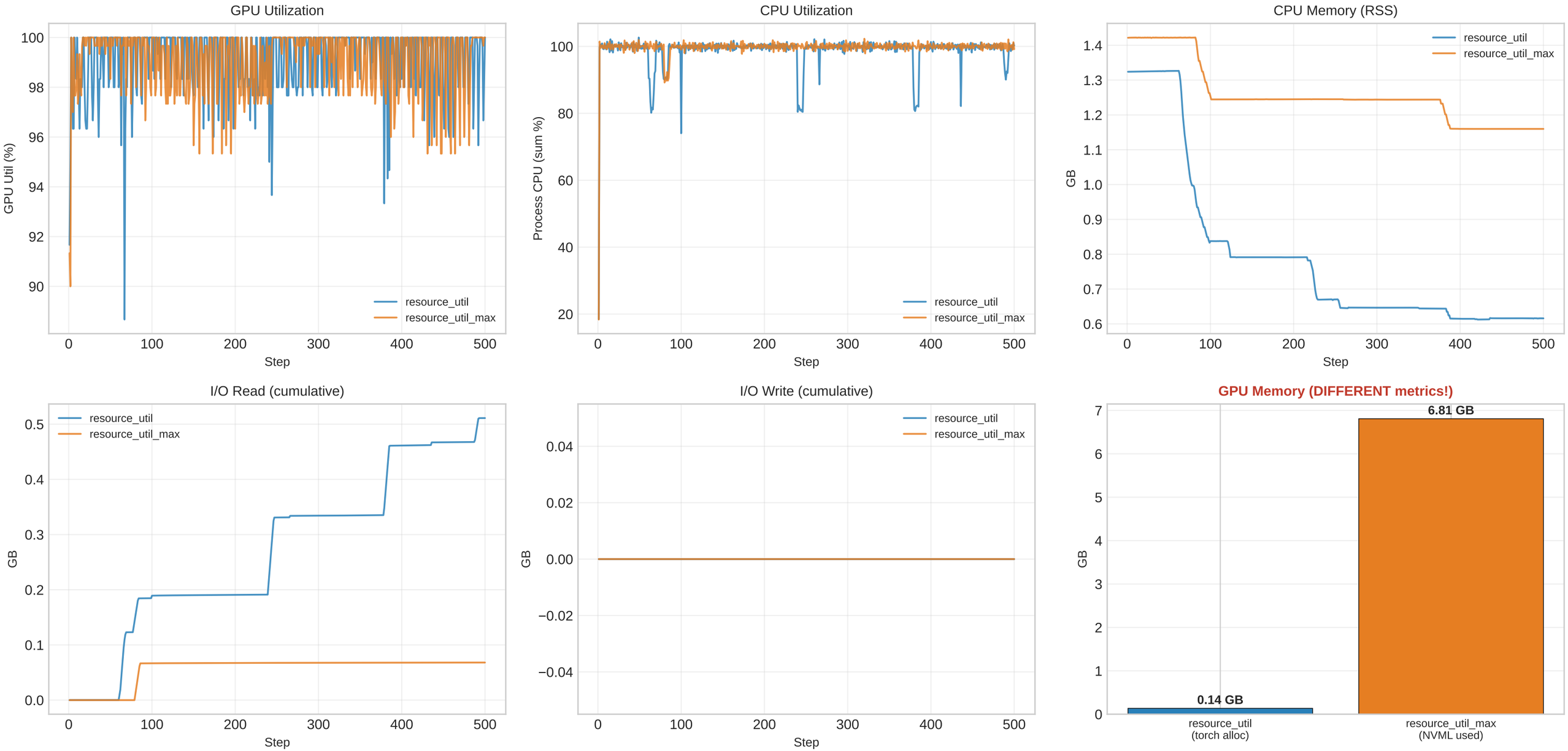
Metric Overlay — Milabench, w=0, B=64
blue = resource_util (CUDA sync) orange = resource_util_max (no sync)



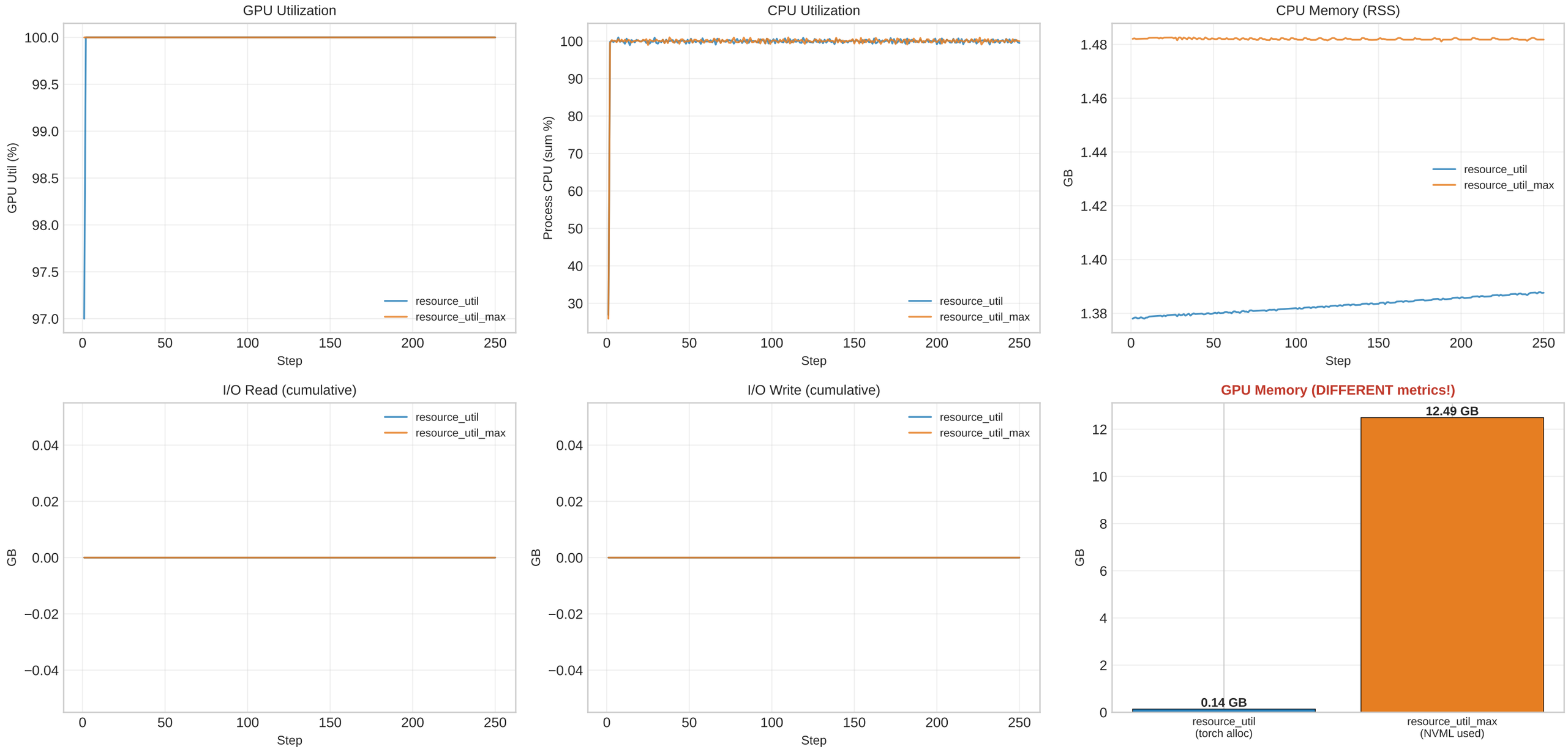
Metric Overlay — Milabench, w=0, B=128
blue = resource_util (CUDA sync) orange = resource_util_max (no sync)

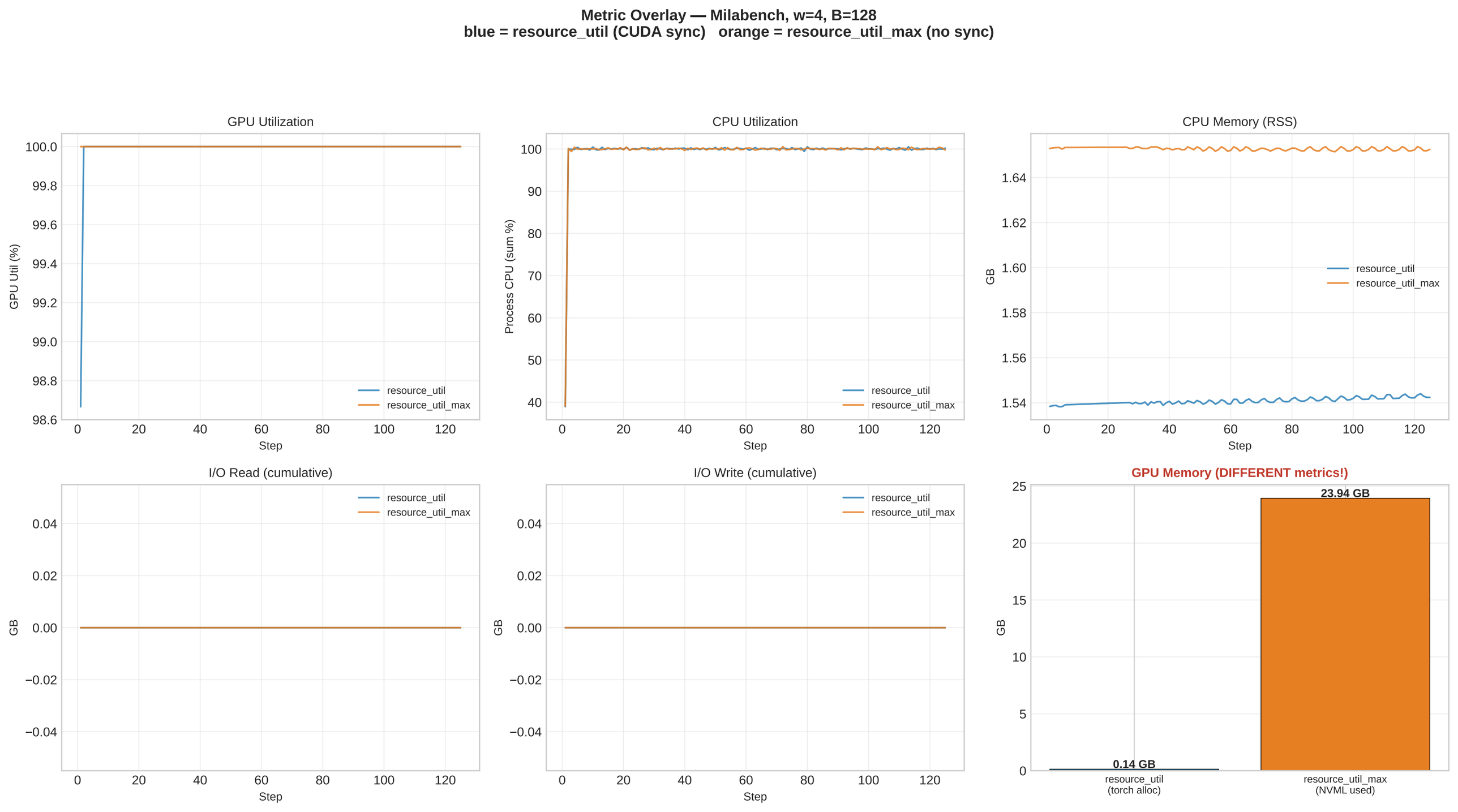


Metric Overlay — Milabench, w=4, B=32
blue = resource_util (CUDA sync) orange = resource_util_max (no sync)



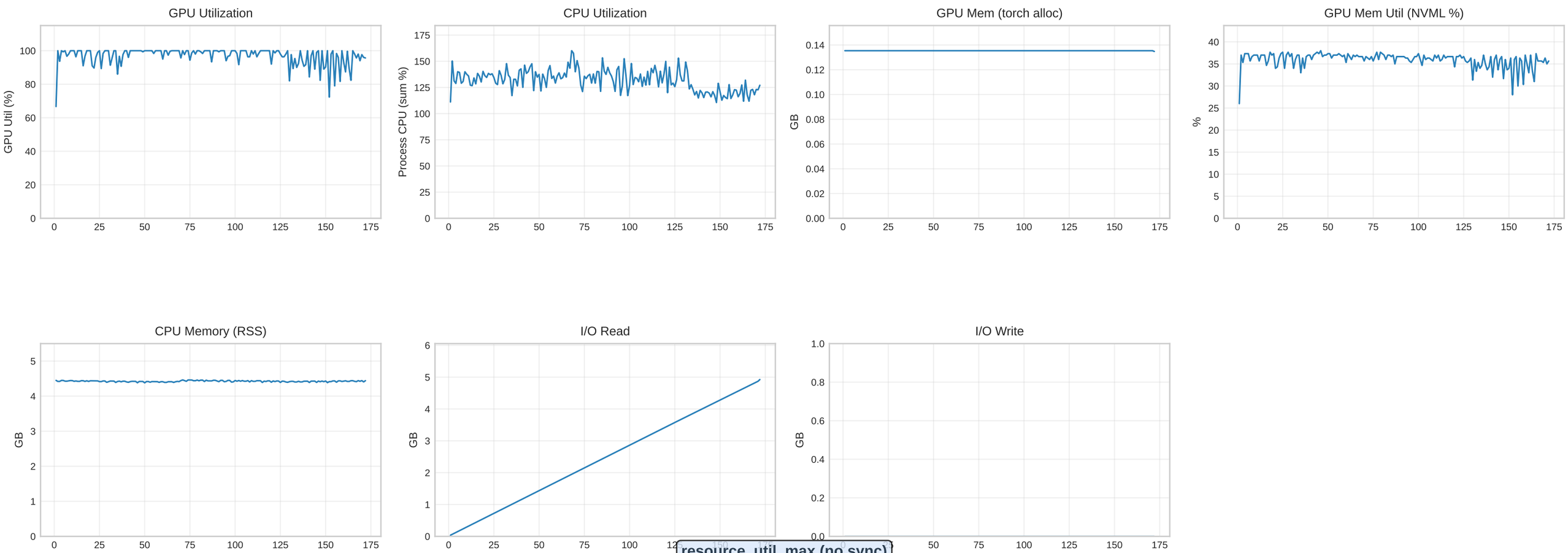
Metric Overlay — Milabench, w=4, B=64
blue = resource_util (CUDA sync) orange = resource_util_max (no sync)



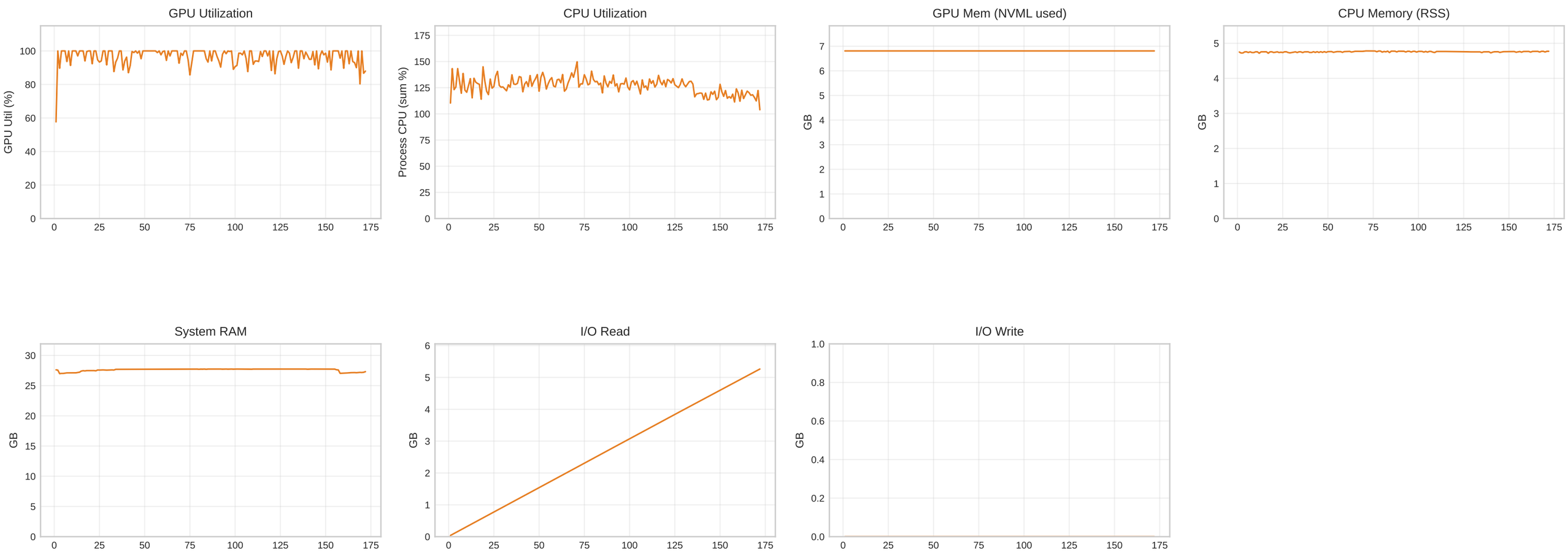

Metric Overlay — Milabench, w=4, B=128
blue = resource_util (CUDA sync) orange = resource_util_max (no sync)
GPU Utilization
GPU Util (%)
Step
resource_util
resource_util_max
CPU Utilization
Process CPU (sum %)
Step
resource_util
resource_util_max
CPU Memory (RSS)
GB
Step
resource_util
resource_util_max
I/O Read (cumulative)
GB
Step
resource_util
resource_util_max
I/O Write (cumulative)
GB
Step
resource_util
resource_util_max
GPU Memory (DIFFERENT metrics!)
GB
23.94 GB
0.14 GB
resource_util
(torch alloc)
resource_util_max
(NVML used)

Resource Timeline — Disk, w=0, B=32

resource_util (with CUDA sync)

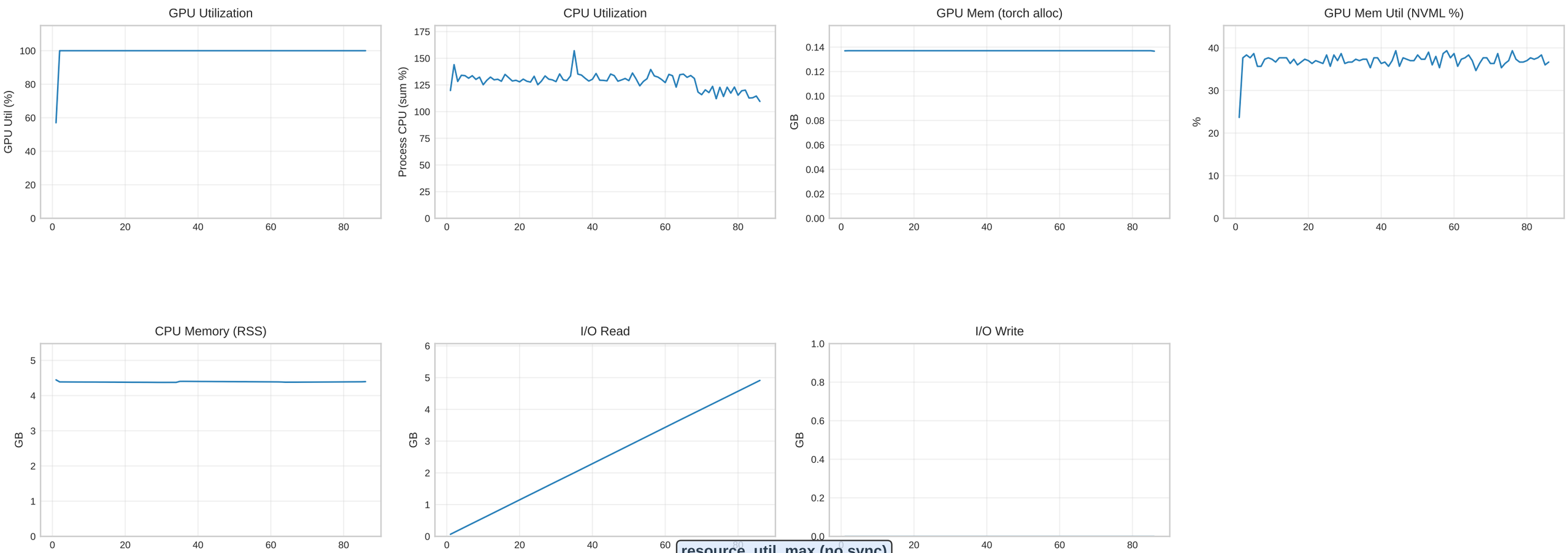


resource_util_max (no sync)

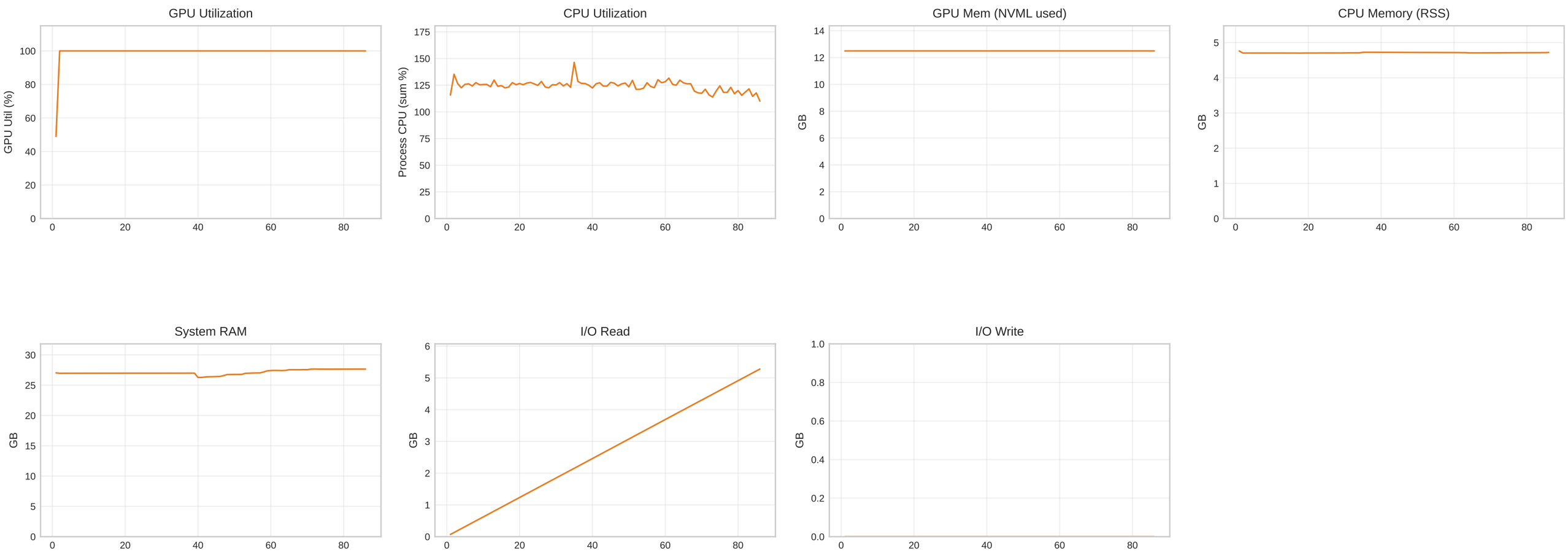


Resource Timeline — Disk, w=0, B=64

resource_util (with CUDA sync)

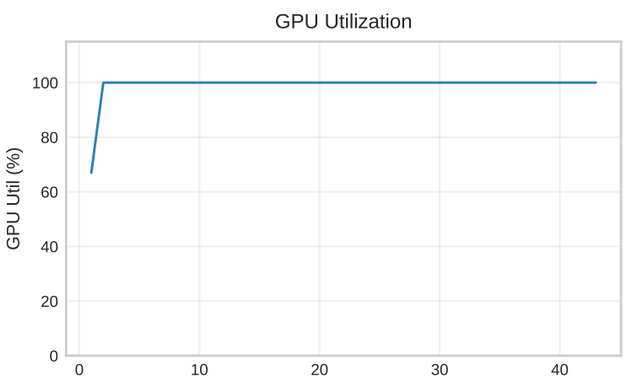


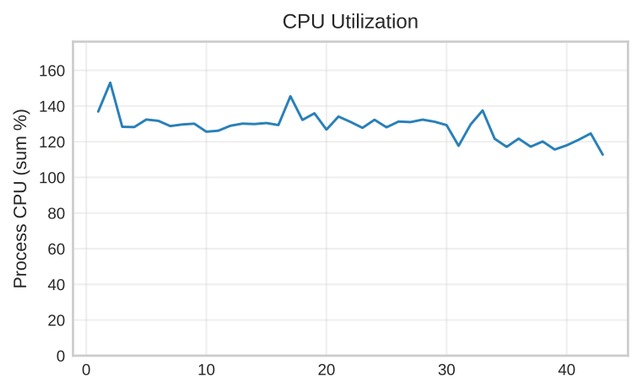
resource_util_max (no sync)

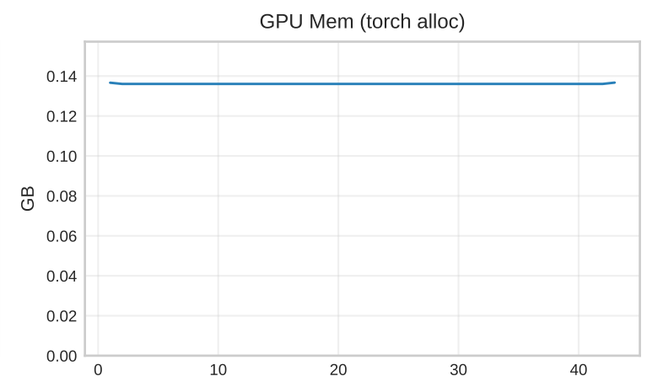


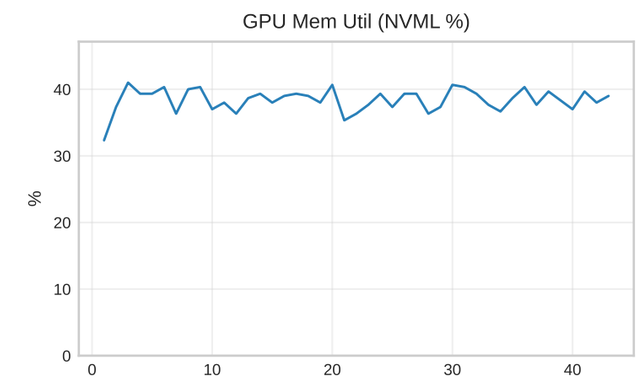
Resource Timeline — Disk, w=0, B=128

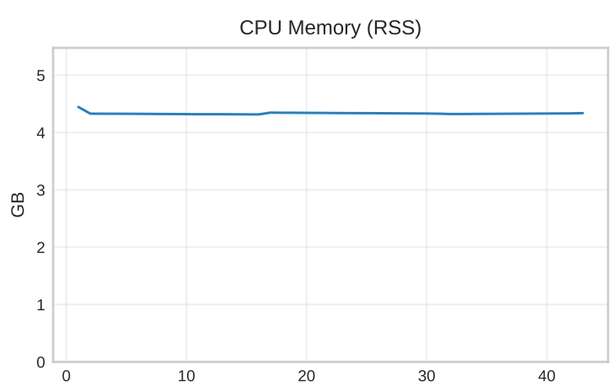
resource_util (with CUDA sync)

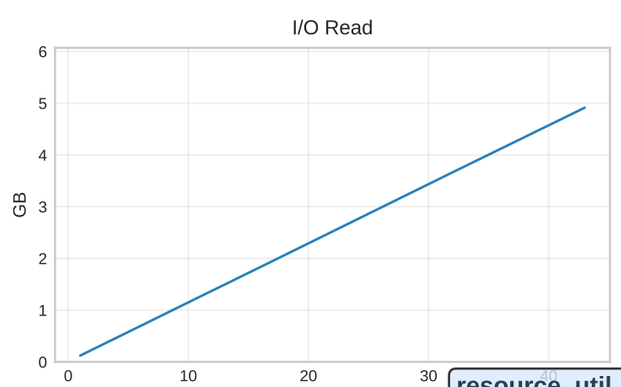


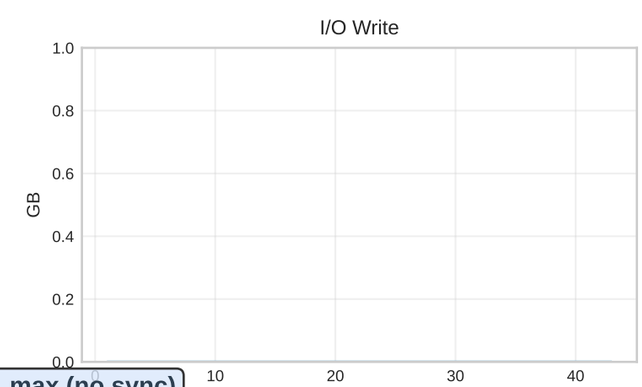




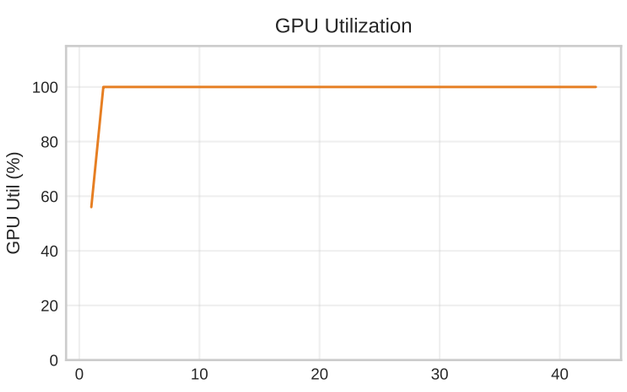


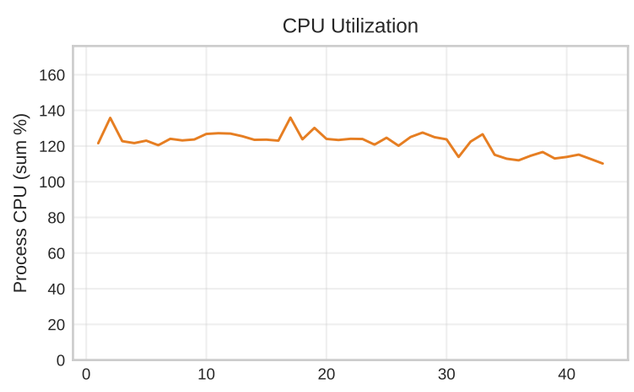


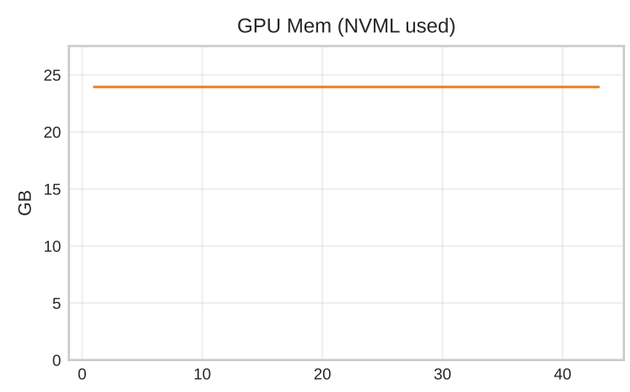


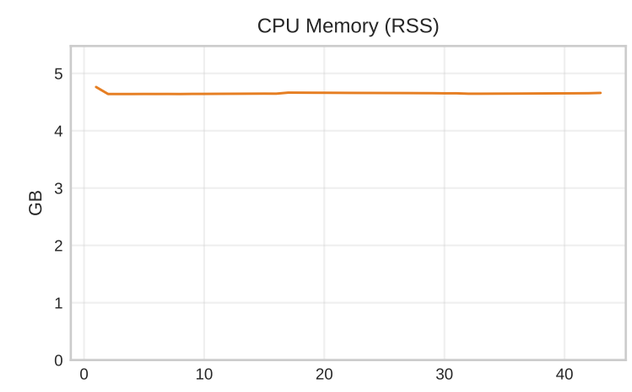


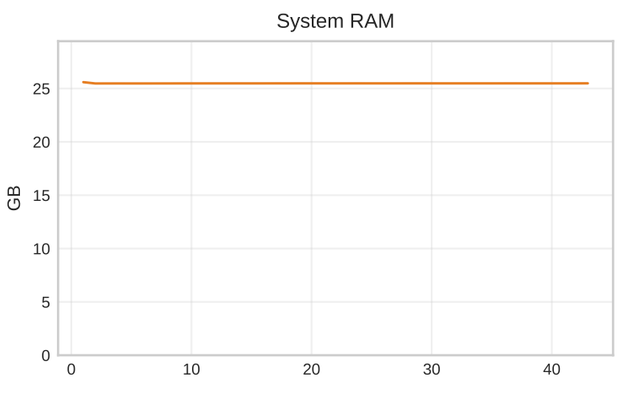
resource_util_max (no sync)

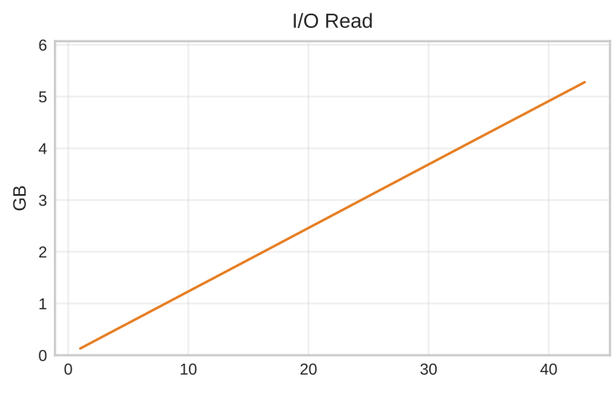


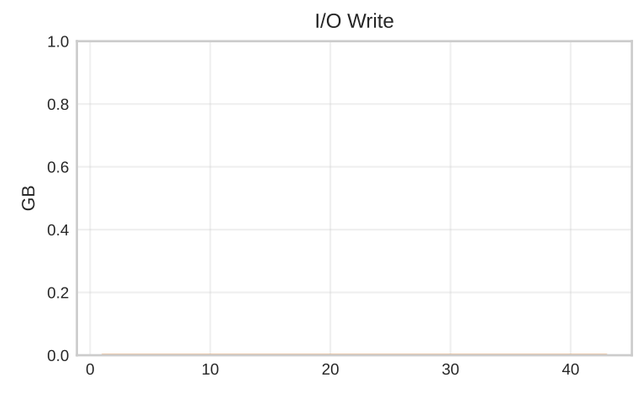






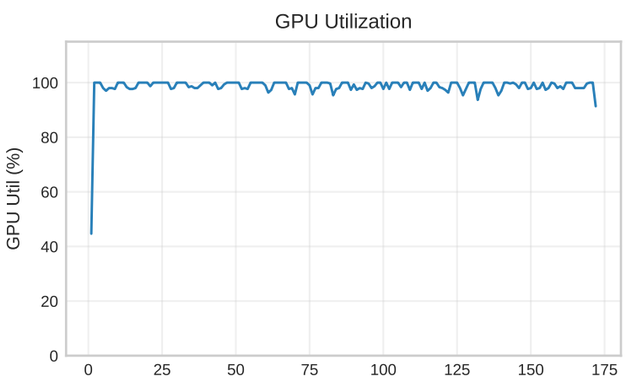


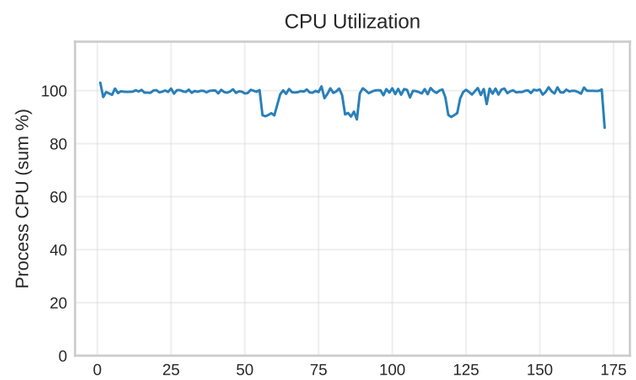


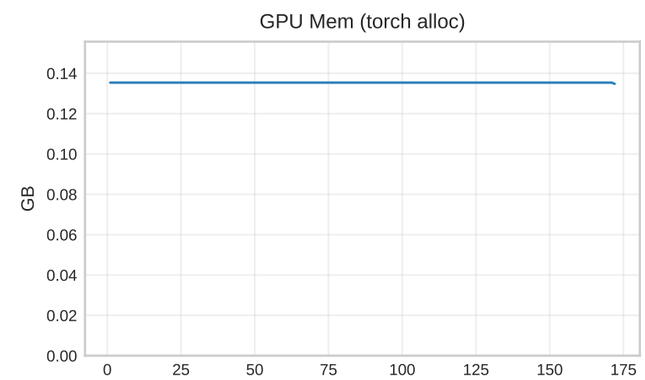


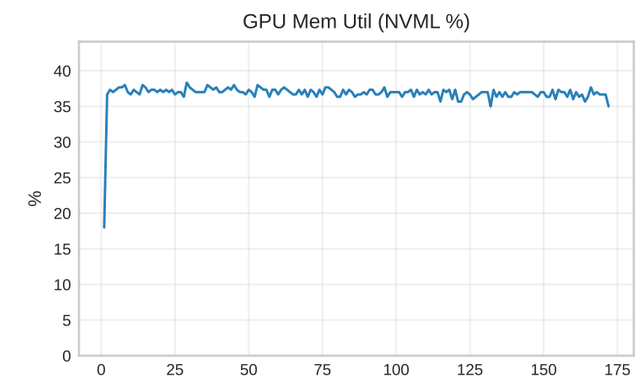
Resource Timeline — Disk, w=4, B=32

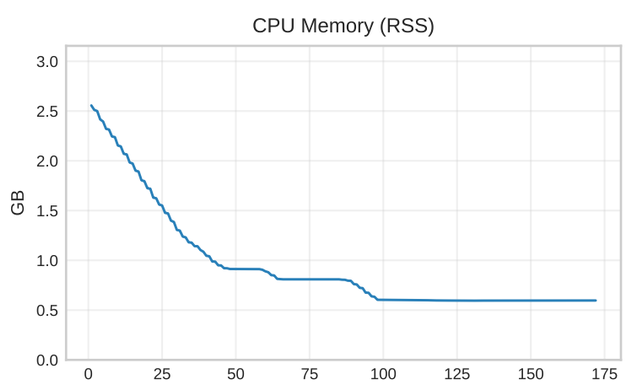
resource_util (with CUDA sync)

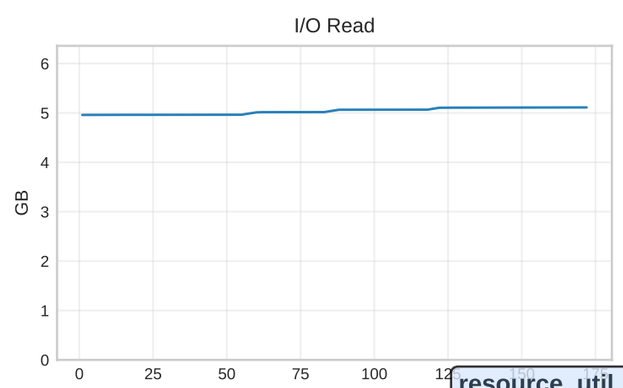


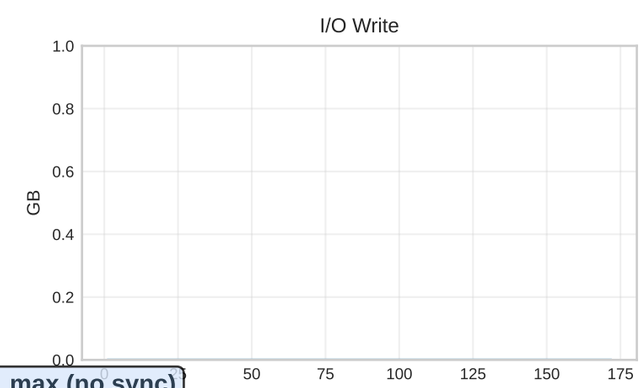




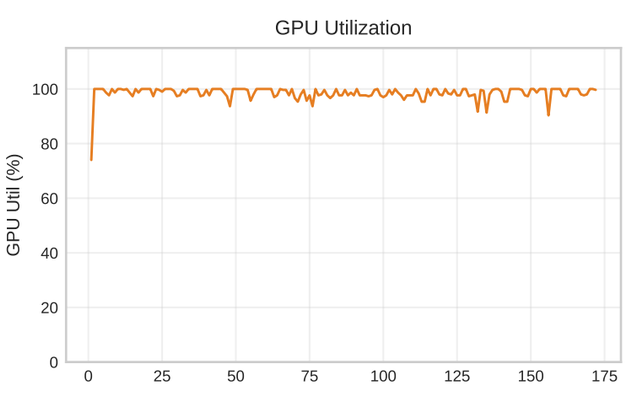


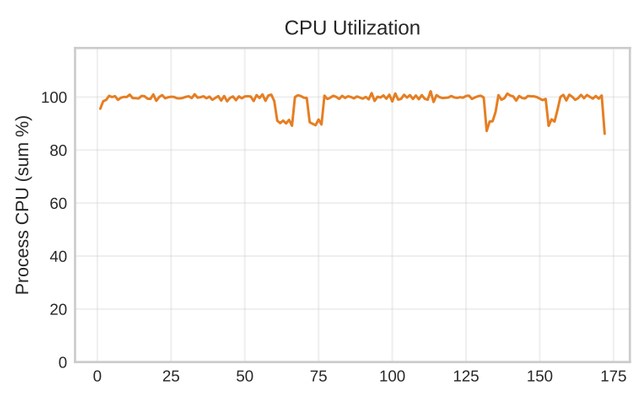


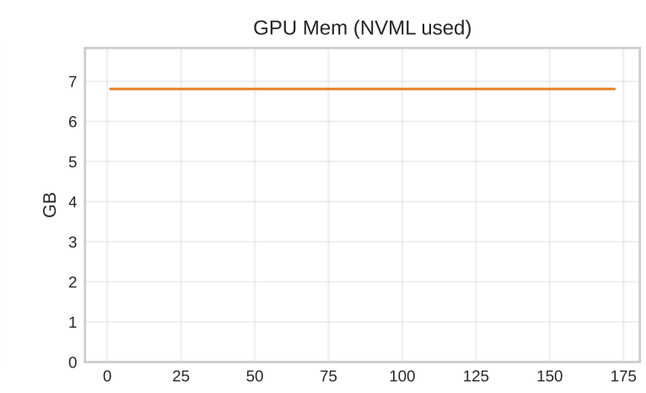


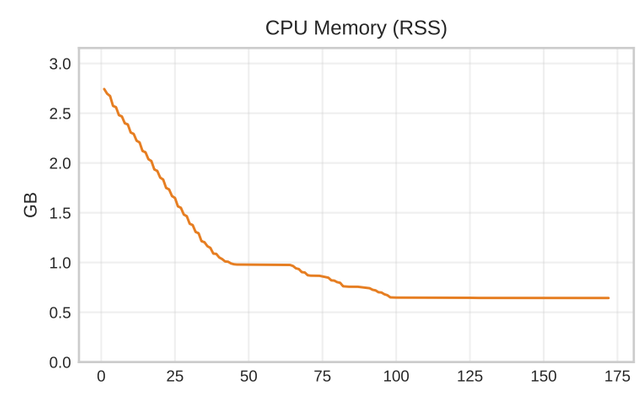


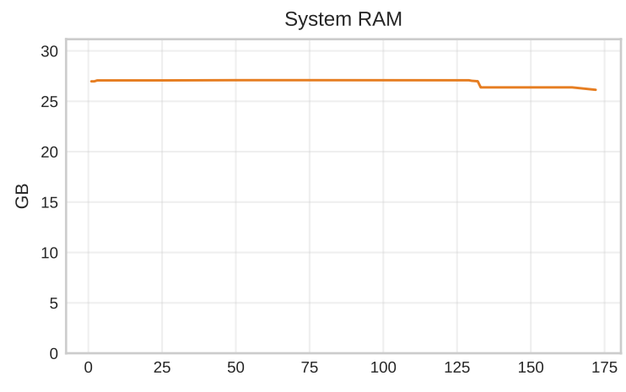
resource_util_max (no sync)

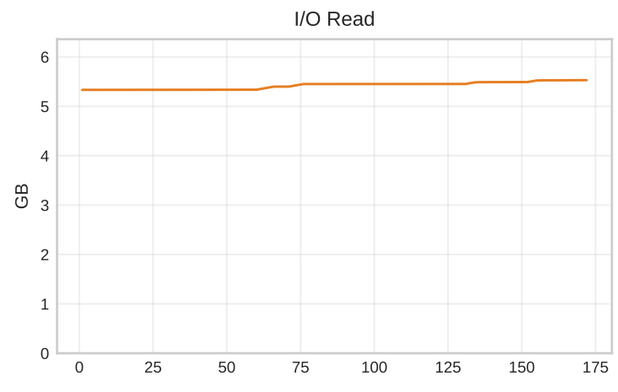


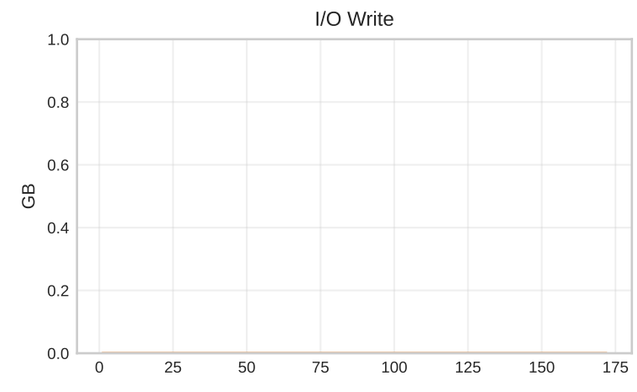






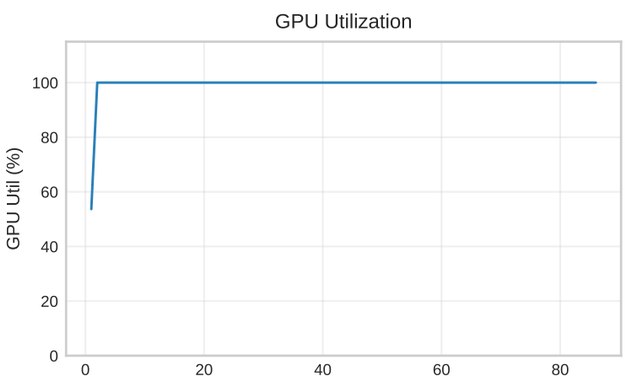


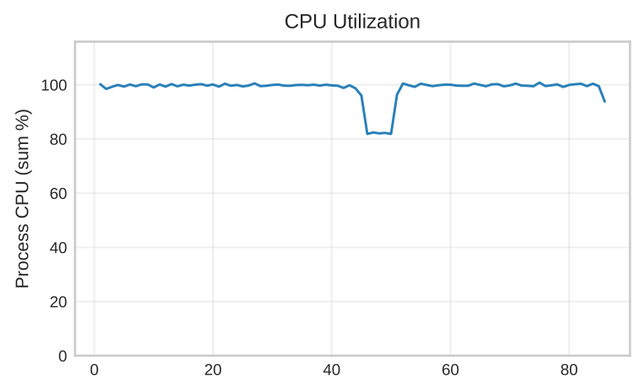


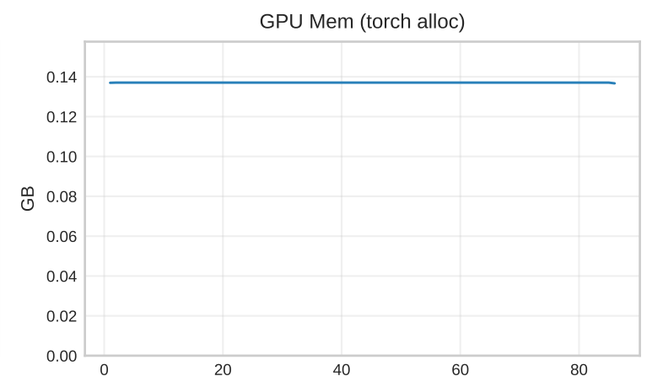


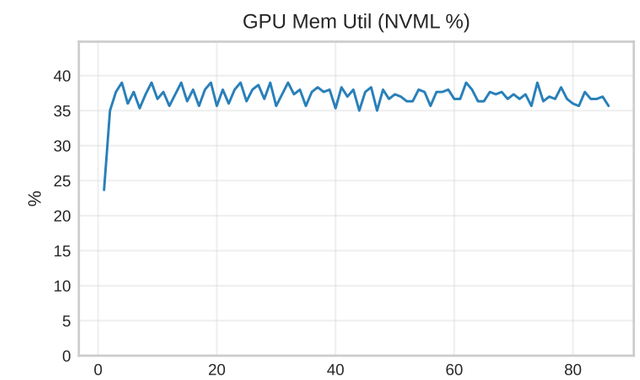
Resource Timeline — Disk, w=4, B=64

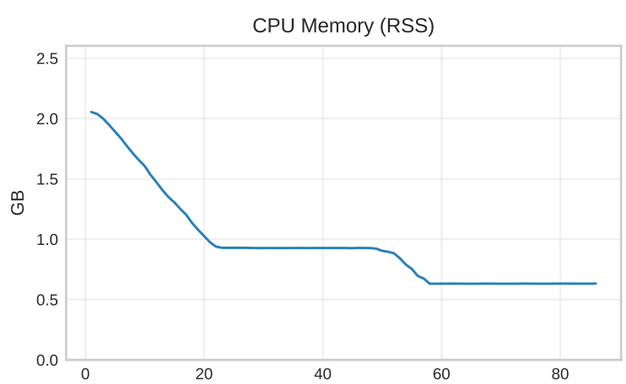
resource_util (with CUDA sync)

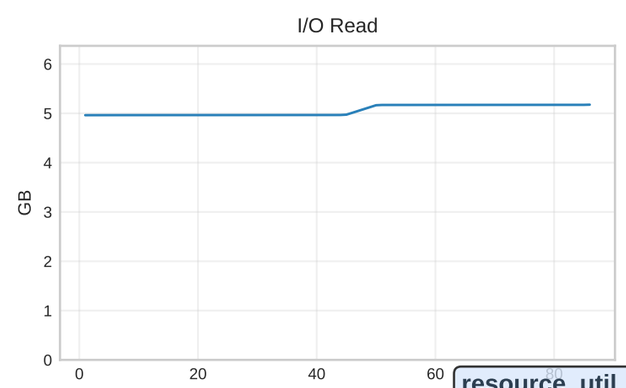


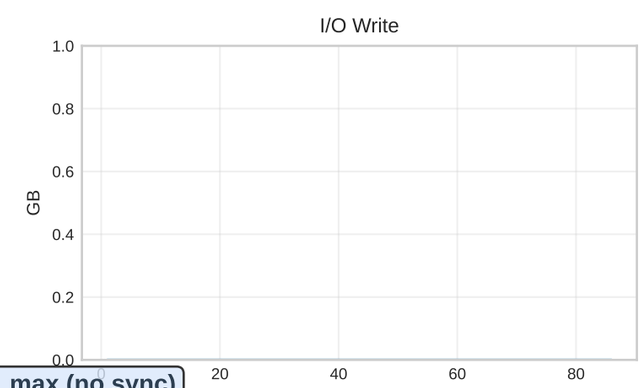




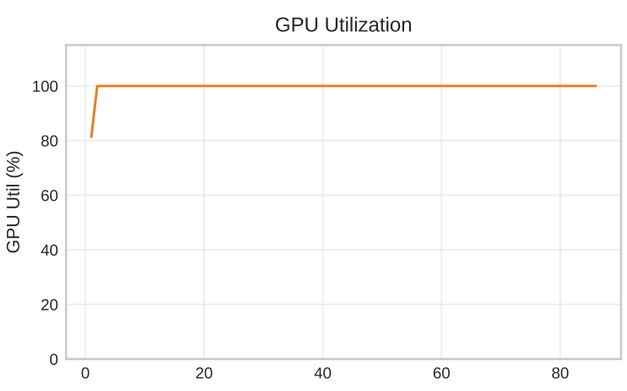


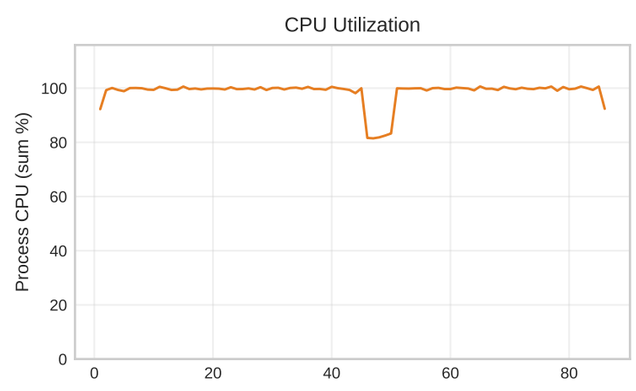


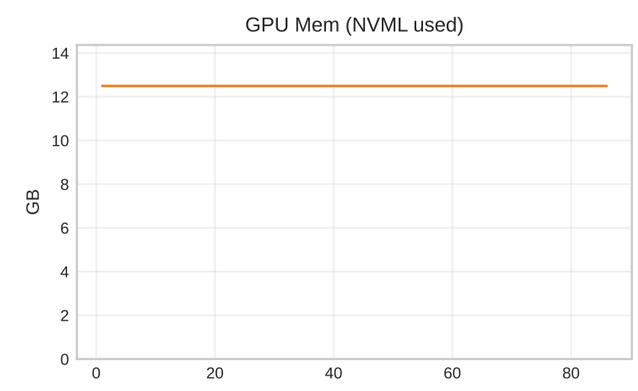


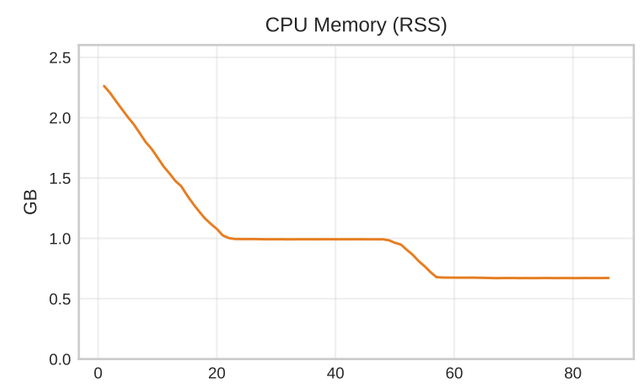


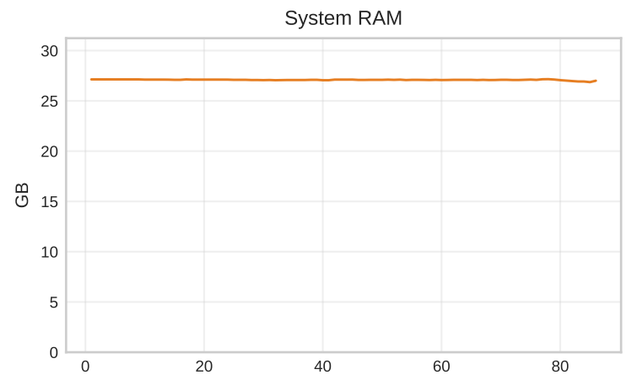
resource_util_max (no sync)

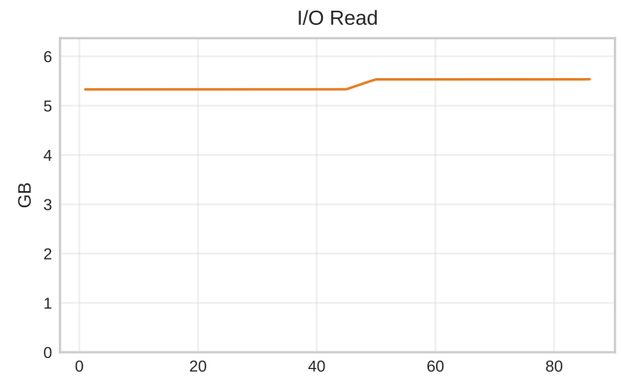


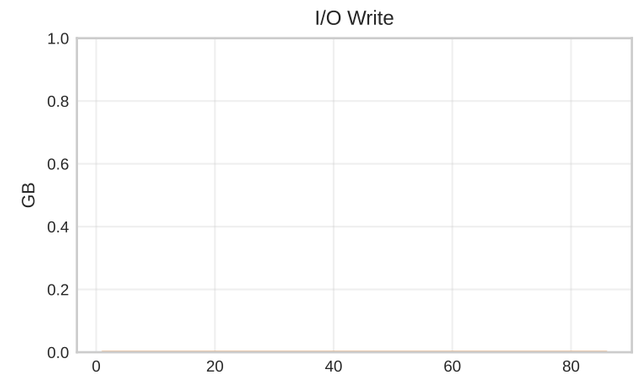






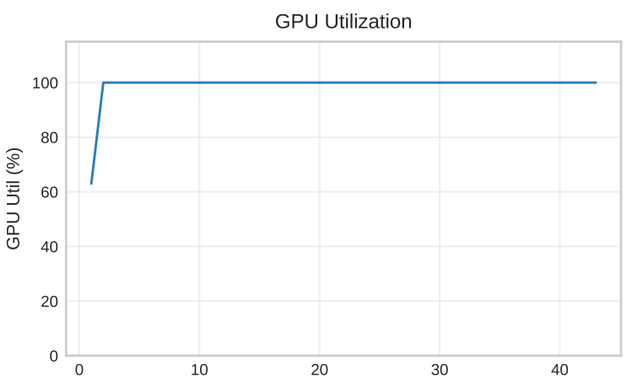


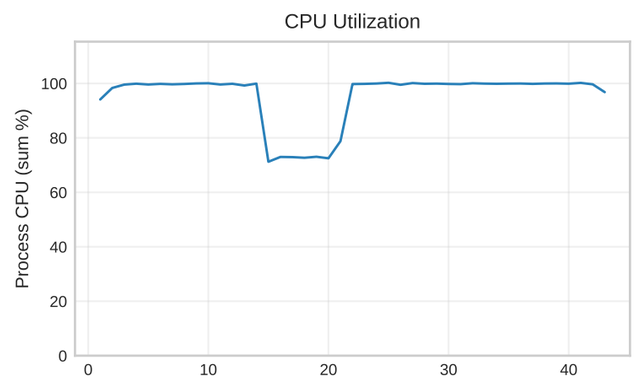


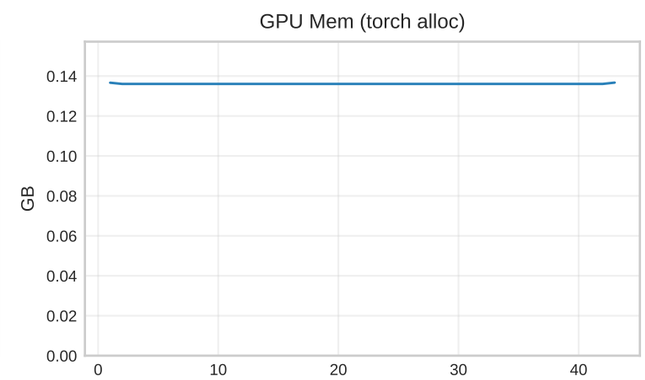


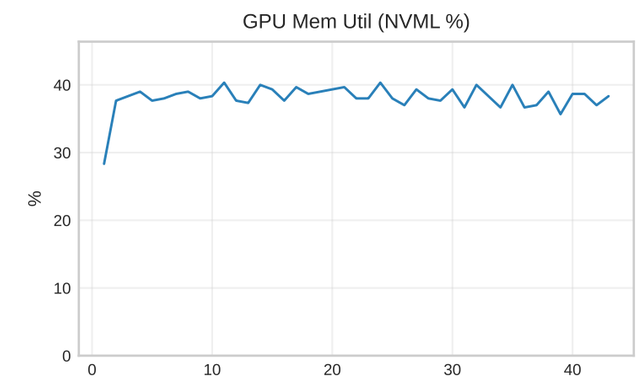
Resource Timeline — Disk, w=4, B=128

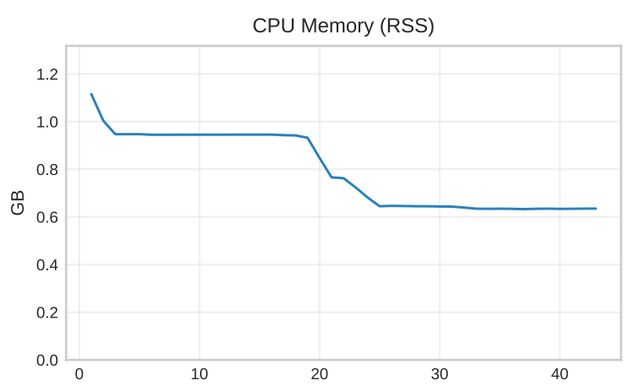
resource_util (with CUDA sync)

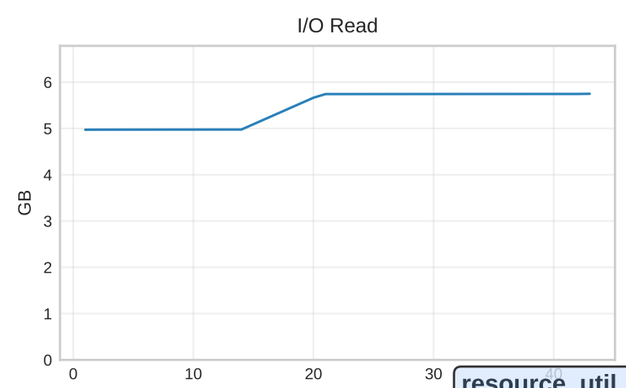


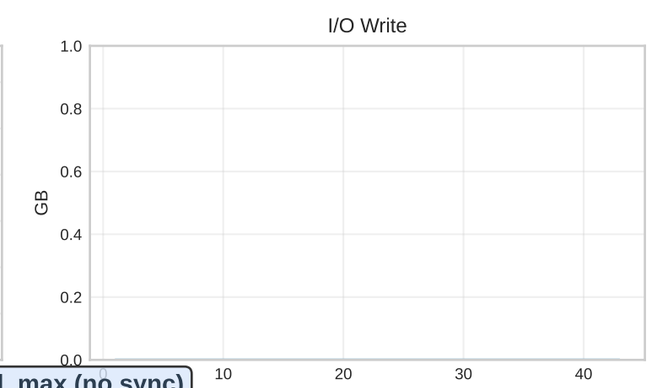




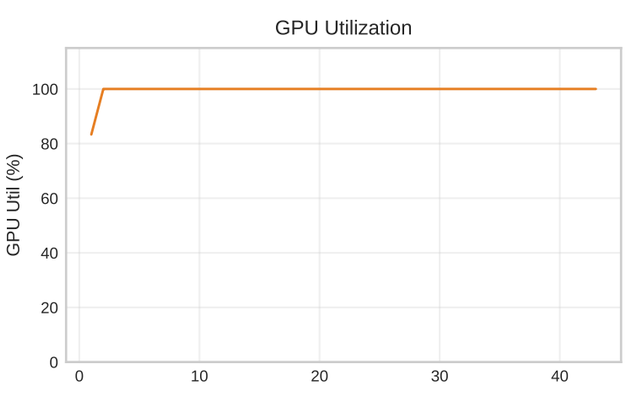


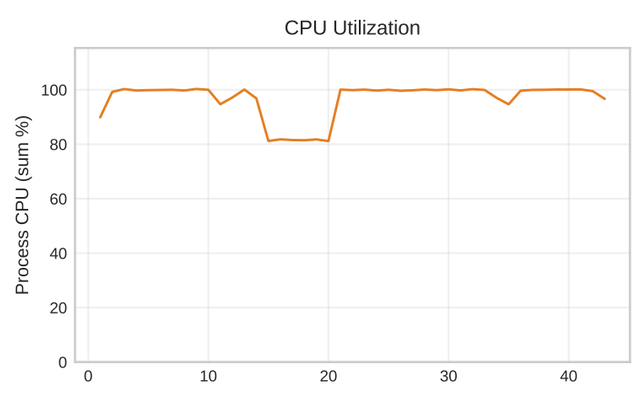


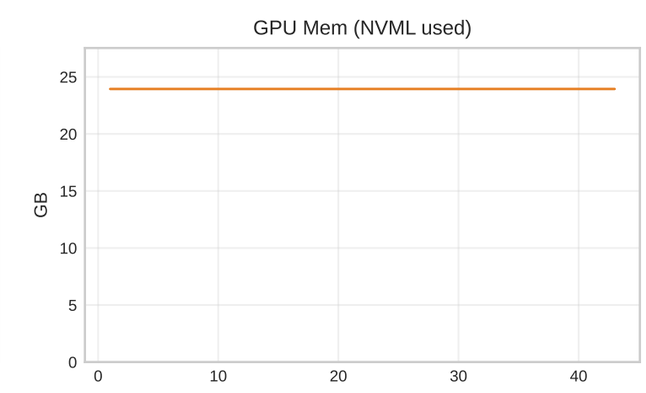


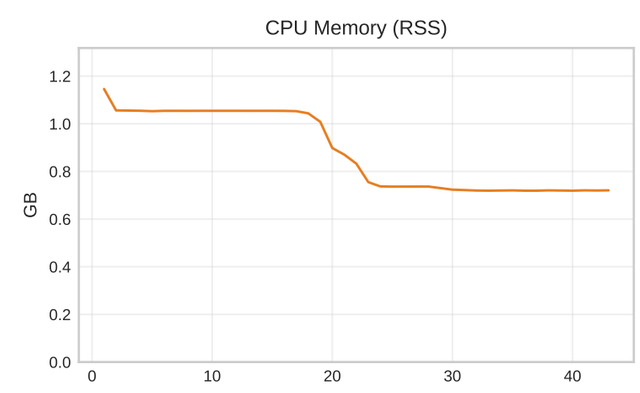


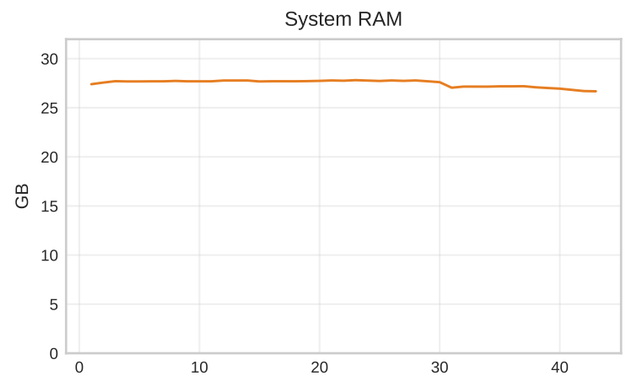
resource_util_max (no sync)

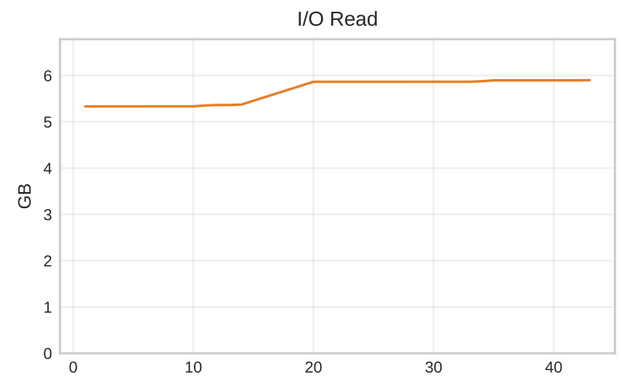


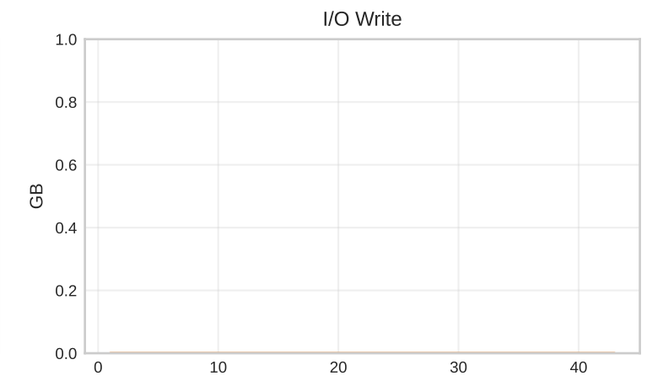






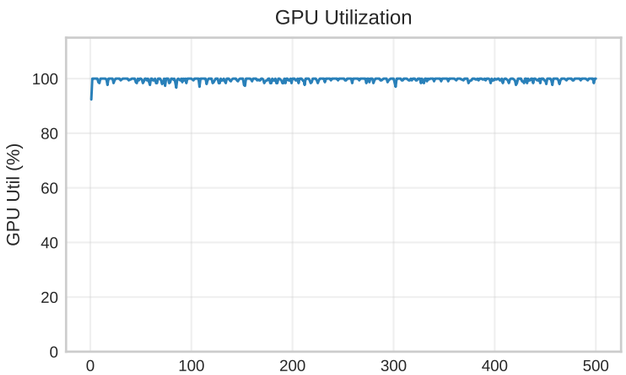


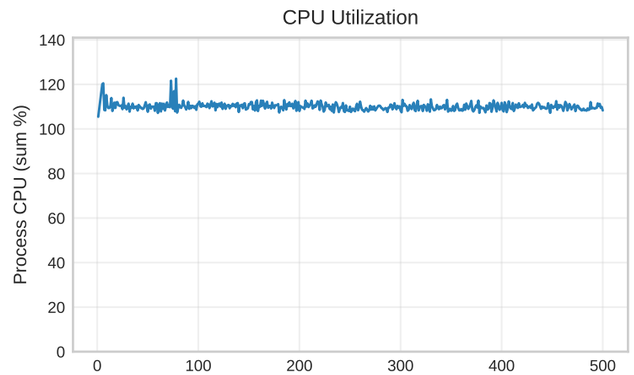


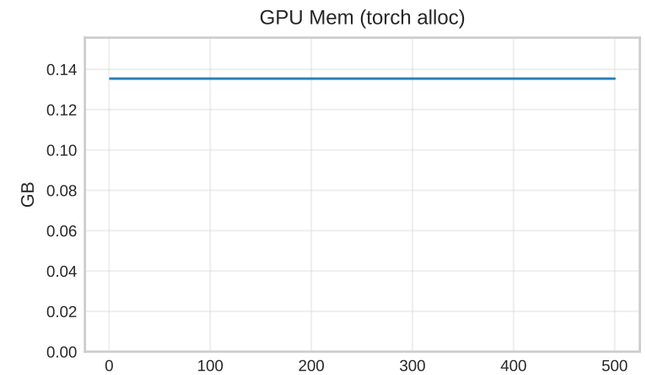


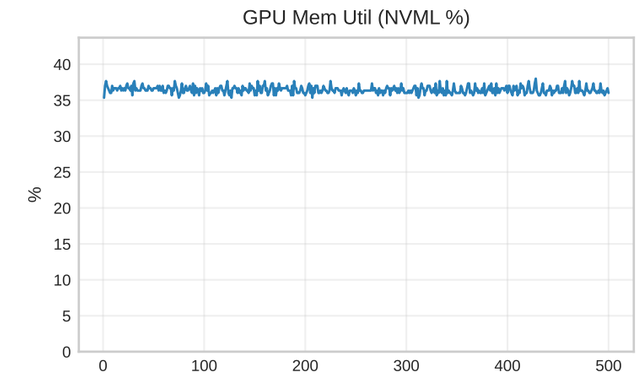
Resource Timeline — Milabench, w=0, B=32

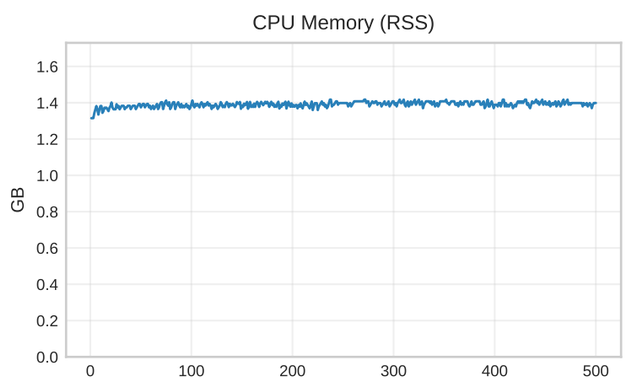
resource_util (with CUDA sync)

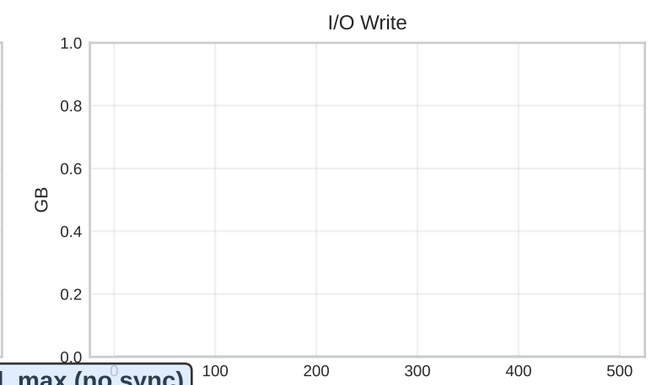




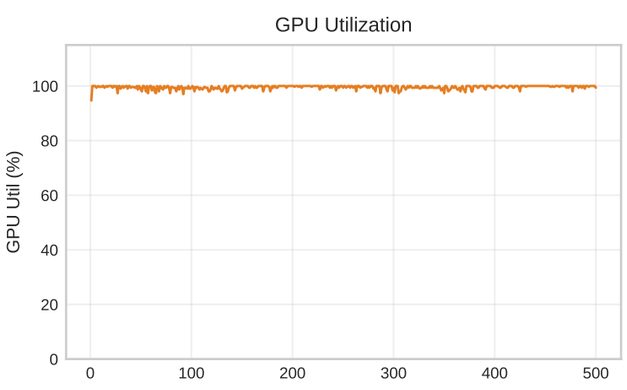


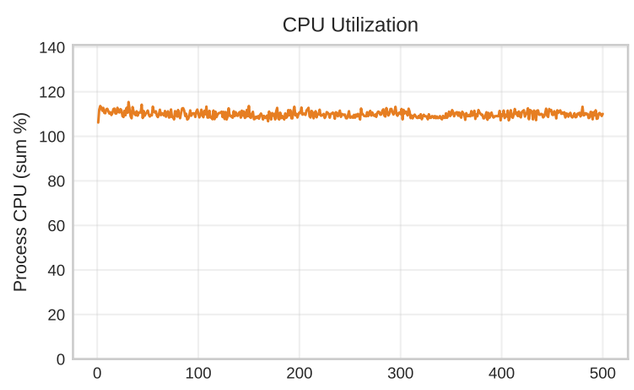


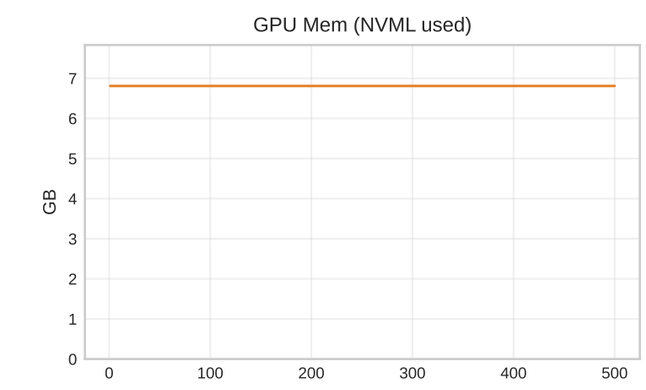


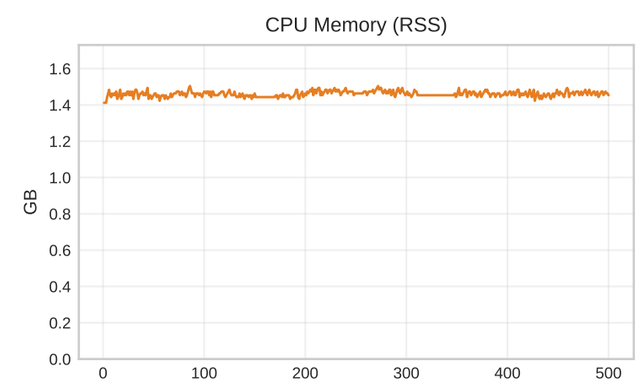


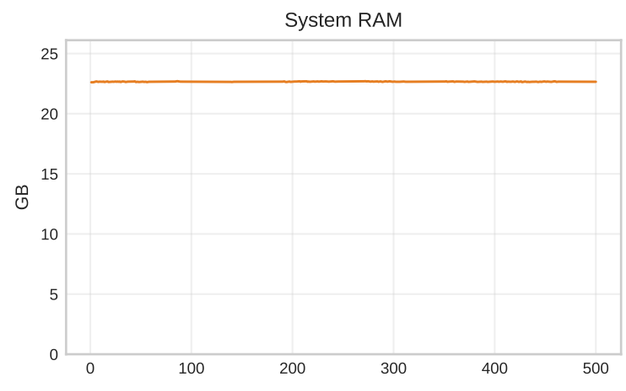
resource_util_max (no sync)

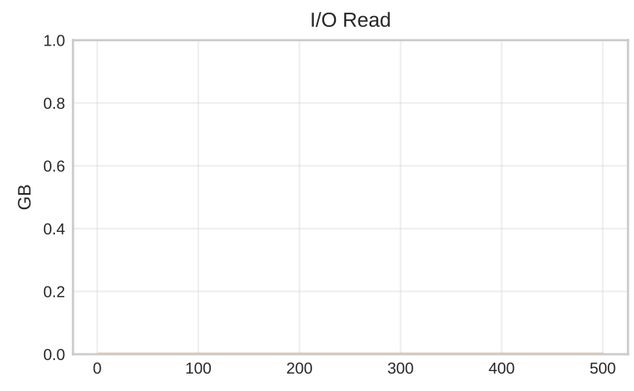


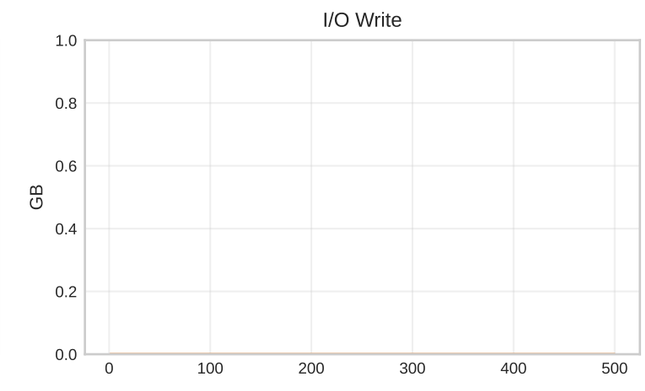






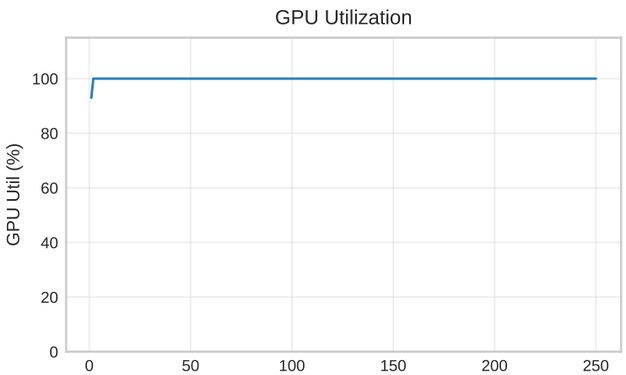


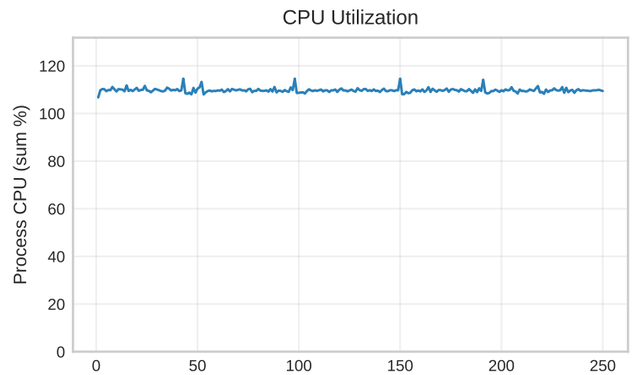


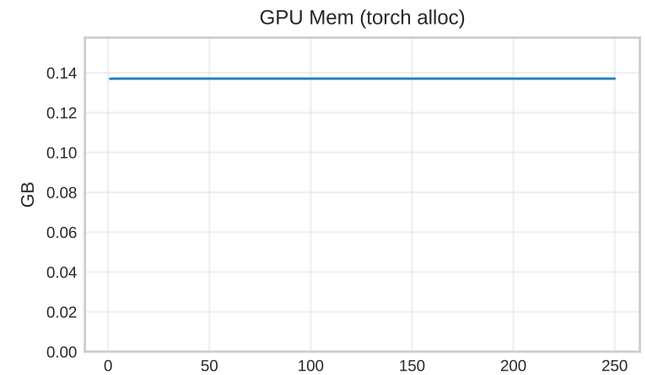


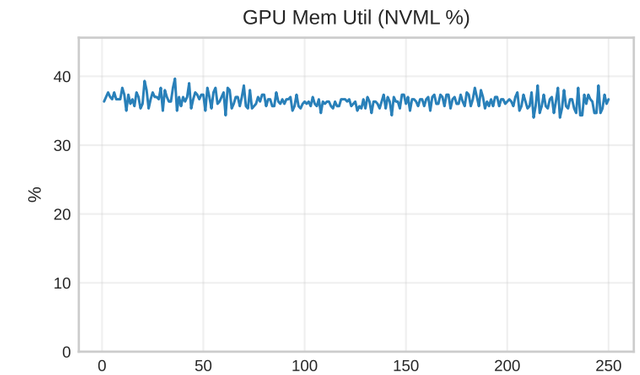
Resource Timeline — Milabench, w=0, B=64

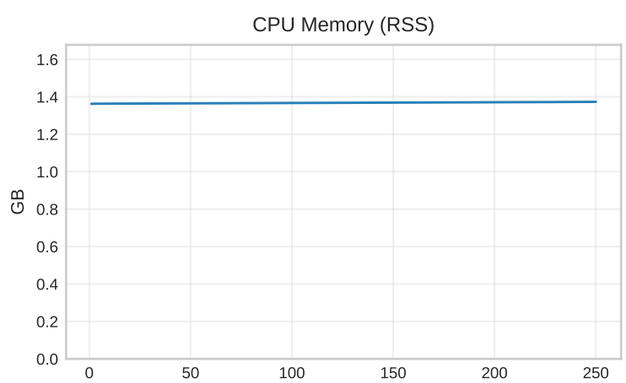
resource_util (with CUDA sync)

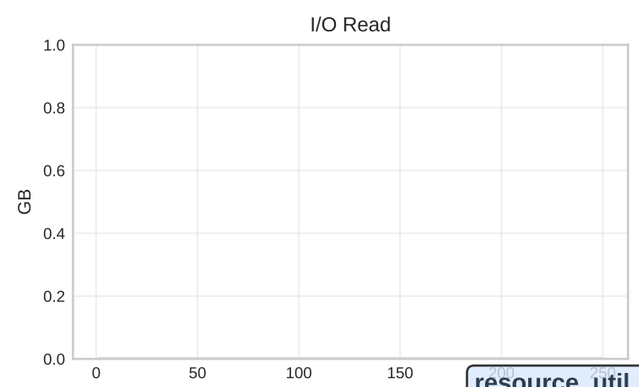


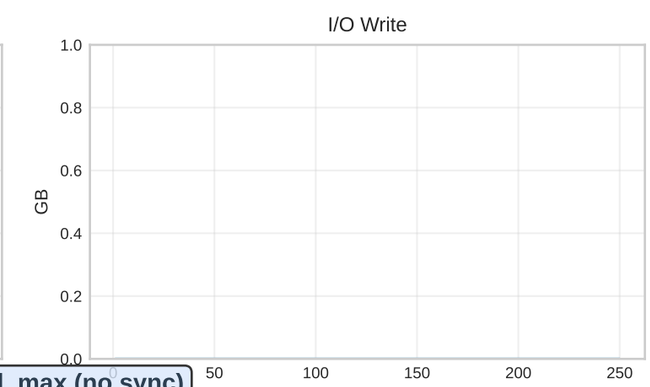




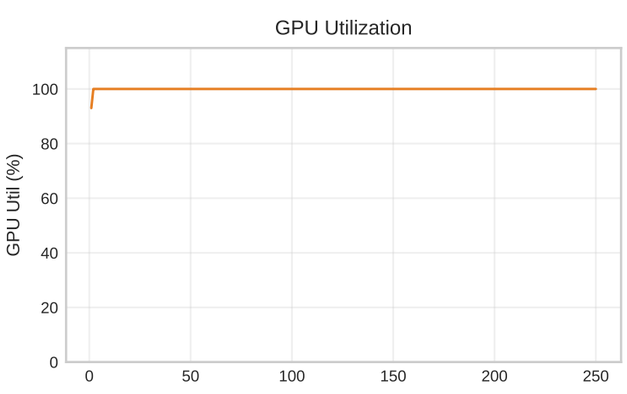


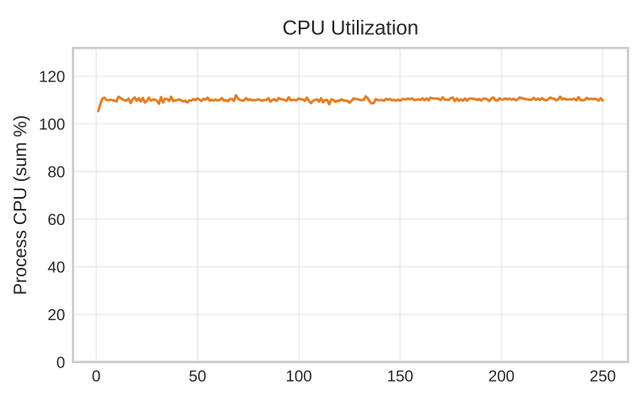


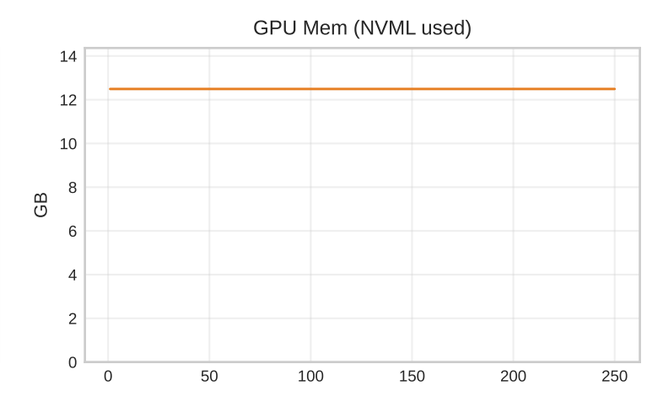


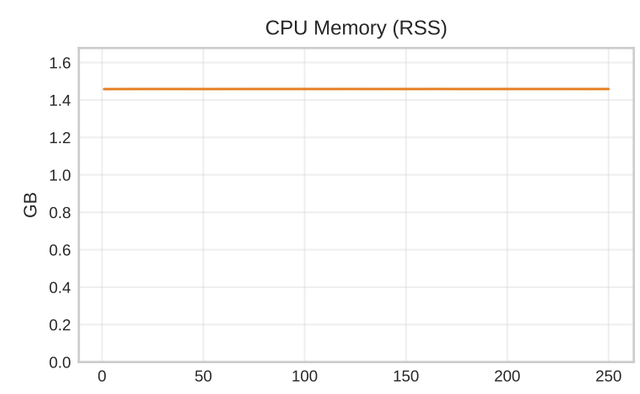


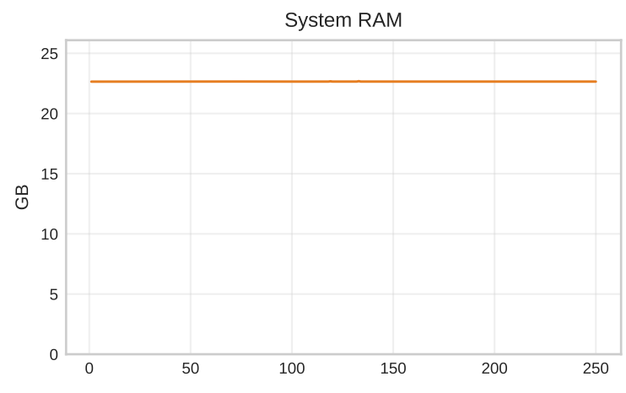
resource_util_max (no sync)

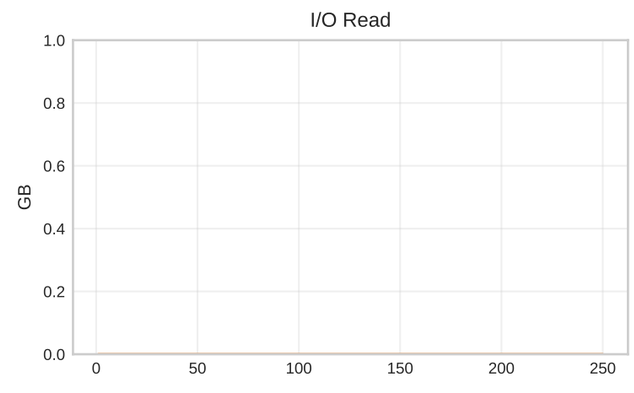


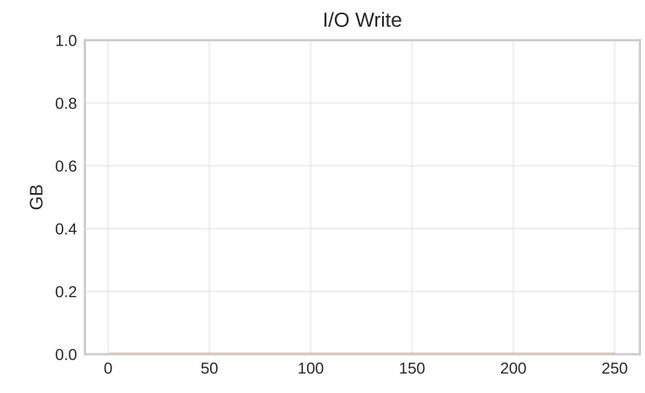






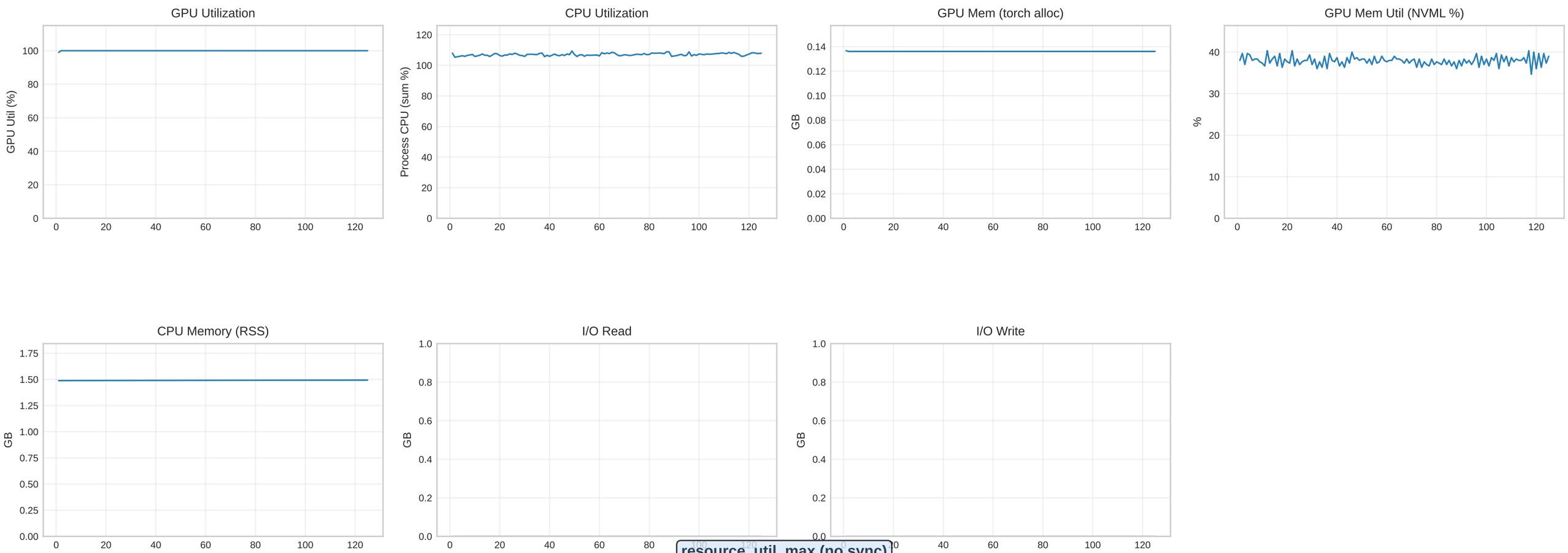




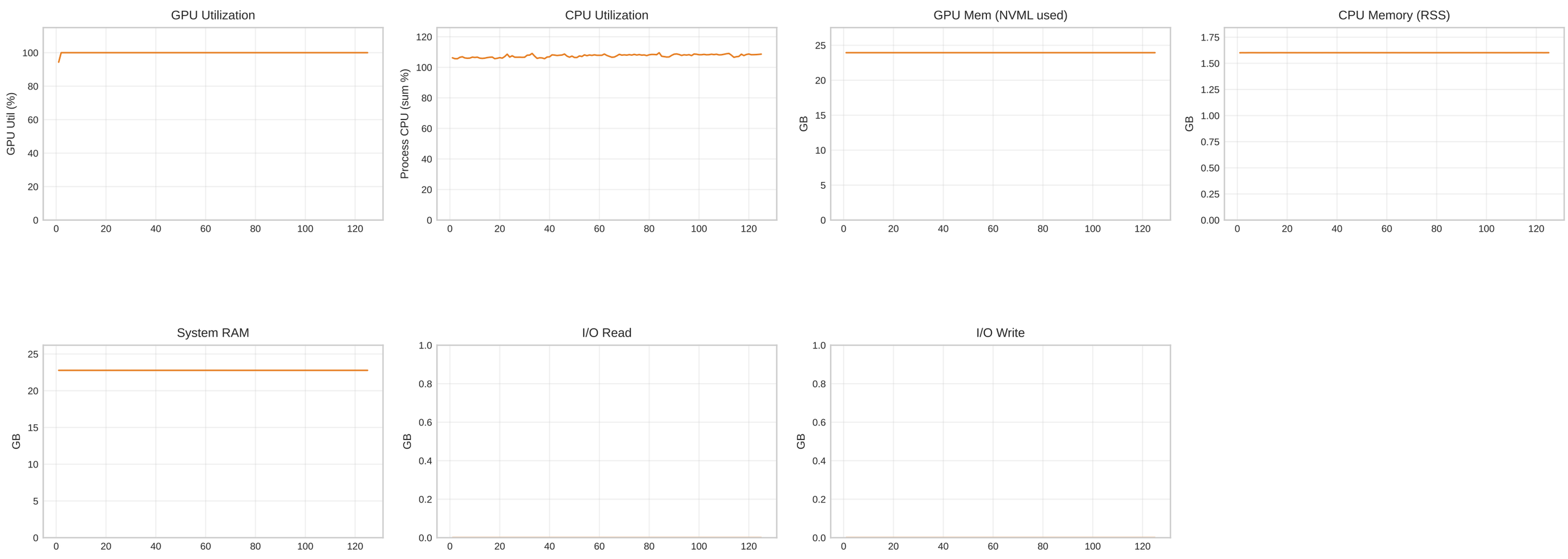


Resource Timeline — Milabench, w=0, B=128

resource_util (with CUDA sync)

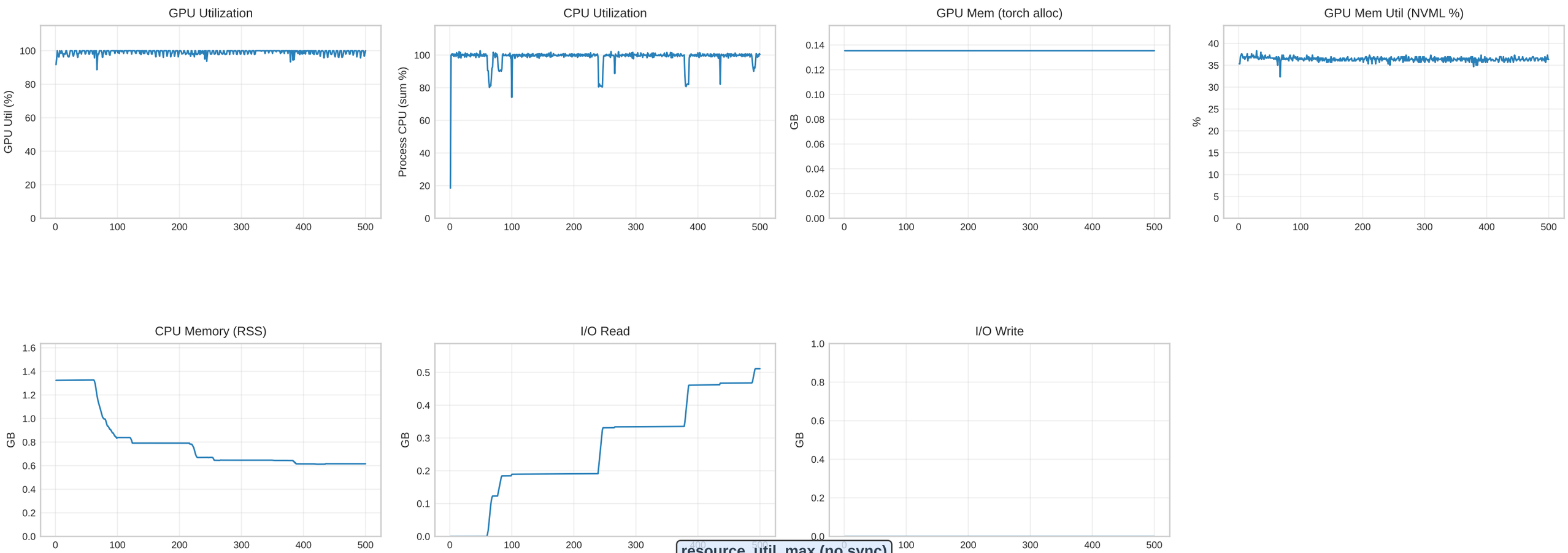


resource_util_max (no sync)

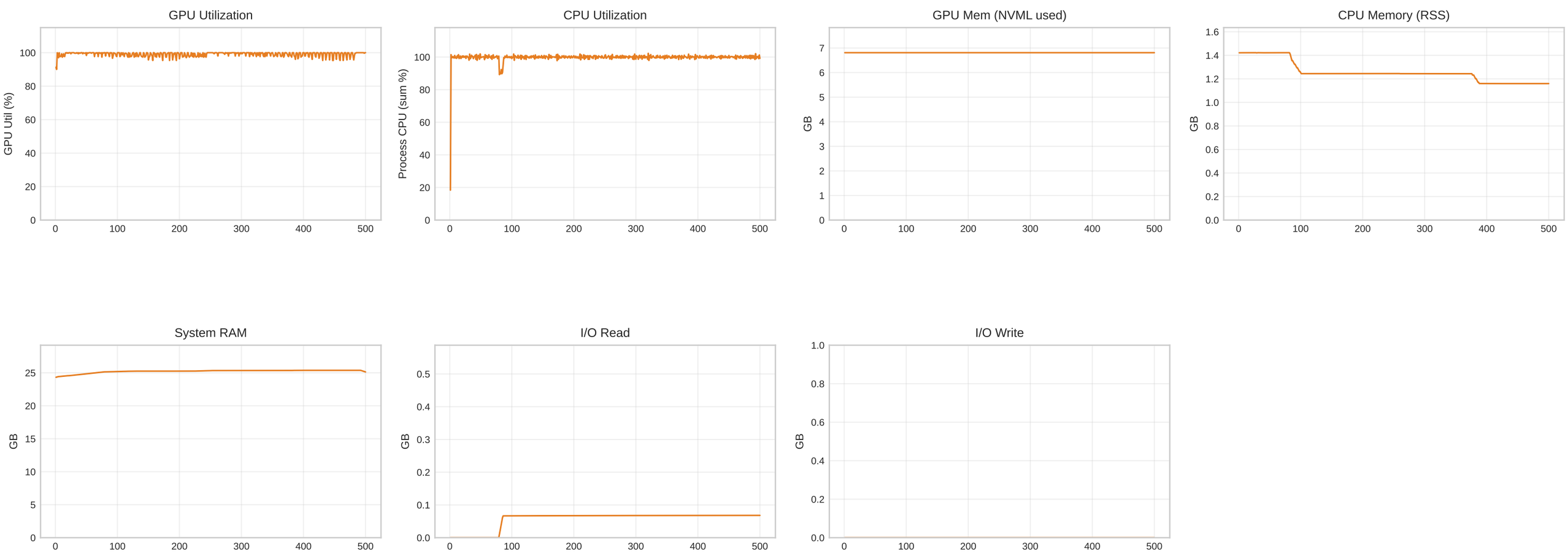


Resource Timeline — Milabench, w=4, B=32

resource_util (with CUDA sync)

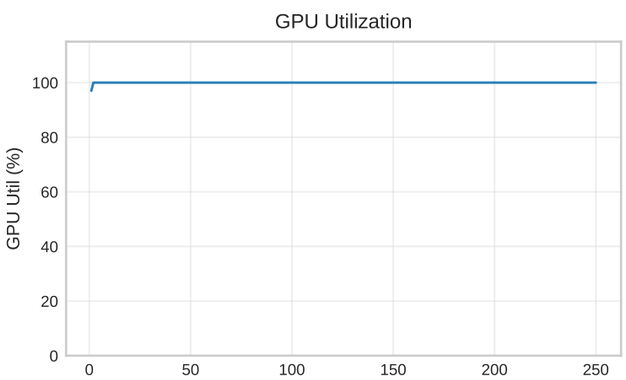


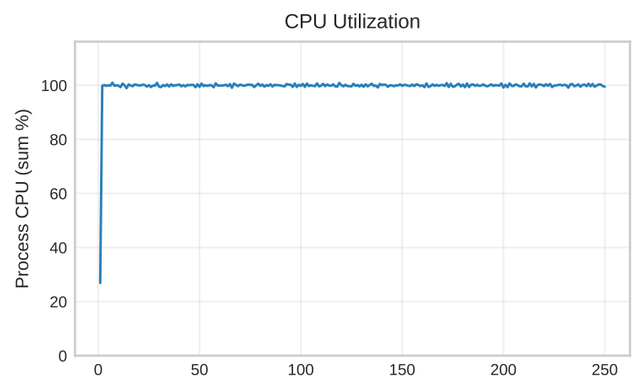
resource_util_max (no sync)

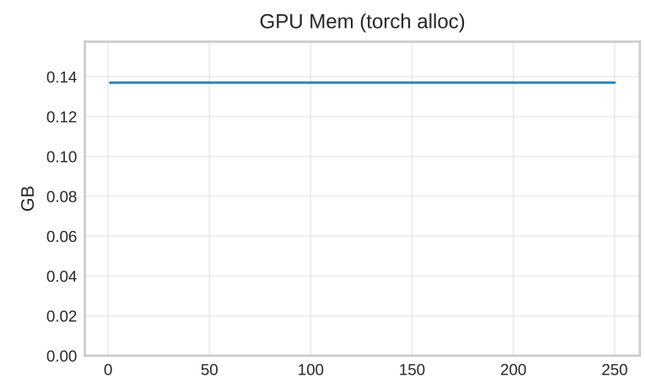


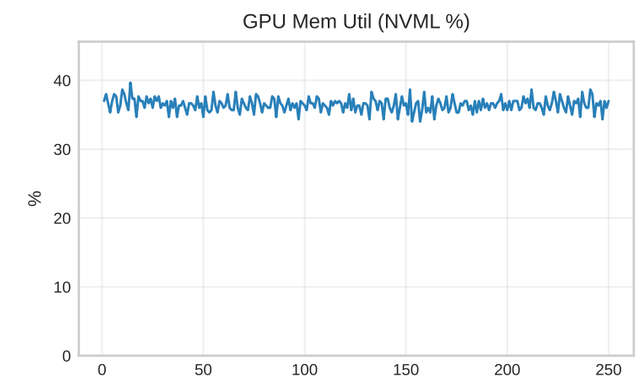
Resource Timeline — Milabench, w=4, B=64

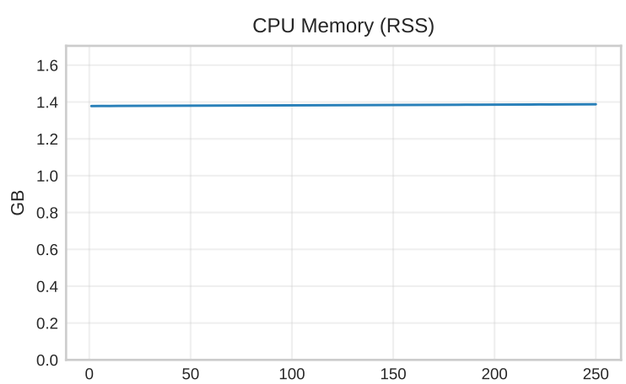
resource_util (with CUDA sync)

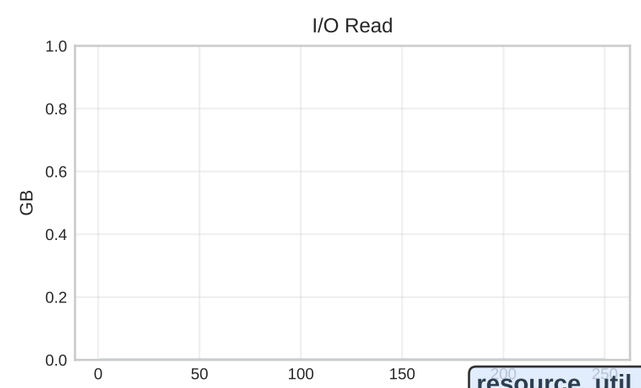


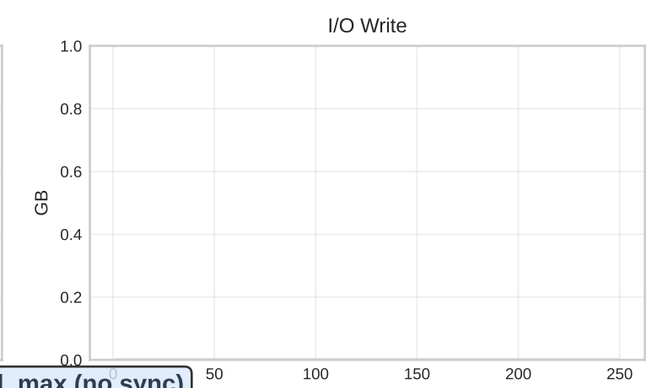




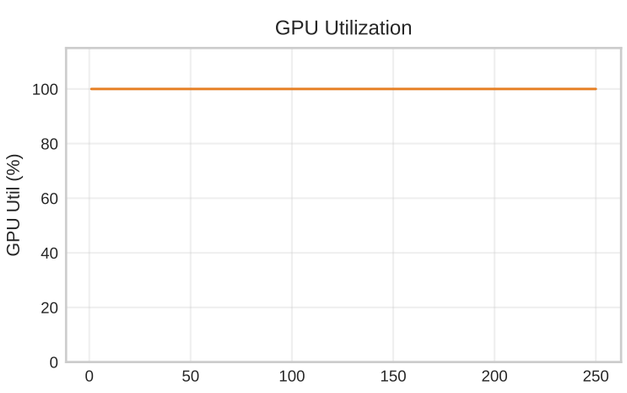


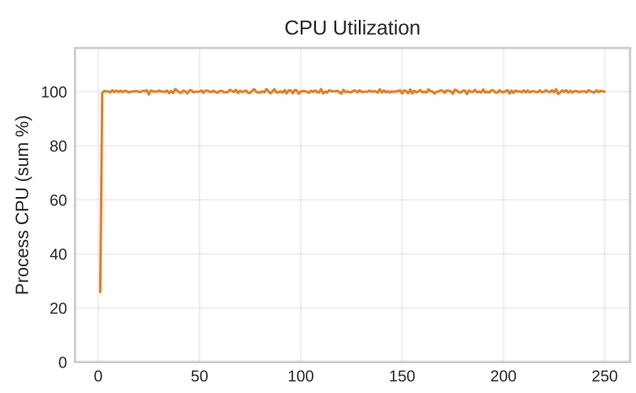


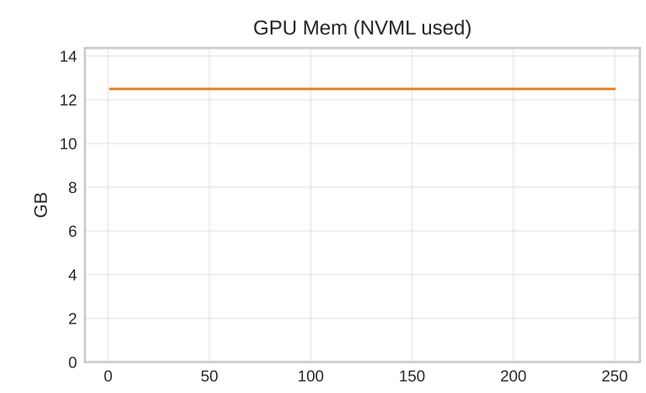


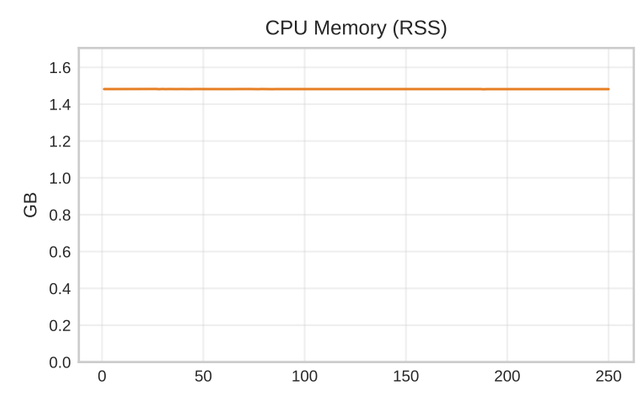


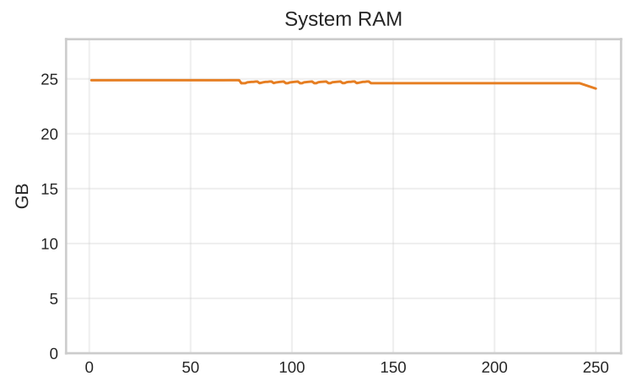
resource_util_max (no sync)

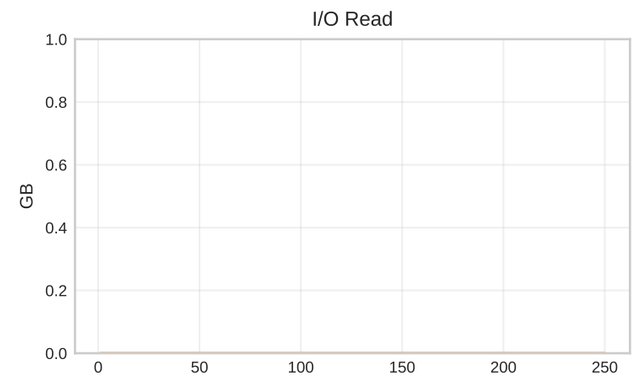


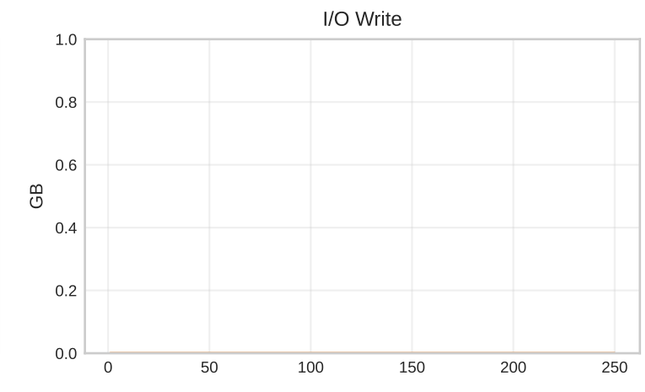






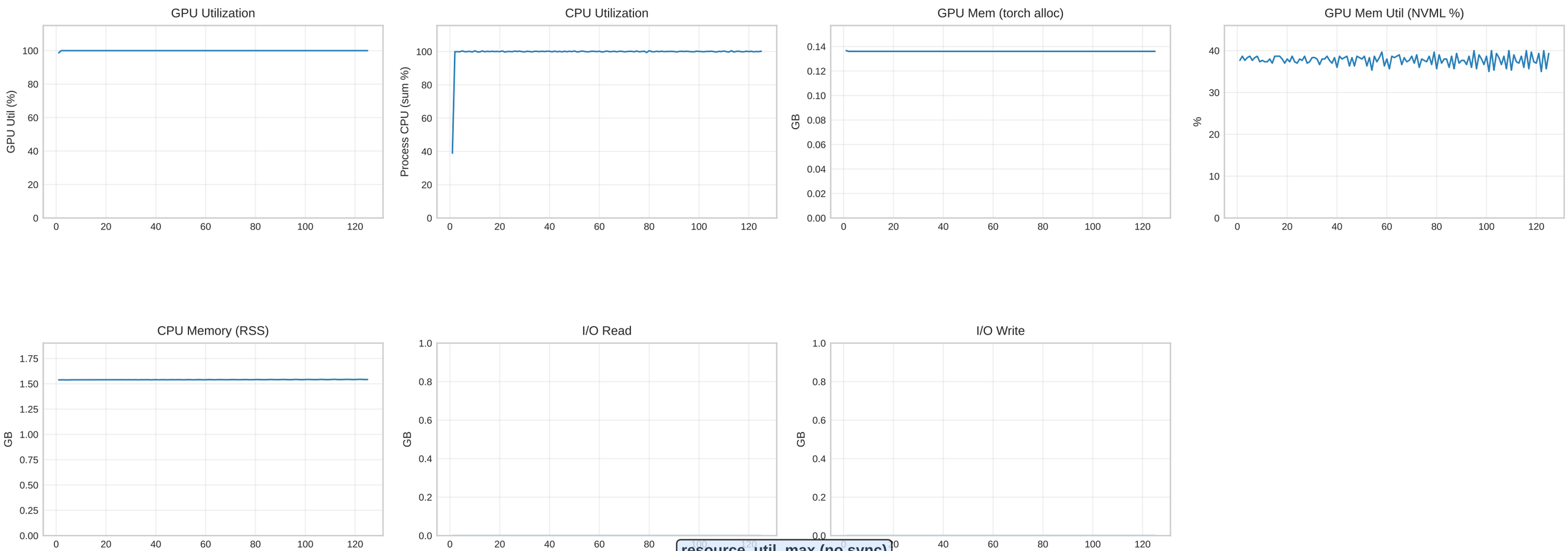
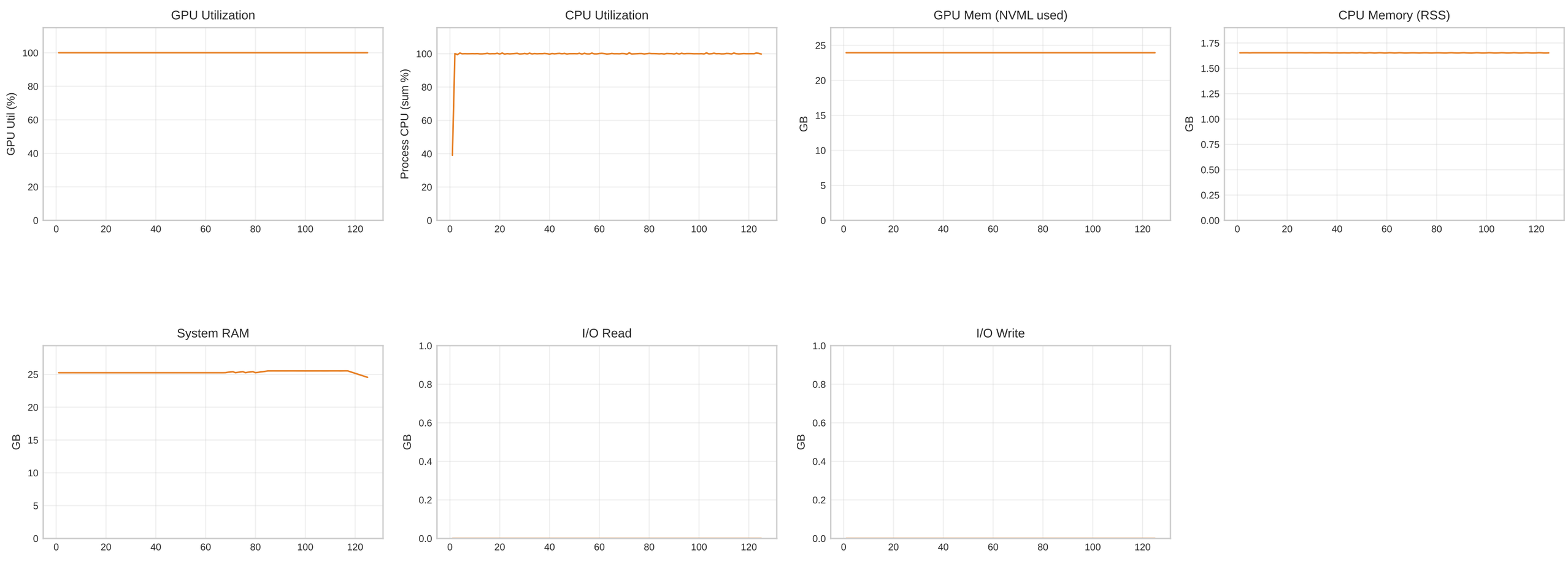






Resource Timeline — Milabench, w=4, B=128

resource_util (with CUDA sync)



resource_util_max (no sync)